PROJECT INDUSTRY PPG: RECOVERABLE ASSETS (RA) & INVENTORY RISK MANAGEMENT (IRM)

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JUN 2026

ACKNOWLEDGEMENT

First and foremost, we would like to express our sincere gratitude to Universiti Teknologi Malaysia (UTM) and the Faculty of Computing for providing us with the opportunity to carry out this project as part of our academic learning and practical development experience. We would also like to extend our appreciation to our lecturer and project supervisor for the guidance, advice, and continuous support provided throughout the completion of this project. The feedback and suggestions given have helped us improve our understanding of data engineering concepts, system design, cloud-based pipeline development, and business intelligence reporting. Our appreciation is also extended to PPG for providing the case study context and business problem related to Recoverable Assets (RA) and Inventory Risk Management (IRM). The case study has allowed us to understand how data engineering solutions can be applied to real industry challenges, especially in inventory risk analysis, financial provision monitoring, and customer sales order impact tracing. In addition, we would like to thank all group members for their cooperation, commitment, and contribution throughout the project. Each member contributed to different parts of the system, including data ingestion, Silver Layer transformation, Gold Layer galaxy schema development, Power BI dashboard creation, testing, and report documentation. The teamwork and shared responsibility among group members were important in ensuring the successful completion of the project. Finally, we would like to thank our classmates, friends, and all individuals who directly or indirectly supported us during the development of this project. Their assistance, encouragement, and feedback were greatly appreciated.

ABSTRACT

This project focuses on the development of an end-to-end data engineering pipeline for PPG Recoverable Assets (RA) and Inventory Risk Management (IRM). The system was designed to address inventory visibility issues, financial write-down risks, and possible stockout impacts on customer sales orders. Six synthetic operational CSV datasets were used, consisting of materials, inventory snapshots, purchase orders, production orders, suppliers, and sales orders. The pipeline was developed using Microsoft Azure services based on the Medallion Architecture, which includes Bronze, Silver, and Gold layers.

Azure Data Factory was used to ingest the source datasets into Azure Data Lake Storage Gen2, where raw and validated data were organized in structured storage layers. Azure Synapse Serverless SQL was then used to perform Silver Layer transformations because the Spark pool could not be executed due to the Azure vCore quota limitation. In the Silver Layer, business logic was applied to identify Recoverable Assets, Magna Carta Dead Stock, Magna Carta Expired materials, stockout-risk materials, and impacted customer sales orders. The Gold Layer was designed using a galaxy schema, with fact_inventory_status as the central fact table and related dimension tables for material, supplier, customer, and date information.

The final outputs were connected to Power BI to provide interactive dashboards for inventory health, financial provision, and operational risk analysis. Overall, the developed Azure-based pipeline successfully transformed raw operational data into reliable business insights for inventory risk monitoring and decision support.

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LIST OF ABBREVIATIONS

ADLS	-	Azure Data Lake Storage
ADF	-	Azure Data Factory
CSV	-	Comma-Separated Value
DAX	-	Data Analysis Expressions
ELT	-	Extract-Load-Transform
FR	-	Functional Requirement
IRM	-	Inventory Risk Management
KPI	-	Key Performance Indicator
MC	-	Magna Carta
NFR	-	Non-Functional Requirement
RA	-	Recoverable Asset
SQL	-	Structured Query Language

CHAPTER 1

INTRODUCTION

1.1 Introduction

In the modern manufacturing landscape, efficient inventory management is a critical driver of financial stability and operational success. This project focuses on the development of an end-to-end data engineering pipeline for PPG, specifically designed to address Recoverable Assets (RA) and Inventory Risk Management (IRM). By utilizing a fully synthetic dataset comprising six operational files, including materials, inventory snapshots and sales orders, this initiative seeks to automate the identification of inventory health. By implementing a three-layer Medallion Architecture (Bronze, Silver, and Gold) within the Microsoft Azure ecosystem, the project transforms raw operational data into actionable business intelligence. The ultimate goal is to provide a centralized analytics platform that empowers stakeholders to optimize stock levels, minimize financial waste and ensure customer satisfaction through data-driven insights.

1.2 Problem Background

Global organizations like PPG face significant challenges in maintaining visibility over vast inventories of raw materials. Without an automated pipeline, it is difficult to identify "Recoverable Assets" which are materials held in excess of 12 months of consumption, leading to inefficient capital allocation. Furthermore, the lack of a standardized system to flag Magna Charta (MC) materials specifically "Dead Stock" with no consumption for nine months or materials that have surpassed their expiry dates presents a major financial risk. These materials often require a 100% financial write-down. Yet without precise tracking, the total provision amount remains unknown and unquantified.

Beyond financial reporting, operational risks such as stockouts pose a direct threat to the supply chain. Currently, the inability to accurately identify materials running lower than their lead time makes it nearly impossible to predict which production orders will fail or which open customer sales orders will be delayed. Manually reconciling these disparate

data sources is prone to error and time-consuming. Therefore, there is a critical need for a robust, automated data engineering solution that can ingest raw data, apply complex business logic and visualize inventory risks in real–time to prevent production failures and financial losses.

1.3 Project Aim

The primary aim of this research project is the design and implementation of a scalable and end-to-end data engineering pipeline using the Medallion Architecture on the Microsoft Azure platform (Codiin., 2024). This system is to automate the identification of inventory obsolescence risks and the quantification of financial provisions for Recoverable Assets (RA) and Magna Carta (MC) materials. Furthermore, a centralized analytical framework is established to evaluate the impact of material shortages on customer sales orders using a high-fidelity Power BI dashboard.

1.4 Project Objectives

To ensure the successful completion of the project aim, the following technical objectives were defined:

1. Ingestion and Data Quality Assurance: A robust ingestion mechanism is developed to extract data from six distinct source files into an Azure Data Lake (Bronze Layer), where automated data quality checks, including null detection and type validation, are rigorously applied.
2. Business Logic Implementation: Complex inventory logic is engineered within the Silver Layer to calculate 12-month consumption rates and to systematically flag materials as Recoverable Assets, Dead Stock, or Expired based on predefined PPG industry criteria.
3. galaxy Schema Optimization: A standardized data warehouse model is constructed in the Gold Layer, where transformed datasets are loaded into a central table to facilitate high-performance analytical querying.
4. Predictive Risk Assessment: Analytical procedures are implemented to identify raw materials at risk of stockout by evaluating current inventory levels against lead times, and the resulting impact on open customer sales orders is traced.

5. Interactive Visualization: An enterprise-level Power BI dashboard is developed to provide stakeholders with clear insights into total financial write-down amounts (Magna Carta provisions) and to answer critical operational questions regarding material reorder levels and customer impact.

1.5 Project Scope

This project focuses on designing a data engineering pipeline using Microsoft Azure and Power BI to support inventory risk analysis. The system processes six structured CSV datasets, including materials, inventory snapshots, purchase orders, production orders, suppliers, and sales orders. These datasets are ingested into a data lake and processed through a three-layer pipeline architecture which consists of Bronze, Silver, and Gold layers.

In the Bronze layer, raw data is ingested with minimal transformation, while basic data quality checks such as null detection, data type validation, and duplicate handling are performed. The Silver layer applies PPG's business logic to transform the data, including calculating 12-month consumption, flag Recoverable Assets, classifying Magna Carta Dead Stock and Expired materials, computing financial provisions, and detecting stockout risks along with their impact on customer orders. In the Gold layer, the processed data is loaded into an analytics-ready format using a star schema model to enhance the efficiency in querying and reporting. The final output is an interactive dashboard developed in Power BI that provides insights into inventory status, financial exposure, and operational risks.

This project is limited to batch data processing using static datasets and does not include real-time data streaming, predictive analytics, or artificial intelligence techniques. It also does not integrate with external enterprise systems, as the project uses synthetic data in a standalone environment.

1.6 Project Importance

Inventory management is crucial in manufacturing and supply chain operations as inefficiencies management can lead to financial and operational issues. Excess inventory

ties up resources and increases the risk of obsolescence, while insufficient stock slows down production and affects customer satisfaction.

The importance of this project is that it provides a structured and data-driven solution to identify and manage the inventory risks faced by organizations. By identifying Recoverable Assets and Magna Carta materials, the organizations can take timely actions such as redistributing excess stock, reducing waste, and performing financial write-downs.

Moreover, the ability to identify stockout risks and analyze their impact on customer orders can improve the operational planning and enhance service reliability. The integration of a modern data engineering pipeline provides an understanding of how raw operational data can be transformed into meaningful insights for better decision-making.

Overall, the project improves inventory visibility, financial control, and operational efficiency across the enterprise environment.

1.7 Report Organization

The report was organized into several chapters to guide from the problem context to the final outcomes of the project. Chapter 1 introduces the overall background about of inventory risk at PPG, outlines problem statement, and presents the project aim, objectives, scope, and importance, focus on why an end-to-end data engineering pipeline is needed for Recoverable Assets (RA) and Inventory Risk Management (IRM). Chapter 2 discusses the literature review, covering previous studies, key data engineering concepts such as the Medallion Architecture, inventory obsolescence theories including RA and Magna Carta classifications, and the main technologies used in this work, namely Microsoft Azure and Power BI. Chapter 3 explains the methodology and system design, describing the data sources, Bronze–Silver–Gold pipeline, data quality checks, transformation logic, and how PPG's business rules are implemented across the layers. Chapter 4 presents the system analysis and design, including the company organization structure, requirements gathering techniques, usecase diagram, functional and non-functional requirements, proposed data architecture, and the database design utilizing a galaxy schema. Chapter 5 discusses the system development and testing, covering the implementation of the Bronze, Silver, and Gold layers, the main coding functions and essential interfaces, the Power BI dashboard deployment, and the validation results from black box, white box, and user testing.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter reviews the theoretical foundations, previous studies, and technologies that support the development of the PPG Recoverable Assets (RA) and Inventory Risk Management (IRM) Analytics System. As inventory management becomes increasingly dependent on data-driven decision-making, organizations require reliable methods to transform large volumes of operational data into meaningful business insights. Therefore, this chapter examines the concepts and practices that underpin modern data engineering, inventory risk management, and business intelligence solutions within manufacturing environments.

The discussion begins with previous studies related to supply chain analytics and data engineering pipelines, followed by the theories that form the foundation of the proposed solution, including Medallion Architecture, inventory obsolescence management, ERP data integration, stockout risk analysis, and the ELT processing approach. In addition, the chapter reviews the key technologies used in this project, namely Microsoft Azure and Power BI, and explains how these technologies support scalable data processing, business rule implementation, and interactive reporting. Together, these studies provide the theoretical and technical justification for the design decisions presented in the subsequent chapters.

2.2 Related Previous Study

The transition from old, inflexible ETL (Extract, Transform, Load) methods to more adaptable, cloud-native ELT (Extract, Load, Transform) frameworks is highlighted in recent industrial case studies in Data Engineering. Prior supply chain analytics pipeline installations have shown that manual inventory tracking frequently overlooks long-term obsolescence, resulting in large financial disparities (Data Warehousing Architecture | Databricks on AWS, 2025).

In particular, research on end-to-end data pipelines emphasizes the use of synthetic datasets to mimic intricate operational systems without jeopardizing private company information. Additionally, research on manufacturing industries, which is comparable to the PPG use case, has demonstrated that detecting "bottleneck" materials that result in production delays requires combining different data sources, such as purchase orders, inventory snapshots, and sales records (The Data Warehouse Toolkit, 2013). According to these studies, computerized "Dead Stock" flagging systems greatly increase the precision of financial write-down provisions.

2.3 Related Theory Used

The implementation of this project is grounded in three core theoretical domains:

2.3.1 Medallion data Architecture

1. Layer 1: Bronze (Raw Ingestion): The idea of preserving unchangeable raw data. In order to do this, raw CSV files must be ingested and basic quality checks such as duplicate handling and null detection must be applied while maintaining the original data.

2. Layer 2-Silver (Transformation): Emphasizes data enrichment and the usage of intricate business logic. Raw consumption records are converted into actionable flags, such as Recoverable Assets or Magna Carta Dead Stock, under this layer.

3. Layer 3-Gold (Analytics-Ready): This layer puts converted data into a data warehouse for high-performance reporting. It is based on the star schema principle. All RA/MC flags are stored in the primary fact table (fact_inventory_status) in preparation for dashboard construction (Heizer et al., 2017).

2.3.2 Supply Chain Theory: Inventory Obsolescence and Risk Management

The business logic finds waste in the six supplied synthetic CSV files by applying accepted inventory management principles:

1. Recoverable Assets (RA): The concept of locating capital invested in surplus raw materials, particularly those that have been consumed for more than a year.
2. Magna Carta (MC) Dead Stock: This term describes items that have not been used for nine months or longer and do not have an expiration date (Christopher, 2011).
3. MC Expired: Materials that need a 100% financial write-down or provision because their lot expiration date has passed.

2.3.3 Enterprise Resource Planning (ERP) Data Integration

The Enterprise Resource Planning (ERP) system is a basic software platform relied on by global manufacturing enterprises like PPG, in which accounting, inventory and supply chain data are integrated into a unified coordination system (Bonthu, 2025). Traditional ERP systems are mainly designed for transaction data to be stored, while a special data engineering process is required by the implementation of this project for real-time processing to be achieved, so that analysis can be performed by PPG when the data arrives without reliance on delayed batch updates (Bonthu, 2025). Six synthetic CSV files (representative materials, inventory snapshots and sales orders) are used by the project to simulate the export of data obtained from the actual PPG operational ERP environment, through which this integration is achieved.

A key theoretical component of this integration is Multi-Domain Master Data Management (MDM), by which consistency of information between products, suppliers and customers within the Medallion Architecture is ensured. This synchronization is essential for Recoverable Assets (RA) and Inventory Risk Management (IRM) logic, as it allows the system to identify materials in excess or at risk of stockout "on the fly" (Bonthu, 2025). Moreover, operational efficiency can be improved by the integration of real-time processing in the ERP system, as replenishment orders are automated and production plans are adjusted, and the original transaction variables are ultimately transformed into strategic insights into managing the terms of the Magna Carta (MC) (Bonthu, 2025).

2.3.4 Predictive Stockout and Fulfillment Theory (Operational Theory)

This theory addresses the impact of material shortages on production and sales:

1. Lead Time vs. Inventory Level: The logic used to flag materials at stockout risk when levels fall critically low relative to their replenishment lead time.
2. Sales Order Impact Tracing: The theoretical link between upstream raw material availability and downstream customer fulfillment, identifying which open sales orders are affected by shortages.

2.3.5 Extract-Transform-Load/Extract-Load-Transform

The data integration approach of the project is based on the theoretical pattern of the ETL (Extract, Transform, Load) model to the new cloud-based ELT (Extract, Transform, Load) model. This is an important division in using a Medallion Architecture on Microsoft Azure.

Within a traditional ETL process, information is dependent on a processing unit and then delivered to the destination for storage. Traditional ETL can be inflexible in modern environments because the transformation logic is linked with the ingestion phase (Kancharla, 2025). When the business logic of the Recoverable Assets or Magna Carta classifications of PPG must be changed, the whole extraction process must be repeated many times. It consumes a lot of time and costs only using this old method.

The project is based on the ELT model that utilizes the high-performance capability of the cloud. Raw data in the model will be removed from the source and loaded directly into the layer before any transformation takes place. The ELT approach is a better choice when integrating the cloud because it is more scalable, and the logic of the transformation can be adjusted (Kancharla, 2025). Extract & Load is used for raw synthetic data in the 6 source CSVs and directly loaded into the Bronze Layer of the Azure Data Lake without adjustment. Next, the Silver Layer does some cleaning like removing duplicate rows, fixing nulls, and standardizing the format. Proceed with Transform to perform calculations to identify 12-month consumption and flag Dead Stock are executed in Gold Layer with

distributed computing. Lastly, the Gold Layer is the final year where it can create the dimension table and be ready for the dashboard.

The Medallion Architecture applied is based on the selection of ELT. The project has a 'Source of Truth' because raw synthetic CSV files are loaded into the Azure Data Lake (Gentyala, 2024). Researchers warn that only transferring data is not the way to handle a lot of care to prevent the so-called data debt. The Medallion Architecture is highly efficient and needs proper validation and control of each stage. It is likely to collect anti-patterns and create broken or unreliable data pipelines (Gentyala, 2024).

2.4 Related Technology Used

This project utilizes the cloud-based data to handle the transforming of the data from raw files to the new updated version of business dashboard. The architecture for Recoverable Assets (RA) and Inventory Risk Management (IRM) is created upon these two technologies which is Microsoft Azure and Microsoft Power BI.

2.4.1 Microsoft Azure

Microsoft Azure provides the enterprise backbone for the Medallion Architecture (Bronze, Silver, and Gold Layer) required for this project. There are a few types of Microsoft Azure that can be used in this project.

- Azure Data Factory (ADF): ADF functions as an orchestration and integration service. ADF is used to create automated, data-driven workflows that can manage the ingestion of synthetic CSV files into cloud environments. By using the ADF, the process will become a repeatable and automated pipeline where it is necessary for professional data engineers (Global, 2025).
- Azure Data Lake Storage (ADLS) Gen2: ADLS Gen2 is specifically designed for big data analytics by integrating the scalability of Blob (Binary Large Object) storage with a hierarchical namespace (On The Use of Hash Functions for Defect Detection in Textures for In-camera Web Inspection Systems, 2002). This allows the project to organize the six operational datasets easily into structured folders. This structure is important for the Medallion Architecture, as it ensures high

throughput and low latency when analytical engines access the data for inventory business.

- Azure Synapse Analytics: It can serve as the Gold Layer by providing a unified that joins the big data processing with enterprise data warehousing. The data is organized into a Galaxy Schema, including the center of the inventory status table linked to relevant tables (Das et al., 2019). This part is also the host of the Galaxy Schema where the fact tables are located.
- Azure Security (Key Vault): Azure Key Vault is to maintain the integrity of a data engineering pipeline; security and identity management need to be considered. It is used to store cryptographic keys, certificates, and secrets. In this project, Key Vault is used to store the connection strings and keys for the Azure Data Lake and SQL database, preventing the credentials from being hard-coded int he Azure Data Factory and SQL script.

2.4.2 Microsoft Power BI

Power BI serves as the final layer of the stack where it can perform Data Visualization and Business Intelligence. It also can do the report or dashboard for better understanding and observation. This software is used to design data models and interactive reports. It connects directly with the Azure Gold Layer to import the galaxy schema with a complete relationship. The final deliverable, like an interactive dashboard, is used to create a dashboard that contains a lot of useful information like Magna Carta. It also provides simple observations to stakeholders to identify the stockout risks and trace the impact on customer sales orders.

2.5 Summary

In this chapter, the theoretical and technical underpinning used for designing and implementing the PPG Recoverable Assets (RA) and Inventory Risk Management (IRM) data engineering pipeline were introduced. The review was conducted in three key areas: firstly, in relation to previous studies that provide a broader contextual background of the project, secondly, in terms of core theories that underpin the business logic and the architectural approach, and thirdly, in terms of the key technologies that facilitate technical implementation of the project.

This collection of literature revealed that there is a well-documented improvement in data engineering practice that's gaining traction as a new paradigm, moving from traditional Extract-Transform-Load (ETL) pipelines to modern cloud-native Extract-Load-Transform (ELT) pipelines. Previous supply chain analytics projects showed that manual inventory systems fail to identify long term obsolescence consistently and create large monitory mismatch between the inventory and financial statements up till the time of periodic audits. Another key finding in manufacturing analytics was that numerous data domains must be woven together in a single analytical model to identify overall bottlenecks in materials and obsolete stock. A third insight from manufacturing analytics was the need to combine multiple data domains into a single analytical model: inventory snapshots, production orders, purchase orders, sales records, etc. The results directly confirmed the multi-source approach used in this project, which uses 6 synthetic CSV files representing the PPG operational environment and combines them into a single Medallion Architecture pipeline.

To inform the project design choices, five theoretical domains were examined. The Medallion Architecture serves as the structure of data that the pipeline uses, with data stored in three increasingly refined layers: Bronze Layer (immu*table* raw data), Silver Layer (business transformation rules applied to the data), and Gold Layer (transformed data structured into a Galaxy Schema for dashboard consumption). The specific business rules for the three core classifications of Recoverable Assets, Magna Carta Dead Stock and MC Expired materials are encoded in the Silver and Gold layers and are directly answering the project's core business questions through the principles of Supply Chain and Inventory Obsolescence Theory. The Enterprise Resource Planning Data Integration Theory described where the six synthetic CSV files came from and explained the concept of Multi Domain Master Data Management data to be consistent across the Material, Supplier, Customer and Inventory Data Domains throughout the pipeline. Predictive Stockout and Fulfillment Theory was created to describe the logic behind determining the materials to watch for a stockout by comparing stock cover days with supplier lead times, and the downstream effect of material shortages on outstanding customer sales orders. Last but not least, in a theory of comparison, the ETL approach proved to be suboptimal, and the ELT approach was deemed as superior, with the decision to load raw data in Azure Data Lake Storage being made before any data transformation, where the business logic like the classification rules used in the Magna Carta and the consumption calculations can be changed without bringing the data back from the data lake.

Microsoft Azure and Microsoft Power BI were identified by the technology review as the two pillars of the platform implementation. In the Microsoft Azure environment, four services were studied according to their specific roles in the pipeline. The orchestration layer in Azure Data Factory is used to automate the ingestion of synthetic CSV files into the cloud environment. Azure Data Lake Storage Gen2 offers the hierarchical structure of storage in which the six operational datasets are organized in the Bronze and Silver layers. As per ELT principle, Azure Synapse Analytics Serverless SQL Pool is hosting the Gold Layer Galaxy Schema that allows creation of SQL views over the validated data in the Bronze layer, without any need for physical data movement. All connection strings and credentials that are used throughout the pipeline are secured in Azure Key Vault, which removes the need to store sensitive information in pipeline scripts or configurations. At the top of the pipeline stack is Microsoft Power BI, where the Galaxy Schema is imported, cross-fact relationships are created, and the interactive dashboard that aggregates all seven answers to the business questions is created and presented in a single reporting interface to PPG stakeholders.

When combined, the previous studies of theory and technology review provide a very solid and clear rationale for each of the bigger design decisions for this project. The ELT model is backed by academic research and the necessity of having the business logic changeable. The Medallion Architecture makes sense when there is a need to maintain the integrity of raw data and then gradually transform this data into outputs that are ready for analysis. The Galaxy Schema, in the Gold Layer, is based on the theory of dimensional modelling and tested with the multi-subject nature of the business questions of the project, spanning both inventory risk analysis and sales order impact. Azure Synapse Analytics SQL Pool and Microsoft Power BI together offer a low-cost, scalable, maintainable platform that directly meets the operational and financial goals of the PPG Recoverable Assets and Inventory Risk Management programmed. All subsequent chapters of the book on system design, development, and testing are based on these theoretical and technological underpinnings.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter introduces the system development methods adopted by the industry case research institute based on PPG. The project is mainly committed to solving problems related to Recoverable Assets (RA) and Inventory Risk Management (IRM), especially for the detection of inventory outage and out-of-stock risks. A structured method is adopted to ensure that the development process can be carried out systematically and in an orderly manner. The method is divided into several stages, namely planning, analysis, design and implementation, and each stage is described in detail to illustrate the process of system development. In addition, the project planning schedule and system requirements, including hardware and software components, were also discussed. Overall, this chapter outlines the methods and processes involved in developing the proposed data engineering solution.

3.2 Introduction of Methodology

The waterfall development method has been selected as the development method of this project. The method is characterized by a linear and sequential pattern, and the next stage will begin only after each stage is completed. The main stages include planning, analysis, design and implementation. This method emphasizes the methodical advancement of activities, so that the development process can be carried out in a clear and clear way, which is suitable for projects with stable and easy-to-understand requirements.

The reason for choosing the waterfall approach is the nature of the PPG case study, because the scope of the problem, data set and business rules have been clearly defined in the initial stage. The project needs to develop a data engineering pipeline to process multiple data sets, apply the business logic of RA and Magna Carta (MC) classification, and generate insights related to inventory risks. Since the demand is predetermined and is unlikely to change significantly, the sequential development method can be effectively adopted. In addition, the method supports comprehensive documentation and systematic verification at

each stage, which is crucial to ensure the accuracy and reliability of data processing. Therefore, the waterfall method is considered to be suitable for guiding the structured development of the proposed system.

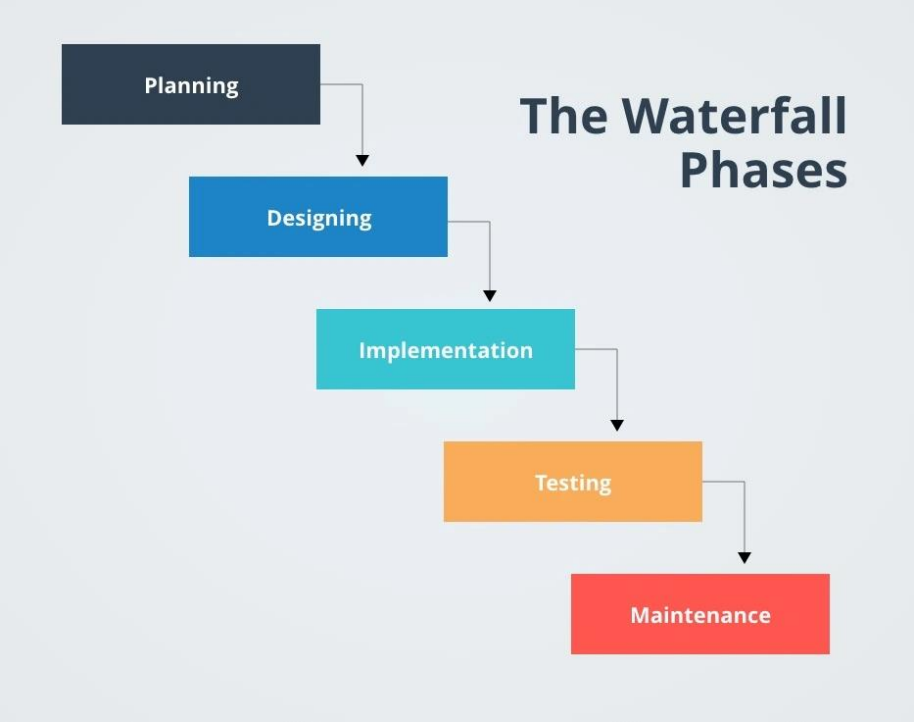


Figure 3.2.1: Waterfall Phases

Figure 3.2.1 shows all five stages waterfall phases.

3.3 Phases of Methodology

This project adopts the Waterfall methodology to provide a structured and systematic approach for developing the PPG Recoverable Assets (RA) and Inventory Risk Management (IRM) Analytics System. The methodology is divided into four sequential phases: Planning, Analysis, Design, and Implementation. Each phase produces a specific set of deliverables that serve as inputs for the next phase, ensuring that project objectives, business requirements, and technical specifications are clearly defined before development activities begin.

The use of Waterfall is appropriate for this project because the business requirements, inventory risk classifications, and operational datasets were already well-defined through industry engagement with PPG. This allows the project to progress through a controlled development process where requirements are analyzed, transformed into system designs,

and subsequently implemented within the Microsoft Azure environment. Through this approach, the project maintains traceability between business problems, system requirements, architectural decisions, and final analytical outputs.

3.3.1 Planning Phase

The planning phase establishes the overall direction of the project by defining the objectives, scope, stakeholders, resources, and expected outcomes of the proposed system. Based on the challenges identified within PPG, the project aims to develop an end-to-end data engineering pipeline capable of automating the detection of Recoverable Assets, Magna Carta inventory classifications, and stockout risks using six structured ERP-derived datasets. Defining these objectives early ensures that all subsequent activities remain aligned with the operational and financial needs of the organization.

During this phase, key stakeholders from supply chain operations, finance and governance, and IT departments are identified to ensure that both business and technical requirements are properly considered. The project scope, technology stack, timeline, and potential risks are also documented to support effective project execution. As a result, the planning phase provides a clear roadmap that guides the development process and reduces uncertainty throughout the remaining phases of the project

3.3.2 Analysis Phase

The Analysis Phase is an important part of the waterfall methodology as it involves breaking down a high-level business problem into specific technical requirements. For this project, the analysis is concerned with converting the business problems faced by PPG Industries into a data engineering specification.

3.3.2.1 Data Collection

This phase included an industry engagement with PPG Industries, a multinational manufacturer based in Malaysia. Through this industry visit, it is crucial to understand the issue of inventory outdated, where the stock on hand is not aligned with the business needs. The lack of a data-driven system at the plant was identified as the reason why stock excesses and financial write-offs are not being identified and quantified effectively. The six core datasets representing PPG's operating systems were obtained: materials.csv, inventory_snapshot.csv, purchase_orders.csv, production_orders.csv, suppliers.csv and

sales_orders.csv. The visit provided the operational insights into PPG's internal risk classifications which are Recoverable Assets (RA) and Magna Carta (MC). The first-hand knowledge of these definitions enables us to identify the data requirements for the next phase of the design process.

3.3.2.2 Data Requirement Analysis

The source files were analyzed in detail to inform the design of the Medallion Architecture. The primary and foreign keys were identified to facilitate the joining of different tables to join the inventory with stock consumption data. The importance of dataset in the pipeline was evaluated. For example, the inventory_snapshot.csv was identified as the key source to detect expiry of materials because of the inclusion of lot_expiry_date field in it. The specification for automated quality checks was defined to be implemented in the Silver layer.

3.3.2.3 Business Logic Formalization

The descriptive business logic provided by PPG was formalized into quantitative logical expressions to be implemented in the Silver, and Gold layers. This guarantees a methodical movement of cleaning information to reasonable business insights.

1. Silver Layer (Data Standardization): This layer was formed where the logic was laid down in a manner to ensure data integrity before business calculations started. This involves elimination of duplicate records, standardization of date formats, and the treatment of null values in the consumption fields to provide a reliable source of truth to the successive layers.
2. Recoverable Assets (RA): Criteria were established to identify raw materials for which the current stock on hand is greater than the expected 12-month consumption.
3. Magna Carta (MC) Dead Stock: Criteria was established to identify materials with no consumption for 9 or more months where there is no expiry.
4. Magna Carta Expired: Logic was defined to identify materials where a lot of expiries has passed, and a 100% financial provision is required.
5. Operational Risk Tracing: Logic was set to trace the effect of material shortages to sales orders to ensure a link from material availability to sales order execution.

6. Gold Layer (Data Modelling): The last phase and it involves formalization of the data into a Galaxy Schema design. Logic was also organizing the transformed data into two central fact tables that are linked with the dimension tables. This layer is optimized to provide the best answers to the seven core business questions using the Power BI visuals dashboard.

3.3.3 Design Phase

The design phase translates the findings from the analysis phase into a structured technical blueprint for the PPG inventory risk management system. Based on the business rules formalized in the analysis phase covering Recoverable Asset classification, Magna Carta Dead Stock and Expired detection, financial provision calculation, and stockout risk tracing, three interconnected design decisions were made which are the data pipeline architecture, the analytical data model, and the reporting layer. These decisions form the technical blueprint that guides the implementation phase.

3.3.3.1 Medallion Architecture Design

Based on the requirements identified during the analysis phase, the Medallion Architecture was selected as the pipeline design approach. This three-layer architecture consisting of Bronze, Silver, and Gold layers was chosen because it provides a clear separation of concerns between raw data preservation, business logic application, and analytical output, which directly addresses PPG's need for traceable and auditable inventory risk classification.

The Bronze layer was designed to serve as the raw ingestion zone. All six source CSV datasets consisting of materials, inventory snapshot, purchase orders, production orders, suppliers, and sales orders are to be ingested into Azure Data Lake Storage Gen2 without any modification to their original structure. This design decision ensures that the source data always remains fully preserved and traceable. Basic data quality checks including null detection, duplicate identification, and data type verification are planned at this stage to separate valid records from those requiring quarantine before downstream processing begins.

The Silver layer was designed to serve as the business logic and standardisation zone. Based on the business logic formalized during the analysis phase, this layer is planned to clean and integrate data across multiple source files before applying PPG's inventory risk rules. The Silver layer is divided into two parallel workstreams. The first workstream is designed to handle RA classification and MC provision logic, producing outputs for Q2, Q3, Q4, and Q5. The second workstream is designed to handle stockout detection and sales order impact tracing, producing outputs for Q1, Q6, and Q7.

The Gold layer was designed to serve as the analytics-ready zone. Transformed outputs from the Silver layer are planned to be consolidated into a structured galaxy schema data model optimised for reporting. Power BI is planned to connect directly to the Gold layer to deliver the final interactive dashboard.

3.3.3.2 Galaxy Schema Data Model Design

Based on the analytical requirements gathered during the analysis phase, the Gold layer data model was designed using a galaxy schema, also known as a fact constellation schema. This design was chosen over a simpler star schema because the PPG system needs to address two distinct analytical domains simultaneously which are inventory risk status and sales order impact, each requiring its own fact structure while sharing common descriptive dimensions.

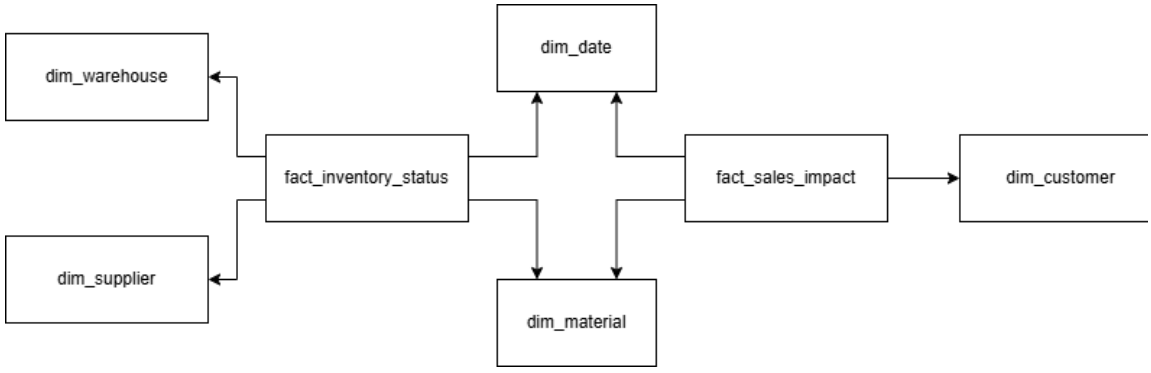


Figure 3.3.3.2.1: Conceptual Star Schema Design of PPG RA and IRM System

As illustrated in Figure 3.3.3.2.1, the galaxy schema is built around two fact tables. The first fact table, fact_inventory_status, is designed to store all computed inventory risk

indicators per material including RA flags, MC classifications, stockout risk status, financial provision values, stock cover days, and consumption metrics. The second fact table, fact_sales_impact, is designed to capture open customer sales orders that are at risk of delayed fulfilment due to material shortages, linking sales order details to the affected materials, finished products, and customers.

The two fact tables share dim_date and dim_material as common dimension tables, allowing consistent time-based and material-level filtering across both analytical domains. Beyond the shared dimensions, each fact table is connected to its own set of dedicated dimension tables based on its analytical scope. The fact_inventory_status table is additionally linked to dim_warehouse and dim_supplier, which provide warehouse location attributes and supplier profile information respectively to support inventory positioning and stockout risk analysis. The fact_sales_impact table is additionally linked to dim_customer, which holds customer identifiers derived from sales order records to enable tracing of which customers are impacted by material shortages.

The galaxy schema design was selected for three reasons. First, the two-fact structure cleanly separates inventory risk analysis from sales order impact analysis while avoiding duplication of shared reference data. Second, sharing dim_date and dim_material across both fact tables ensures consistent filtering and cross-domain analysis within Power BI. Third, the design is scalable and can accommodate additional fact tables in the future without restructuring the existing shared dimensions.

3.3.3.3 Power BI Analytics Dashboard Design

Based on the seven business questions defined during the analysis phase, the reporting layer was designed using Power BI Desktop connected directly to the Gold layer in Azure Synapse. Power BI was selected because it is compatible with Azure Synapse that supports the galaxy schema data model through its relationship engine, and provides an interactive reporting experience suitable for business stakeholders.

The dashboard was designed across 2 reporting pages, each addressing a distinct group of business questions. The first page is planned to cover materials management and inventory stock health by presenting Q1, Q2, Q3, Q4 and Q5 findings related to reorder levels,

Recoverable Assets, MC Dead Stock, MC Expired materials, and the total financial write-down amount. The second page is planned to focus on operational risk by presenting Q6, and Q7 results related to stockout risk materials and impacted customer sales orders.

The dashboard design includes key performance indicator cards for high-level metrics, detailed material tables and suitable charts for drill-down analysis to allow stakeholders to explore inventory risk across different material groups. All visuals are planned to connect directly to the Gold layer to ensure that every figure displayed on the dashboard is derived from consistent, validated, and fully traceable data that has passed through all three layers of the Medallion Architecture.

3.3.4 Implementation Phase

The implementation phase translates the technical blueprints established in the design phase into a functional, end-to-end data engineering pipeline and reporting system on Microsoft Azure and MS Power BI. This phase covers the physical creation of storage architectures, orchestration of data flows, programmatic execution of business rules, data warehouse modeling, and dashboard construction. The implementation is executed across four main stages to reflect the Medallion Architecture:

3.3.4.1 Storage and Ingestion Environment Setup (Bronze Layer)

The foundation of the pipeline is established using **Azure Data Lake Storage (ADLS) Gen2**, organized into containers reflecting the pipeline layers.

1. **ADF Pipeline Orchestration: Azure Data Factory (ADF)** is configured as the central management engine. Six separate *Copy Data* activities are constructed within an ADF pipeline to ingest the synthetic CSV operational files (materials.csv, inventory_snapshot.csv, purchase_orders.csv, production_orders.csv, suppliers.csv, and sales_orders.csv) from the landing zone directly into the Bronze Layer container.

2. **Raw Data Preservation:** To strictly comply with the requirement of keeping source data as-is, the files are copied into the Bronze container without changing their schema or formats, ensuring perfect data lineage.

3. **Basic Ingestion Checks:** Basic quality validation steps, such as checking for completely empty files or missing columns, are implemented at the ADF level before triggering downstream processing.

3.3.4.2 Cleansing and Standardization Implementation (Silver Layer)

Data processing shifts from ingestion to transformation by executing Azure Synapse Notebooks programmatically triggered by an ADF Notebook Activity.

1. **Data Laundering & Deduplication:** Using **SQL** queries within the Synapse Notebooks, scripts are executed to parse the raw Bronze data, detect and drop duplicate records, and enforce strict type validation.

2. **Schema Standardization:** Varied date string representations across source files are standardized into a unified YYYY-MM-DD date format to enable seamless relational joining.

3. **Null Handling for Consumption:** For fields tracking material usage and production volumes, null values are programmatically isolated and converted to zero (0) using SQL functions like COALESCE or IFNULL. This critical data engineering treatment prevents calculation errors and null-propagation when aggregating historical consumption in the subsequent Gold phase.

3.3.4.3 Business Logic & Semantic Transformation (Gold Layer)

The core PPG business rules and risk management logic are codified inside specialized Azure Synapse SQL Notebooks, processing the cleaned Silver datasets into analytics-ready metrics:

1. **Recoverable Assets (RA) Scripting:** Queries calculate a rolling 12-month historical consumption per material by joining inventory data with consumption records. Conditional flags are coded to categorize a material as an RA if the current on-hand stock exceeds this 12-month consumption figure.

2. **Magna Carta (MC) Dead Stock & Expired Rules:** Multi-conditional logic is established to filter items:

    I. *Dead Stock:* Flagged where a material shows zero consumption for a period of 9 months and does not contain an active lot expiry date.

    II. *Expired:* Flagged automatically if the lot_expiry_date has passed relative to the reporting snapshot period.

3. **Financial Provisioning & Stockout Risk:** For all flagged MC components, a calculation field is constructed to output a 100% financial write-down provision. Concurrently, inventory replenishment parameters evaluate stock levels against vendor lead times to calculate stockout risks and flag impacted open customer sales orders.

3.3.4.4 Power BI Dashboard Deployment

1. **Power BI Integration: MS Power BI Desktop** is linked directly to the Azure SQL data warehouse containing the Gold star schema. Data Analysis Expressions (DAX) are written to build highly responsive, dynamic measures for cumulative financial provisions, stock-to-consumption ratios, and order counts.

2. **Visualizing Insights:** An interactive multi-tab dashboard is deployed with cross-filtering components, KPI blocks, and alert charts. This visualization layer is engineered specifically to address the 7 core business questions—enabling stakeholders to seamlessly trace inventory risk from high-level corporate financial write-downs down to specific impacted sales orders and customers.

3.4 Project Planning Schedule

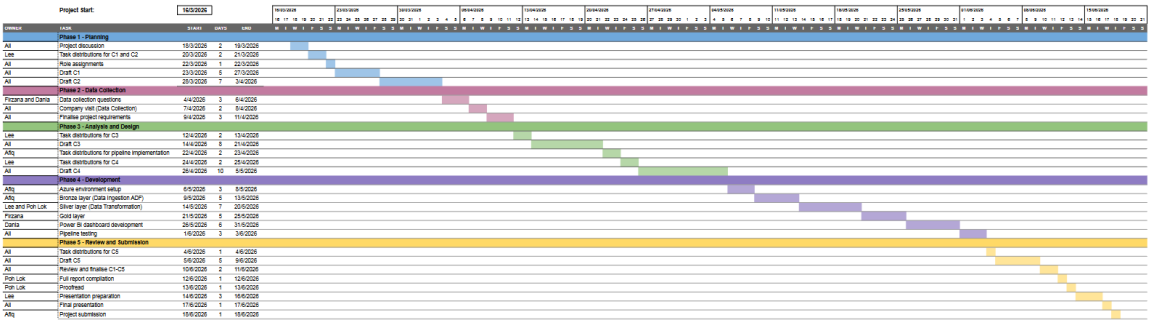


Figure 3.4.1: Gantt Chart of the Project (Source Link: https://docs.google.com/spreadsheets/d/12Vp29ctoFKUH4rvy9-zIv1IJVezrK8v5kXG_VTk3ajs/edit?usp=sharing)

Figure 3.4.1 above illustrates the overall project schedule spanning from 18 March 2026 to 18 June 2026. The project is divided into five phases, each represented by a distinct color to differentiate the phases and their respective timelines. Each horizontal bar represents the duration of a task with the length of the bar corresponding to the number of days allocated for that particular activity. The project is broken down into several phases as follows:

1. **Phase 1 - Planning (18 March – 3 April)** is represented in blue. This phase marks the beginning of the project where the team conducts an initial project discussion to understand the scope and objectives. This is followed by the distribution of tasks for Chapter 1 and Chapter 2 among team members as well as the assignment of roles and responsibilities. Once the planning is complete, team members begin drafting Chapter 1 and Chapter 2 simultaneously as both chapters cover the introduction and literature review which can be done before the data collection phase.

2. **Phase 2 - Data Collection (4 April – 11 April)** is represented in pink. This phase focuses on gathering real-world data from the client organisation, PPG. The team begins by preparing data collection questions to be brought to the company visit. On 7 and 8 April, the team conducted an on-site visit to PPG's premises to collect the required operational data including the six CSV files used in the pipeline. Following the visit, the team finalises the project requirements based on the information and data gathered during the visit.

3. **Phase 3 - Analysis and Design (12 April – 5 May)** is represented in green. This phase involves the planning and design of the system before development begins. Task distributions are made for Chapter 3, after which the team proceeds to draft

Chapter 3 which covers the system analysis. Simultaneously, task distributions are made for the pipeline implementation and Chapter 4 to prepare for the development phase. The phase concludes with the drafting of Chapter 4 which documents the system design including the data architecture, database design, and system requirements.

4. **Phase 4 - Development (6 May – 3 June)** is represented in purple. This is the most technical and intensive phase of the project where the actual data pipeline is built on Microsoft Azure. The phase begins with setting up the Azure environment including the Data Lake, Data Factory and Synapse Analytics services. The team then proceeds to implement the three-layer Medallion architecture in sequence starting with the first layer which is Bronze layer for raw data ingestion using Azure Data Factory, followed by the Silver layer for data transformation and business logic implementation using Azure Databricks, and then the Gold layer for loading the analytics-ready star schema into Azure Synapse Analytics. Once the pipeline layers are complete, the Power BI dashboard is developed to answer the seven business questions outlined in the project brief. The phase concludes with pipeline testing and debugging to ensure data accuracy and system reliability.

5. **Phase 5 - Review and Submission (4 June – 18 June)** is represented in yellow. This final phase focuses on completing the written report and preparing for the final presentation. Task distributions for Chapter 5 are made at the start of this phase, after which the team drafts Chapter 5 covering the implementation and results. All chapters from C1 to C5 are then reviewed and finalised to ensure consistency and quality across the report. The full report is compiled and formatted according to the FYPi report format, followed by a proofreading session. Then, the team prepares the presentation slides before delivering the final presentation on 17 June. The completed project report is submitted on 18 June which marks the conclusion of the project.

3.5 Summary

This chapter has presented the methodology and project planning schedule for the development of the PPG Recoverable Assets and Inventory Risk Management project. The Waterfall methodology was selected as the development approach for this project due to its

structured and sequential nature which is well-suited for a project with clearly defined requirements and deliverables.

The methodology is carried out across four sequential phases. The first phase, Planning, involves establishing the project scope, distributing tasks among team members, assigning roles and responsibilities and developing the project timeline as illustrated in the Gantt chart. The second phase, Analysis, focuses on gathering requirements through an on-site visit to PPG where the six operational CSV datasets were collected, followed by a thorough analysis of the business rules governing Recoverable Assets and Magna Carta classifications. The third phase, Design, covers the system architecture design including the three-layer Medallion Architecture on Microsoft Azure, the galaxy schema database design for the Gold layer and the Power BI dashboard design. The fourth phase, Implementation, involves the actual development of the end-to-end data pipeline encompassing the Bronze, Silver, and Gold layers, followed by the development of the Power BI dashboard to answer the seven key business questions defined by PPG.

The Gantt chart presented in this chapter provides a visual overview of the project schedule for three months, ensuring that all tasks are completed within the stated deadlines. Overall, the structured approach of the Waterfall methodology ensures that each phase is completed thoroughly before proceeding to the next, thereby maintaining the quality and integrity of the final deliverable.

CHAPTER 4

ANALYSIS AND DESIGN

4.1 Introduction

This chapter presents the analysis and design of the proposed PPG Recoverable Assets (RA) and Inventory Risk Management (IRM) Analytics System. Building upon the problem background, objectives, and theoretical foundations discussed in the previous chapters, this chapter translates business requirements into a structured technical solution that can be implemented within a cloud-based data engineering environment. The primary focus is to demonstrate how inventory risks, financial exposure, and operational disruptions can be systematically identified through an integrated analytics framework.

The chapter progresses from understanding the business context to defining the proposed system architecture, data pipeline, analytical data model, and reporting components. Particular attention is given to how the Medallion Architecture supports data quality, traceability, and scalability throughout the data lifecycle. The outputs of this chapter serve as the blueprint for implementation and provide the technical foundation required to answer the business questions related to Recoverable Assets, Magna Carta classifications, financial provisions, stockout risks, and customer order impacts.

4.2 System Analysis

The system analysis examines how PPG's current inventory risk challenges can be translated into a structured data engineering solution that supports Recoverable Assets (RA) and Inventory Risk Management (IRM). At a business level, PPG needs to systematically identify excess materials held beyond 12 months of consumption, classify Magna Carta (MC) Dead Stock and Expired items that require financial write-downs, and detect stockout risks that may delay production or open customer sales orders. At a technical level, these needs are mapped into clear functional requirements for data ingestion, quality checks, business rule implementation, schema modeling, and interactive reporting, using six synthetic ERP-like datasets hosted within a Medallion Architecture on Microsoft Azure.

This analysis links organizational roles, business rules, and data structures into one coherent view. Supply chain and warehousing define the operational scenarios for stock coverage and lead times, finance establishes the rules for RA, MC Dead Stock, and MC Expired provisioning, and the IT and data engineering unit provides the cloud platform, governance, and security required for an automated pipeline. Through industry visits, interviews, and dataset profiling, these perspectives are consolidated into functional and non-functional requirements that guide the design of the Bronze, Silver, and Gold layers, ensuring that the resulting system not only automates existing manual processes but also delivers reliable, analytics-ready outputs for decision makers across the company.

4.2.1 Case Study

This project is based on a real-world case study involving PPG Industries, a global manufacturing company that manages large volumes of raw materials to support continuous production operations. The company faces challenges in maintaining visibility over excess inventory, obsolete materials, and stockout risks due to the reliance on manual monitoring processes and spreadsheet-based reporting. As a result, identifying financial exposure and operational risks becomes time-consuming, inconsistent, and prone to human error.

The project proposes an automated data engineering solution that processes six ERP-derived synthetic datasets through a Medallion Architecture implemented on Microsoft Azure. The solution applies business rules to identify Recoverable Assets, Magna Carta Dead Stock, Magna Carta Expired Materials, financial provisioning requirements, and stockout risks that may affect customer orders. The requirements and business logic used throughout the project were derived from industry visits, stakeholder discussions, and analysis of operational processes, ensuring that the developed system accurately reflects PPG's inventory risk management practices while maintaining data privacy through the use of synthetic datasets.

4.2.2 Company Organization Structure

The company organization needs to be structurally aligned from legacy inventory methods to an automated analytical data-driven architecture. This system deployment has an organizational structure consisting of the following functional divisions:

1. Supply Chain and Warehousing Operations: This division is responsible for ensuring the physical inventory, always keeping track of replenishment of lead times and as the business owner for material availability. The platform allows the operational employees to monitor the stockout risk, track safety of stock levels, and position the raw materials in relation to production schedules. Having this operational-to-analytical alignment is critical to help the people working on the job clearly understand at risk resources much faster than legacy manual operations (Krystofik et al., 2020).

2. Finance and Governance Department: This used the outputs from the automated pipeline to audit the inventory valuations as a responsibility that it also falls within the scope of finance and governance role. It heavily depends on the systematic extraction of Magna Carta classifications to estimate the correct monthly write-down provisions and ensure that the corporate accounting standards are met.

3. Technical Unit: The Unit that handles the infrastructure of the enterprise is called Information Technology (IT) and Data Engineering Unit. As part of this architecture, the responsibility for supplying the cloud ecosystem is held, maintaining the connection security through key vaults, setting up the pipeline triggers and ensuring the smooth flow of transactional data from data source environments towards the analytical layer.

4.2.3 Company Requirements Gathering Techniques

A variety of approaches to requirements was used to define a complete an accurate set of functional and non-functional requirements. This framework was based on qualitative and quantitative methods to connect operational reality and technical system specifications.

Primary data collection involved an official industry visit to PPG manufacturing plant in Malaysia and Semi-Structured interviews. Through this field observation, the physical stock management process was audited, and the flaws of the manual assessment process were discovered. Qualitative business rules and guidelines were gathered through talk and semi-structured interviews with domain experts, including the rules for what constitutes Dead Stock after 9 months and dependency rules between the raw materials and the open customer sales orders.

An in-depth review of Enterprise spreadsheets, the historical reporting templates, and material definitions were completed. This technical review also detailed the exact formulas to be used to convert business rules that were not explicit in the text, into mathematical expressions. This is to understand the inventory variables before formalizing the logic.

The requirement discovery method involved auditing the structure attributes of the six core operational datasets from enterprise resource planning (ERP) environment to gain the quantitative requirements. Profiling these files (materials.csv, inventory_snapshot.csv, purchase_orders.csv, production_orders.csv, suppliers.csv, and sales_orders.csv) enabled primary and foreign key constraints to be validated, data quality issues to be expected and structure specifications for the Data Lakehouse refinement layers to be formalized.

4.2.4 Usecase Diagram

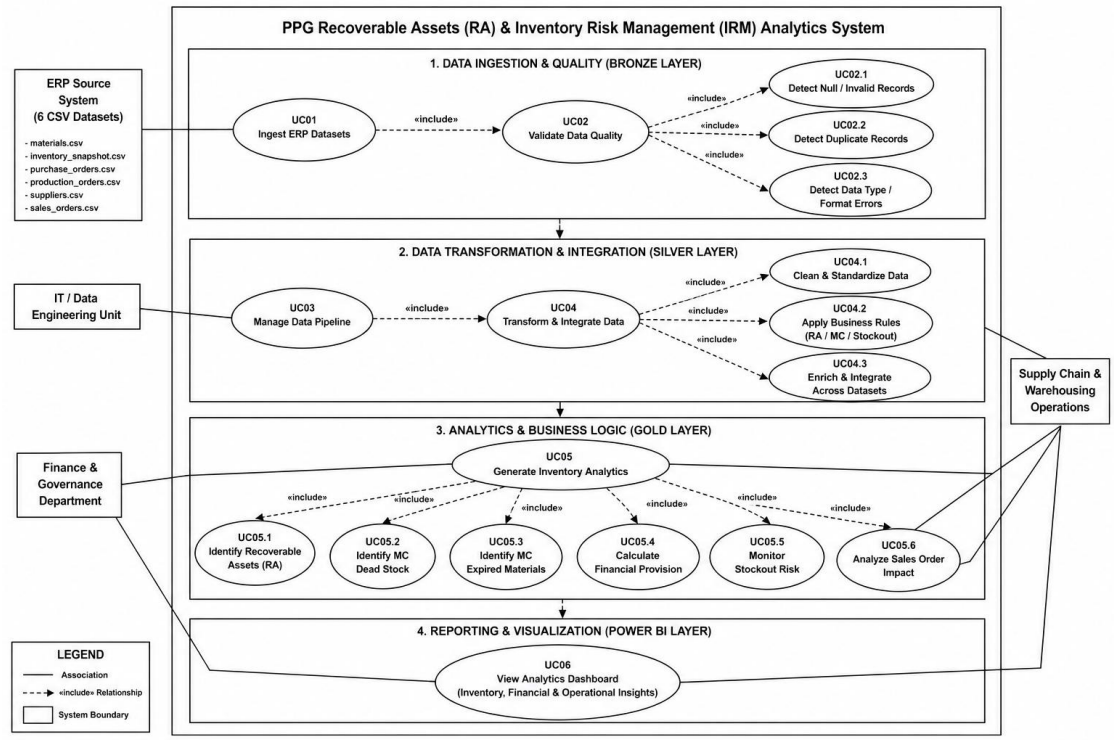


Figure 4.2.4.1: Conceptual Star Schema Design of PPG RA and IRM System

The Use Case Diagram illustrates the end-to-end workflow of the PPG Recoverable Assets (RA) and Inventory Risk Management (IRM) Analytics System, which is designed to transform raw ERP operational data into actionable business insights. The system integrates six datasets, namely materials, inventory snapshots, purchase orders, production orders, suppliers, and sales orders, within a cloud-based Medallion Architecture. The process involves four main actors: the ERP Source System, IT and Data Engineering Unit, Finance and Governance Department, and Supply Chain & Warehousing Operations. Together, these actors support inventory monitoring, financial risk assessment, and operational decision-making through a centralized analytics platform.

The first stage consists of the Bronze Layer and Silver Layer, where data is ingested, validated, transformed, and integrated. Raw datasets are initially loaded into the system through the *Ingest ERP Datasets* use case and then subjected to quality validation processes, including null detection, duplicate identification, and data type verification to ensure data accuracy and reliability. After validation, the IT and Data Engineering Unit manages the pipeline by cleaning, standardizing, and integrating data from multiple operational domains. Business rules related to Recoverable Assets, Magna Carta

classifications, and stockout analysis are also applied during this stage, ensuring that the data is consistent, trustworthy, and ready for advanced analytics.

The Gold Layer serves as the analytical core of the system, where inventory risk intelligence is generated to answer PPG's key business questions. The system identifies Recoverable Assets (RA), detects Magna Carta Dead Stock and Expired Materials, calculates financial provisions for inventory write-downs, monitors stockout risks, and evaluates the impact of material shortages on customer sales orders. These analytical outputs are then delivered through the Power BI Dashboard, providing stakeholders with a unified view of inventory health, financial exposure, and operational risks. By combining automated data processing, business rule implementation, and interactive reporting, the system enables PPG to improve inventory visibility, strengthen governance, reduce operational risks, and support faster, data-driven decision-making across the organization.

4.3 System Requirements

This section defines the system requirements for the PPG Recoverable Assets and Inventory Risk Management pipeline. The requirements are categorized into functional requirements, which describe the specific capabilities the system must deliver, and non-functional requirements, which define the quality standards and operational constraints the system must satisfy. These requirements were derived from the business rules gathered during the industry visit to PPG and the dataset profiling conducted during the analysis phase.

4.3.1 Functional Requirements

Functional requirements describe the specific capabilities that the system must deliver to meet the project's objectives. The functional requirements for this project are shown below.

FR1 – The system shall ingest all six operation CSV datasets into the Bronze layer of data lake without modifying the original file structure.

FR2: The system shall automatically perform data quality checks on all ingested datasets, including detection of missing values in critical fields, identification of incorrect data types, and removal or flagging of duplicate records.

FR 3: The system shall calculate the total quantity consumed for each raw material over the past 12 months using historical production order records. This calculation forms the basis for identifying whether a material is held in excess relative to actual operational demand.

FR 4: The system shall flag a material as a Recoverable Asset if its current quantity on hand, derived from the most recent inventory snapshot, exceeds its calculated 12-month consumption. The excess quantity shall be computed and stored for each flagged material.

FR 5: The system shall flag a material as MC Dead Stock if it has not been consumed for nine or more consecutive months and does not have a lot expiry date recorded in the inventory snapshot.

FR 6: The system shall flag a material as MC Expired if its lot expiry date field is populated and the date has passed relative to the snapshot reporting date.

FR 7: The system shall calculate a 100% financial write-down provision amount equal to the material's unit cost multiplied by its quantity on hand for all materials classified as either MC Dead Stock or MC Expired, The system shall also aggregate the total provision amount across all MC materials.

FR 8: The system shall flag materials whose current quantity on hand falls below their defined reorder level, as specified in the materials dataset, to support replenishment planning.

FR9: The system shall calculate the average daily consumption rate per material and the number of stock cover days remaining. A material shall be flagged as having stockout risk if its stock cover days are less than its supplier lead time in days.

FR10: The system shall identify which open customer sales orders are at risk of delayed fulfilment due to stockout risk in their required raw materials. This is achieved by linking at-risk materials through production orders to finished products, and finished products to open sales order records.

FR11: The system shall organise all transformed and classified outputs into a galaxy schema data warehouse in Azure Synapse Analytics SQL.

FR12: The system shall provide an interactive Power BI dashboard that visualises all seven business questions, including materials below reorder level, Recoverable Asset listings, MC Dead Stock and Expired material tables, total financial provision amounts, stockout risk materials, and impacted customer sales orders.

FR13: The system shall orchestrate all pipeline stages includes Bronze ingestion, Silver transformation, and Gold loading through a master Azure Data Factory pipeline that enforces sequential execution with dependency control, ensuring each stage completes successfully before downstream stages begin.

4.3.2 Non-Functional Requirements

Non-functional requirements define the quality standards and constraints the system must satisfy beyond its core functions. The non-functional requirements for this project are shown below.

NFR1: The system shall include error handling at each pipeline stage so that failures are detected and reported without corrupting downstream data.

NFR2: The system shall ensure that raw source data in the Bronze layer is never modified or overwritten. All business logic transformations shall be applied only in the Silver and Gold layers, maintaining a clear and auditable lineage from source to final output.

NFR3: The pipeline architecture shall be designed in a way that can accommodate larger volumes of data in the future without requiring fundamental changes to its structure.

NFR4: The system shall be built in a modular manner, where each pipeline layer operates independently and can be updated or rerun without affecting the others.

NFR5: The system shall protect all connection credentials and access keys by storing them securely, rather than embedding them directly in pipeline scripts or configurations.

NFR6: The Power BI dashboard shall be designed to load and respond to user interactions within a reasonable time. Data shall be pre-processed and structured in the Gold layer so that the dashboard does not need to perform heavy computations at query time.

NFR7: The system is designed exclusively for batch processing of static synthetic datasets.

4.4 System Design

This section presents the system design of the PPG Recoverable Assets and Inventory Risk Management pipeline. The design covers the overall system architecture, the proposed data pipeline, the database model, and the scheduling and error handling mechanisms. Each subsection describes a specific design component and explains the rationale behind the design decisions made to support the seven business questions defined in the project scope.

4.4.1 Overview of the Proposed System Design

The proposed system is designed as an end-to-end data engineering pipeline for the PPG Recoverable Assets (RA) and Inventory Risk Management (IRM) case study. The system is intended to process six operational datasets, apply inventory-related business rules, and produce analytics-ready data for reporting through Power BI. The overall design combines Microsoft Azure services with the Medallion Architecture approach, where data is processed progressively through Bronze, Silver, and Gold stages. This structure allows raw data to be preserved, cleaned, transformed, modelled, and visualized in a systematic manner.

4.4.2 Data Architecture

The proposed data architecture is designed based on Microsoft Azure services and the Medallion Architecture approach. It is structured to support the PPG Recoverable Assets (RA) and Inventory Risk Management (IRM) case study by organizing the data engineering workflow from raw data ingestion to dashboard reporting. The architecture is divided into six main components, namely data sources, data ingestion, data storage, data processing, data governance and security, and data consumption. (Krantz, T., 2024).

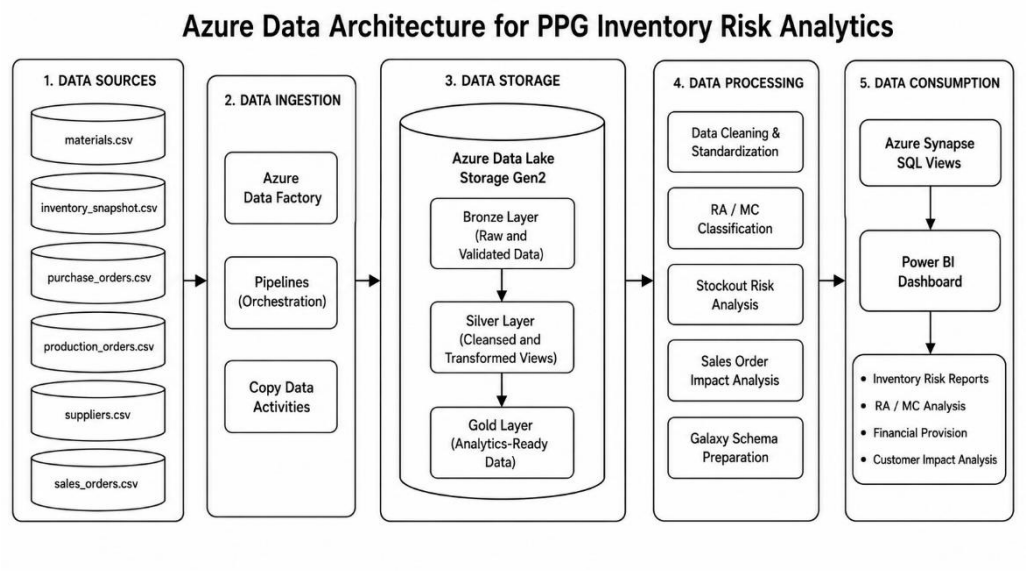


Figure 4.4.2.1: Proposed Azure Data Architecture for PPG Inventory Risk Analytics

Figure 4.4.2.1 shows the proposed Azure data architecture for the PPG inventory risk analytics system. The architecture was designed based on a simplified data warehouse flow, consisting of data sources, data ingestion, data storage, data processing, and data consumption. It illustrates how the six CSV datasets are ingested using Azure Data Factory, stored in Azure Data Lake Storage Gen2, processed using Azure Synapse Serverless SQL, and finally consumed through Azure Synapse SQL views and Power BI dashboard.

Table 4.4.2.1: Components of the Proposed Data Architecture

Component	Tool / Layer Used	Description

Data Sources	Six CSV datasets	Operational datasets are provided for material, inventory, supplier, production, purchase, and sales order analysis.
Data Ingestion	Azure Data Factory	CSV files are ingested into the Azure environment through Copy Data activities and pipeline orchestration.
Data Storage	Azure Data Lake Storage Gen2	A centralized storage platform is used to organize the data into Bronze, Silver, and Gold layers.
Data Processing	Azure Synapse Serverless SQL	Data cleaning, standardization, RA/MC classification, stockout risk analysis, sales order impact analysis, and star schema preparation are performed.
Data Consumption	Azure Synapse SQL views and Power BI	Business-ready outputs and dashboard visualizations are provided for inventory risk analysis, RA/MC analysis, financial provision, and customer impact analysis.

Table 4.4.2.1 summarizes the main components of the proposed data architecture. It shows the tools and layers used in each component, including data sources, data ingestion, data storage, data processing, and data consumption.

The data source component consists of six synthetic CSV files, namely materials.csv, inventory_snapshot.csv, purchase_orders.csv, production_orders.csv, suppliers.csv, and sales_orders.csv. These datasets are used to support the identification of Recoverable Assets, Magna Carta Dead Stock, Magna Carta Expired materials, stockout risks, and impacted customer sales orders.

The data ingestion component is handled using Azure Data Factory. The datasets are transferred into Azure Data Lake Storage Gen2 through pipeline orchestration and Copy

Data activities. Azure Data Lake Storage Gen2 acts as the central storage environment, where the data is organized into Bronze, Silver, and Gold layers. The Bronze layer stores raw and validated data, the Silver layer stores cleansed and transformed views, and the Gold layer stores analytics-ready data for reporting.

Data processing is performed using Azure Synapse Serverless SQL. This tool is used to clean and standardize the data, apply RA and MC classification logic, calculate stockout risk, trace sales order impact, and prepare the analytical data model. In the updated implementation, the analytical model is designed as a Galaxy Schema because two fact views are used. The fact_inventory_status view supports Q1 to Q6, while the fact_sales_impact view supports Q7. These fact views share common dimensions such as dim_material, dim_date, and dim_warehouse, while dim_customer is used to support customer impact analysis.

Finally, Azure Synapse SQL views and Power BI are used as the data consumption layer. The processed outputs are consumed through dashboard visualizations and reports, including inventory risk reports, RA/MC analysis, financial provision analysis, stockout risk analysis, and customer impact analysis. Although the governance layer is not shown as a separate component in the logical architecture diagram, data quality checks, validation rules, access control, and traceability are still applied throughout the pipeline implementation.

4.4.3 Explanation of Proposed Data Architecture

The proposed architecture is designed to support a structured data engineering workflow from raw data ingestion to business intelligence reporting. The process begins with six CSV datasets that represent the operational data required for the PPG RA and IRM case study. These datasets include material master data, inventory snapshots, purchase orders, production orders, supplier information, and sales orders.

Azure Data Factory is used as the ingestion and orchestration tool. Each CSV file is loaded into Azure Data Lake Storage Gen2 using Copy Data activities. This allows the ingestion process to be repeated and managed systematically. The use of Azure Data Factory also

supports pipeline control, where ingestion activities can be monitored and validated before downstream processing is performed.

The data storage component is implemented using Azure Data Lake Storage Gen2. The storage design follows a three-layer Medallion Architecture consisting of Bronze, Silver, and Gold layers. The Bronze layer stores the raw and validated CSV datasets. Raw data is preserved to support traceability, while validated data is prepared after basic data quality checks. These checks include missing value detection, duplicate checking, data type validation, and column structure verification. Invalid or problematic records may be separated into a quarantine folder to prevent them from affecting later processing stages.

After the Bronze layer, the data is processed into the Silver layer. The Silver layer is responsible for cleaning, standardization, and transformation. In this project, Azure Synapse Serverless SQL is used to implement the Silver layer processing. This approach is used because the Synapse Spark pool could not be executed due to workspace vCore quota limitations. Therefore, SQL views are used to perform the transformation logic while maintaining the same business rules and expected outputs.

In the Silver layer, the RA and MC classification logic is applied. Production order data is used to calculate material consumption, while inventory snapshot data is used to determine quantity on hand. Recoverable Assets are identified by comparing current stock with expected consumption. Magna Carta Dead Stock is identified by checking materials with no consumption for a defined period, while Magna Carta Expired materials are identified using the lot_expiry_date field. These classifications are required because MC materials may require full financial provision.

Stockout risk analysis is also performed in the Silver layer. The quantity on hand is compared with the reorder level to identify materials below the required threshold. Average daily consumption and stock cover days are then calculated and compared with material lead time. If the remaining stock cannot cover the lead time, the material is flagged as having stockout risk. At-risk materials are further linked to finished products and open sales orders, allowing affected customer sales orders to be identified.

The Gold layer is used to organize the processed outputs into an analytics-ready model. Azure Synapse SQL is used to prepare the Gold layer tables and views. A star schema design is applied, where the central fact_inventory_status table is linked to dimension tables such as dim_material, dim_supplier, dim_customer, and dim_date. The fact table stores calculated measures and flags, including inventory quantity, consumption values, RA and MC indicators, stockout risk status, and financial provision amount. This structure supports efficient filtering and reporting in Power BI.

Data governance and security are included throughout the architecture. Access control is supported through role-based access control, while data quality checks and validation rules are applied to improve data reliability. The quarantine folder provides a location for invalid records, and data lineage is maintained by separating the workflow into clear pipeline layers. These controls help ensure that inventory risk calculations and financial provision outputs are based on reliable data.

Finally, the data consumption layer is supported by Azure Synapse SQL views and Power BI. Power BI connects to the prepared analytical outputs to create dashboards and reports. The dashboard presents inventory risk reports, RA and MC analysis, financial provision values, and customer impact analysis. Through this architecture, raw operational data is transformed into business-ready insights in a structured, traceable, and scalable manner.

4.4.4 Scheduling and Error Handling

To make sure the system runs reliably without needing human intervention, it includes automated schedules and error management:

1. **Automated Trigger:** The pipeline is attached to an ADF **Schedule Trigger**. This trigger is set to run automatically at the end of every month or operational cycle, right when the factory generates new inventory records and order sheets.

2. **Error Tracking:** If something goes wrong inside a Synapse Notebook, for example, if a file has a misspelled column name or a mathematical calculation crash,

the notebook will stop running immediately. It sends a failure message back to Azure Data Factory.

3. **Data Rollback:** The system uses safe transactional database commands when writing data to the Gold layer. This means if a notebook crashes halfway through updating the final database, it will undo its partial changes and roll back to the last safe state. This stops incomplete or corrupted data from showing up on the Power BI dashboards.

4.4.5 Database Design

The Gold layer adopts a galaxy schema, also known as a fact constellation as its analytical data model, a design pattern used in dimensional modelling for data warehouses that require multiple fact tables to represent distinct business processes. In a galaxy schema, two or more fact tables share one or more common dimension tables, allowing related business processes to be analysed both independently and in combination through their shared dimensions. This structure was adopted because the project requires two distinct analytical perspectives which are inventory status and sales order impact that share common reference data but capture different grain and measures.

Figure 4.4.5.1 below shows the schema implemented in this project, consisting of two fact tables and five dimension tables. The first fact table, fact_inventory_status, captures the monthly inventory position of each material, including Recoverable Asset and Magna Carta flags, provision amounts, reorder status, and stockout risk metrics. The second fact table, fact_sales_impact, captures open sales orders affected by stockout-risk materials, including delivery risk and stock cover details at the point of impact.

The five dimension tables are dim_material, dim_date, dim_supplier, dim_warehouse, and dim_customer. Of these, dim_material and dim_customer serve as conformed dimensions, each connected to both fact tables, enabling cross-fact analysis such as linking a customer's affected sales order back to the material driving the stockout. dim_date connects to fact_inventory_status via snapshot_date, while dim_supplier and dim_warehouse connect exclusively to fact_inventory_status via supplier_id and warehouse_location respectively, as these attributes are relevant only to the inventory dimension of the business process.

To support standalone querying of fact_sales_impact without requiring a join back to dim_material for every analytical query, selected material attributes (material_name, category, quantity_on_hand, reorder_level, avg_daily_consumption, stock_cover_days, lead_time_days, is_stockout_risk) are denormalised directly into the fact table at the time of order impact assessment. This reflects the snapshot nature of fact_sales_impact as it records the inventory condition of the material at the moment the sales order was identified

as impacted, rather than its current state, which may differ from the corresponding row in fact_inventory_status at query time.

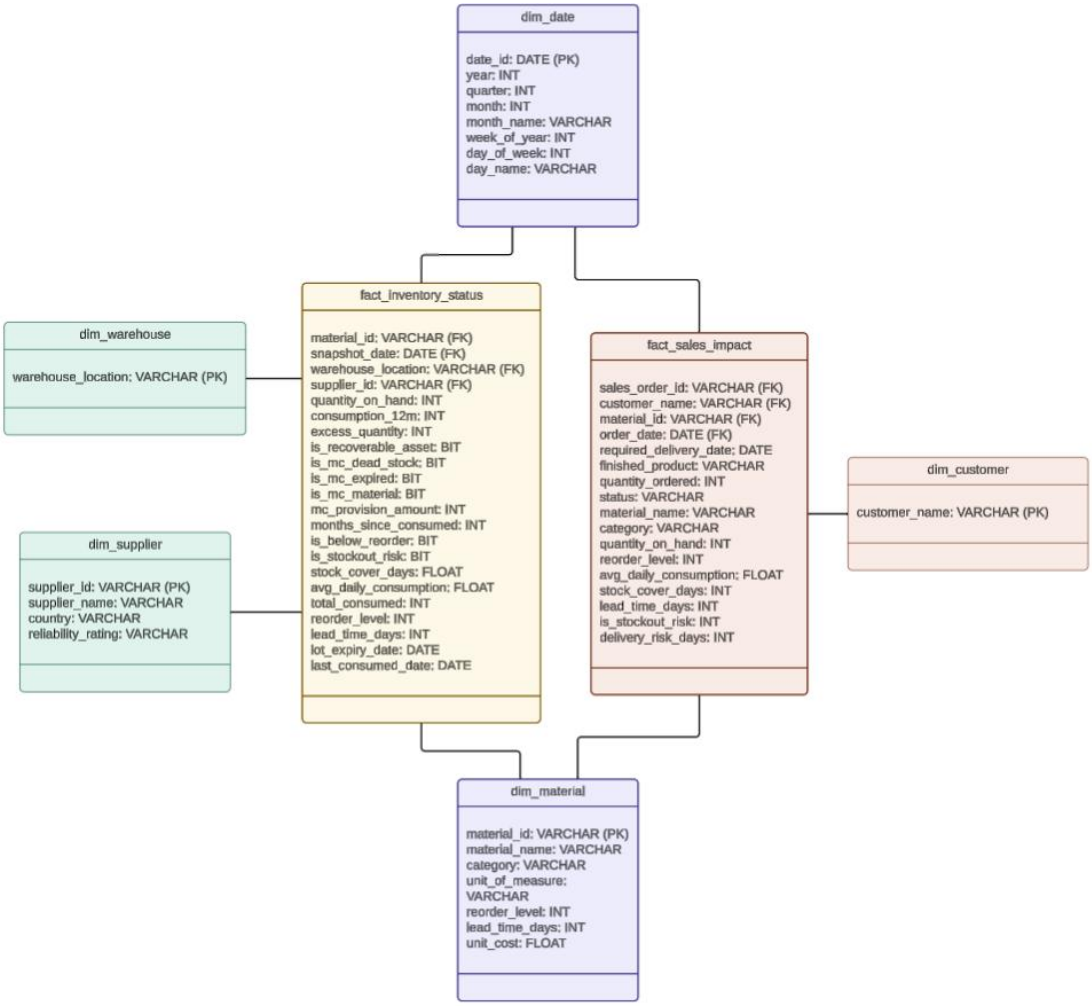


Figure 4.4.5.1: Galaxy Schema of PPG RA and IRM System

4.4.5.1 Fact Table

The fact_inventory_status table serves as the central fact table of the star schema and is the primary analytical table of the entire pipeline. Each row in this table represents the inventory status of a single material at a specific point in time, capturing the outcome of all business logic applied during the Silver layer transformations. The table is identified by a surrogate primary key, inventory_status_id, which uniquely distinguishes each record. It also carries four foreign keys which are material_id, supplier_id, customer_id, and date_id to establish the relationships to each of the surrounding dimension tables.

Beyond the key columns, the fact table holds a comprehensive set of measures and derived indicators. The snapshot_date and warehouse_location fields provide operational context for each record. Quantitative measures include quantity_on_hand, consumption_12m, excess_quantity, and months_since_consumed which together capture the stock position and consumption behaviour of each material. The fields lot_expiry_date and last_consumed_date are retained as supporting date attributes to preserve traceability back to the Silver layer.

4.4.5.2 Dimension Tables

The dimension tables are created as follows:

1. **dim_material -** A dimension table that holds the reference attributes for all raw materials tracked within the pipeline. It is keyed on material_id as its primary key and contains descriptive attributes including material_name, category, unit_of_measure, reorder_level, lead_time_days, and unit_cost. These attributes provide the contextual information needed to interpret the measures in the fact table — for example, reorder_level is used in conjunction with quantity_on_hand to determine whether a material has fallen below its replenishment threshold, and unit_cost is the basis for calculating the Magna Carta provision amount. The category field, which classifies materials into groups such as Pigment, Resin, Solvent, and Packaging, also serves as a key filtering dimension in the Power BI dashboard, allowing analysts to slice inventory risk results by material type.
2. **dim_date –** A dimesion table that provides the time dimension for the star schema, enabling date-based filtering and aggregation within the Power BI dashboard. It is keyed on date_id as its primary key and contains the derived calendar attributes date, month, quarter, and year. Each row in this table corresponds to a distinct snapshot date present in the inventory data, with the current dataset spanning twelve monthly entries from January to December 2025. By separating date intelligence into its own dimension, the schema enables analysts to filter dashboard visuals by time period without embedding date logic directly into the fact table or the reporting layer. This also allows future expansion of the date range without structural changes to the fact table.

3. **dim_supplier –** A dimension table that holds reference information about the raw material suppliers engaged by PPG. It is keyed on supplier_id as its primary key and contains supplier_name, country, and reliability_rating as descriptive attributes. The supplier_id foreign key in the fact table establishes a direct relationship between each inventory record and its associated supplier, enabling supply chain analysis within the dashboard. The reliability_rating field, which categorises suppliers by their delivery performance, is particularly relevant in the context of stockout risk — materials sourced from lower-reliability suppliers may carry a higher risk of supply disruption, which can be surfaced through Power BI filters applied against this dimension.
4. **dim_customer –** A dimension table that captures the distinct set of customers whose open sales orders are potentially affected by material shortages. It is keyed on customer_id as its primary key, with customer_name as the identifying attribute. This table is populated from the sales orders data and is linked to the fact table through the customer_id foreign key, establishing the connection between inventory risk outcomes and downstream customer impact. In the context of the dashboard, dim_customer supports the Q7 analysis — together with the analytical view vw_q7_impacted_orders, it enables the identification of which specific customers are exposed to fulfilment risk as a result of stockout conditions at the material level.

CHAPTER 5

IMPLEMENTATION AND TESTING

5.1 Introduction

This chapter presents the practical implementation of the PPG Recoverable Assets (RA) and Inventory Risk Management (IRM) data engineering pipeline based on the Medallion Architecture. It explains how the Bronze, Silver, and Gold layers were developed on Microsoft Azure to ingest six synthetic CSV datasets, enforce data quality checks, apply RA and Magna Carta business rules, detect stockout risks, and load the transformed data into a star schema centered on the factinventorystatus table. The chapter also describes the integration of this Gold layer with a Power BI dashboard that delivers interactive visualizations for financial provisions, inventory classifications, and customer order impacts.

This chapter explained details the testing activities carried out to ensure the correctness and reliability of the system. Black Box Testing validates the outputs produced by each layer and the Power BI dashboard against the expected business results, while White Box Testing verifies the internal logic of the ingestion flows, transformation scripts, and star schema loading processes. User Testing is then conducted to confirm that stakeholders can interpret the dashboards effectively and use the insights to support inventory risk decisions.

5.2 System Development

This section presents the technical design of the PPG RA and Inventory Risk Management IRM system into functional cloud infrastructure on Microsoft Azure by detailing the practical implementation of the three-tier Medallion Architecture which include the Bronze, Silver, and Gold layers and its final integration with the Power BI visualization layer. It shows how raw datasets are systematically ingested, cleansed, enriched with complex business rules, and structured into a galaxy schema database. By documenting the exact configurations, data models, and scripts deployed across each layer, this section provides an end-to-end technical solutions of how raw operational data is transformed into analytics-ready business intelligence.

5.2.1 Bronze Layer

The Bronze Layer constitutes the first stage of the three-tier Medallion Architecture implemented in this project. Its primary responsibility is the automated ingestion of raw source data from six Comma-Separated Values (CSV) files into Azure Data Lake Storage Gen2 (ADLS Gen2), followed by systematic data quality and duplicate detection checks to ensure only validated data progresses to the Silver Layer. The Bronze Layer was developed entirely within the Azure Data Factory (ADF).

Prior to pipeline construction, the foundational Azure infrastructure was provisioned under the resource group rg-ppg-pipeline in the Southeast Asia region. The storage account ppgdatalake was created with the hierarchical namespace enabled, configuring it as an ADLS Gen2 instance. Three containers which are bronze, silver, and gold were created within the storage account, corresponding to each layer of the Medallion Architecture. The ADF instance adf-ppg-pipeline and the Synapse Analytics workspace synapse-ppg were provisioned within the same resource group to centralise all data engineering resources, as illustrated in Figure 5.2.1.1.

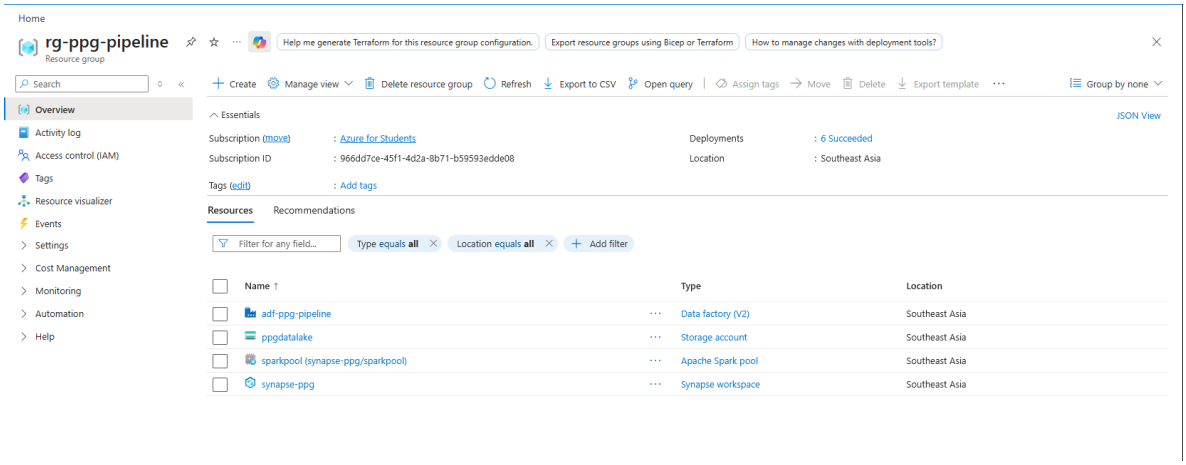


Figure 5.2.1.1: Azure Resource Group rg-ppg-pipeline

The storage account was configured with three dedicated containers corresponding to each layer of the Medallion Architecture, as shown in Figure 5.2.1.2.

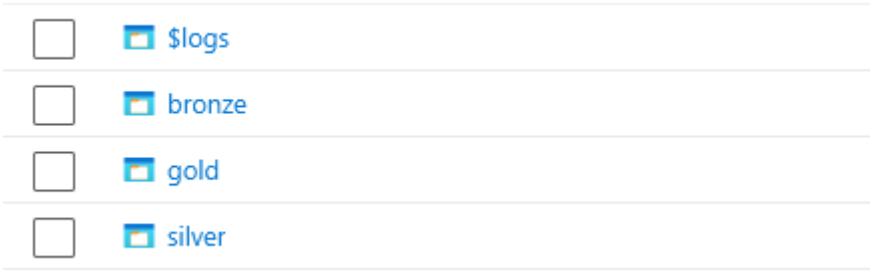


Figure 5.2.1.2: ADLS Gen2 Containers

Two Linked Services were configured in ADF Studio to establish authenticated connections to the project's data stores. The first, ls_adls_ppg, connects to ppgdatalake using System-Assigned Managed Identity authentication and serves as the connection reference for all Bronze Layer datasets. The second, ls_synapse_ppg, connects to the Synapse Analytics Serverless SQL Pool via the endpoint synapse-ppg-ondemand.sql.azuresynapse.net, targeting the ppg_gold database using SQL authentication. Both linked services were validated using the Test Connection function prior to publication, as shown in Figure 5.2.1.3.

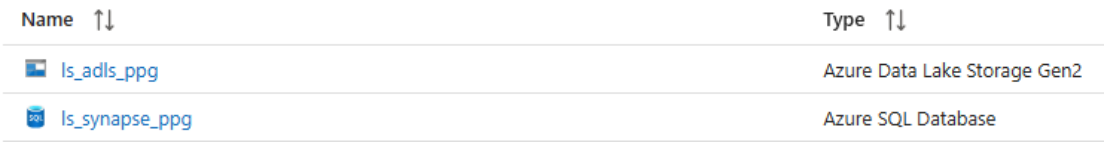


Figure 5.2.1.3: Configured Linked Services in Azure Data Factory

5.2.2 Silver Layer — RA/MC Logic

The Silver layer RA and MC component was implemented using Azure Synapse Serverless SQL by replacing the originally planned PySpark notebook due to Spark pool vCore quota constraints in the project environment. It was designed to identify materials where stock levels exceeded actual operational demand, detect materials that had been idle for an extended period or had passed their lot expiry date, and calculate the total financial write-down provision required for all affected materials. This implementation is contained in a single SQL script named M2_silver_ra_mc_logic which creates a reusable SQL view called vw_silver_ra_mc_logic in the ppg_gold database ad shown in Figure 5.2.2.1.

```sql
CREATE OR ALTER VIEW vw_silver_ra_mc_logic AS
WITH materials AS (
    SELECT DISTINCT *
    FROM OPENROWSET(
        BULK 'https://ppgdatalake.dfs.core.windows.net/bronze/validated/materials/*.csv',
        FORMAT = 'CSV',
        PARSER_VERSION = '2.0',
        FIRSTROW = 2
    )
    WITH (
        material_id VARCHAR(20) COLLATE Latin1_General_100_BIN2_UTF8,
        material_name VARCHAR(100) COLLATE Latin1_General_100_BIN2_UTF8,
        category VARCHAR(50) COLLATE Latin1_General_100_BIN2_UTF8,
        unit_of_measure VARCHAR(20) COLLATE Latin1_General_100_BIN2_UTF8,
        reorder_level INT,
        lead_time_days INT,
        unit_cost FLOAT
    ) AS m
),
inventory AS (
    SELECT
        TRY_CONVERT(DATE, snapshot_date_raw, 103) AS snapshot_date,
        material_id,
        warehouse_location,
        quantity_on_hand,
        TRY_CONVERT(DATE, lot_expiry_date_raw, 103) AS lot_expiry_date
    FROM OPENROWSET(
        BULK 'https://ppgdatalake.dfs.core.windows.net/bronze/validated/inventory_snapshot/*.csv',
        FORMAT = 'CSV',
        PARSER_VERSION = '2.0',
        FIRSTROW = 2
    )
    WITH (
        snapshot_date_raw VARCHAR(20) COLLATE Latin1_General_100_BIN2_UTF8,
        material_id VARCHAR(20) COLLATE Latin1_General_100_BIN2_UTF8,
        warehouse_location VARCHAR(50) COLLATE Latin1_General_100_BIN2_UTF8,
        quantity_on_hand INT,
        lot_expiry_date_raw VARCHAR(30) COLLATE Latin1_General_100_BIN2_UTF8
    ) AS i
),
production AS (
    SELECT DISTINCT *
    FROM OPENROWSET(
        BULK 'https://ppgdatalake.dfs.core.windows.net/bronze/validated/production_orders/*.csv',
        FORMAT = 'CSV',
        PARSER_VERSION = '2.0',
        FIRSTROW = 2
    )
    WITH (
        production_order_id VARCHAR(20) COLLATE Latin1_General_100_BIN2_UTF8,
        production_date DATE,
        finished_product VARCHAR(100) COLLATE Latin1_General_100_BIN2_UTF8,
        material_id VARCHAR(20) COLLATE Latin1_General_100_BIN2_UTF8,
        quantity_consumed INT
    ) AS p
```

Figure 5.2.2.1: SQL view creation for vw_silver_rc_mc_logic

The view reads directly from the three Bronze validated CSV files includes **materials.csv**, **inventory_snapshot.csv**, and **production_orders.csv** that stored in Azure Data Lake Storage Gen2. The materials dataset was used to obtain unit cost, reorder level, lead time days, and material category. The inventory snapshot dataset was used to obtain the latest quantity on hand and lot expiry date for each material. The production orders dataset was used to calculate 12-month material consumption and identify the last date each material was consumed.

Although the Bronze layer had already applied data quality checks, additional safeguards were retained in the Silver layer as a precaution. Date fields in the inventory snapshot were

stored in DD/MM/YYYY string format and required conversion to DATE using TRY_CONVERT with style code 103, where empty or invalid values safely return NULL. UTF-8 collation and SELECT DISTINCT were also applied to prevent encoding errors and remove any residual duplicates.

5.2.3 Silver Layer — Stockout & Sales Impact

The stockout and sales impact process was developed as part of the Silver Layer implementation. This process was designed to identify raw materials that were below their reorder level, detect materials that were at risk of stockout, and trace the possible impact of material shortages on open customer sales orders. This component was important because it connected upstream inventory availability with downstream customer order fulfilment.

The implementation was carried out using Azure Synapse Serverless SQL. The initial implementation approach was planned using Synapse PySpark Notebook. However, the Spark notebook could not be executed because the Azure workspace had a zero vCore quota limitation. Therefore, the approach was changed to SQL-based processing. Although the tool was changed, the original business logic was maintained.

After the mock data was increased, the updated validated Bronze Layer datasets were used as the source for this process. The inventory snapshot dataset increased from 240 rows to 300 rows after January, February, and March 2026 inventory records were added for all 20 materials. The production order dataset increased from 336 rows to 384 rows, while the sales order dataset increased from 48 rows to 67 rows. The purchase order dataset also increased from 40 rows to 50 rows, although it was not directly used in this Silver Layer stockout and sales impact logic.

The main datasets involved in this component were materials.csv, inventory_snapshot.csv, production_orders.csv, and sales_orders.csv. The material dataset was used to obtain reorder level, lead time days, unit cost, and material category. The inventory snapshot dataset was used to obtain the latest quantity on hand for each raw material. The production order dataset was used to calculate material consumption and link raw materials to finished products. The sales order dataset was used to identify open customer orders that could be affected by shortage risks.

A fixed reference date of 31 March 2026 was applied in the SQL logic because the updated dataset was anchored to March 2026. This ensured that the latest inventory snapshot and the 12-month production consumption window were calculated consistently. If the real current date was used, the calculation window would shift and the stockout results might become inconsistent with the updated mock dataset.

Two SQL views were created in Azure Synapse. The first view, vw_silver_stockout_risk, was created to calculate reorder status, average daily consumption, stock cover days, and stockout risk. The second view, vw_silver_impacted_sales_orders, was created to trace affected open sales orders based on stockout-risk materials.

```
CREATE OR ALTER VIEW vw_silver_stockout_risk AS
WITH params AS (
    SELECT CAST('2026-03-31' AS DATE) AS reference_date
),
materials_raw AS (
    SELECT DISTINCT *
    FROM OPENROWSET(
        BULK 'https://ppgdatalake.dfs.core.windows.net/bronze/validated/materials/*.csv',
        FORMAT = 'CSV',
        PARSER_VERSION = '2.0',
        FIRSTROW = 2
    )
    WITH (
        material_id VARCHAR(20) COLLATE Latin1_General_100_BIN2_UTF8,
        material_name VARCHAR(100) COLLATE Latin1_General_100_BIN2_UTF8,
        category VARCHAR(50) COLLATE Latin1_General_100_BIN2_UTF8,
        unit_of_measure VARCHAR(20) COLLATE Latin1_General_100_BIN2_UTF8,
        reorder_level VARCHAR(50) COLLATE Latin1_General_100_BIN2_UTF8,
        lead_time_days VARCHAR(50) COLLATE Latin1_General_100_BIN2_UTF8,
        unit_cost VARCHAR(50) COLLATE Latin1_General_100_BIN2_UTF8,
        lot_expiry_date VARCHAR(50) COLLATE Latin1_General_100_BIN2_UTF8
    ) AS m
),
materials AS (
    SELECT
        LTRIM(RTRIM(material_id)) AS material_id,
        LTRIM(RTRIM(material_name)) AS material_name,
        LTRIM(RTRIM(category)) AS category,
        LTRIM(RTRIM(unit_of_measure)) AS unit_of_measure,
        TRY_CAST(reorder_level AS INT) AS reorder_level,
        TRY_CAST(lead_time_days AS INT) AS lead_time_days,
        TRY_CAST(unit_cost AS FLOAT) AS unit_cost,
        COALESCE(
            TRY_CONVERT(DATE, lot_expiry_date, 103),
            TRY_CONVERT(DATE, lot_expiry_date, 23),
            TRY_CONVERT(DATE, lot_expiry_date, 120)
```

Figure 5.2.3.1: SQL view creation for vw_silver_stockout_risk

Figure 5.2.3.1 shows the SQL view creation for vw_silver_stockout_risk. The view reads the validated Bronze Layer datasets from Azure Data Lake Storage and prepares the required fields for stockout risk analysis.

```sql
CREATE OR ALTER VIEW vw_silver_impacted_sales_orders AS
WITH params AS (
    SELECT CAST('2026-03-31' AS DATE) AS reference_date
),
production_raw AS (
    SELECT DISTINCT *
    FROM OPENROWSET(
        BULK 'https://ppgdatalake.dfs.core.windows.net/bronze/validated/production_orders/*.csv',
        FORMAT = 'CSV',
        PARSER_VERSION = '2.0',
        FIRSTROW = 2
    )
    WITH (
        production_order_id VARCHAR(20) COLLATE Latin1_General_100_BIN2_UTF8,
        production_date VARCHAR(50) COLLATE Latin1_General_100_BIN2_UTF8,
        finished_product VARCHAR(100) COLLATE Latin1_General_100_BIN2_UTF8,
        material_id VARCHAR(20) COLLATE Latin1_General_100_BIN2_UTF8,
        quantity_consumed VARCHAR(50) COLLATE Latin1_General_100_BIN2_UTF8
    ) AS p
),
production AS (
    SELECT
        LTRIM(RTRIM(production_order_id)) AS production_order_id,
        COALESCE(
            TRY_CONVERT(DATE, production_date, 103),
            TRY_CONVERT(DATE, production_date, 23),
            TRY_CONVERT(DATE, production_date, 120)
        ) AS production_date,
        LTRIM(RTRIM(finished_product)) AS finished_product,
        LTRIM(RTRIM(material_id)) AS material_id,
        TRY_CAST(quantity_consumed AS INT) AS quantity_consumed
    FROM production_raw
    WHERE production_order_id <> 'production_order_id'
      AND production_date <> 'production_date'
      AND material_id <> 'material_id'
      AND finished_product IS NOT NULL
      AND material_id IS NOT NULL
```

Figure 5.2.3.2: SQL view creation for vw_silver_impacted_sales_orders

Figure 5.2.3.2 shows the SQL view creation for vw_silver_impacted_sales_orders. This view was created to connect stockout-risk materials with finished products and open customer sales orders.

5.2.4 Gold Layer – Galaxy Schema

The Gold Layer is the last and most advanced layer of the data pipeline in the PPG Recoverable Assets (RA) and Inventory Risk Management (IRM) project. It is the job of this layer to convert the cleaned, enriched data in the Silver Layer into a structured, query-friendly data format that can be used directly for business reporting and analytical decision making.

Azure Synapse SQL was selected to build the Gold Layer, since SQL-based views can be created and queried on top of Silver Layer views without needing to move any physical

data or store in the cloud. This method is efficient, cost-effective, and appropriate for data in the project.

The overall design of the Gold Layer is based on the Galaxy Schema design, well-known as Fact Constellation Schema. A Galaxy Schema is a way to better support the multi-subject nature of the project's business questions. This allows different business subject like inventory risk analysis and sales order impact to be modelled independently while reusing the same descriptive dimensions, where it can reduce redundancy and enable cross-subject analysis in Power BI.

The Gold Layer consists of seven keys database objects, with five dimension views and two fact views, where all had been created within ppg_gold in Azure Synapse Analytics as shown in Table 5.2.4.1.

Table 5.4.2.1: Gold Layer Overview

Object Name	Type	Business Subject	Purpose
dim_material	View	Shared	Material attributes
dim_supplier	View	Shared	Supplier details
dim_customer	View	Sales Orders	Unique customer names from sales orders
dim_warehouse	View	Shared	Warehouse locations
dim_date	View	Shared	Date breakdown by year, quarter, month and week
fact_inventory_status	View	Inventory Risk	Central fact views that answer all Q1 - Q6
fact_sales_impact	View	Sales Impact	Central fact views that answers Q7

Both fact views share dim_material and dim_date, forming the galaxy structure. The fact_sales_impact additionally shares dim_customer for customer-level filtering in Power BI. The dim_warehouse is shared across both facts to enable warehouse-level drill-down analysis.

5.2.5 Power BI Dashboard

The Power BI Dashboard constitutes the final presentation layer of the PPG inventory analytics pipeline. It connects directly to the Gold Layer views hosted on Azure Synapse Analytics and presents the results of all seven business questions in an interactive, visual format. The dashboard was developed using Power BI Desktop and is organised into two report pages, each addressing a distinct category of inventory risk.

Prior to building the dashboard visuals, a data model was established within Power BI by importing the Gold Layer views from the ppg_gold database. The imported tables included fact_inventory_status, fact_sales_impact, dim_material, dim_supplier, dim_customer, dim_date, dim_warehouse, vw_silver_ra_mc_logic, vw_silver_stockout_risk, and vw_silver_impacted_sales. Due to the presence of multiple analytical views with overlapping material identifiers, standard Power BI relationships were insufficient to propagate filter context correctly across all visuals. This was resolved through the application of TREATAS-based DAX measures, which explicitly map filter context from dim_material into each view table without relying on model relationships.

The dashboard is structured across two pages. The first page, titled Inventory Health, addresses Q1 through Q5 by presenting materials below reorder level, recoverable assets, MC dead stock, MC expired materials, and the total financial provision amount. The second page, titled Operational Risk, addresses Q6 and Q7 by presenting materials at stockout risk and the customer orders impacted by those shortages.

5.3 Coding of system's main functions

This section moves from abstract designs to the actual code, scripts, and configurations that power the automated pipeline. The following segments detail the end-to-end technical implementation across all layers of the cloud framework. This includes the parallel orchestration pipelines inside Azure Data Factory, data cleansing logic, multi-conditional SQL views for business rules, and the custom DAX measures used to power the reporting dashboard.

5.3.1 Bronze Ingestion & Quality Check

This section describes the implementation of the two core components of the Bronze Layer: the ingestion pipeline pl_bronze_ingestion and the Data Flow df_bronze_quality. Together, these components automate the transfer of raw source data from the landing zone into a validated, quality-assured state ready for Silver Layer processing.

5.3.1.1 Source Data Upload

Prior to pipeline execution, the six synthetic CSV source files were manually uploaded to the bronze container of ppgdatalake under the structured directory path bronze/raw/<entity_name>/. A dedicated subdirectory was created per file to isolate each data source and support individual dataset referencing within ADF. Table 5.3.1.1 summarises the uploaded source files and their respective characteristics.

Table 5.3.1.1.1 Source CSV File in Bronze Container

File	Directory Path	Primary Key
materials.csv	bronze/raw/materials/	material_id
inventory_snapshot.csv	bronze/raw/inventory_snapshot/	snapshot_date + material_id
purchase_orders.csv	bronze/raw/purchase_orders/	po_id
production_orders.csv	bronze/raw/production_orders/	production_order_id
suppliers.csv	bronze/raw/suppliers/	supplier_id
sales_orders.csv	bronze/raw/sales_orders/	sales_order_id

It is worth noting that inventory_snapshot requires a **composite primary key** comprising both snapshot_date and material_id as neither field alone uniquely identifies a record which the same material appears across multiple monthly snapshots. All six files were verified to be present in the portal before pipeline construction commenced, as shown in Figure 5.3.1.1.1.

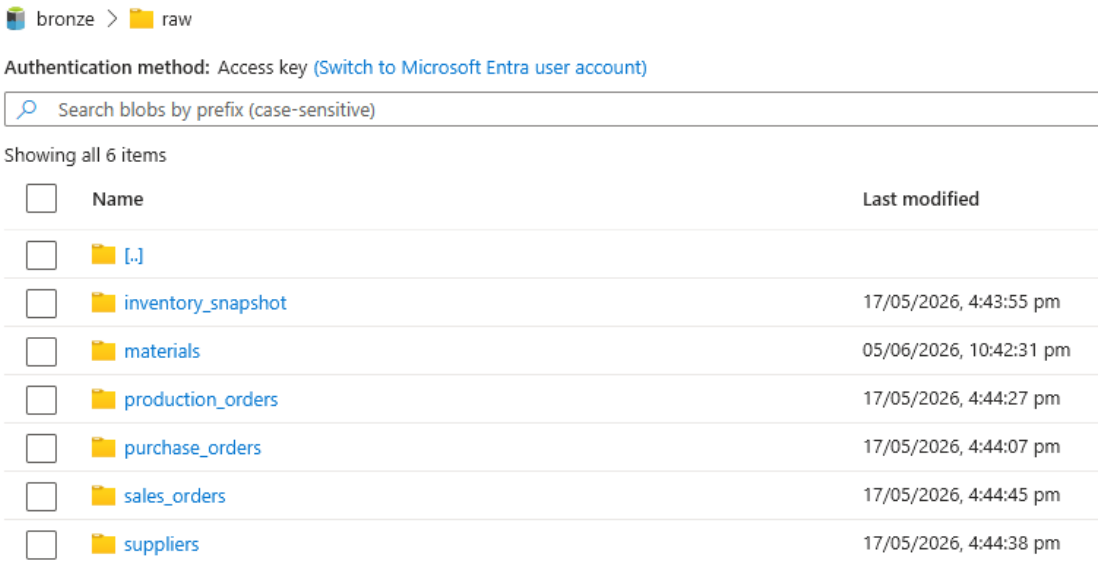


Figure 5.3.1.1.1: Six Source CSV Files in Raw Directory

5.3.1.2 Bronze Ingestion Pipeline

The pipeline pl_bronze_ingestion was constructed in ADF Studio to automate the transfer of all six raw CSV files from bronze/raw/ to bronze/ingested/. The pipeline contains six Copy Data activities, one per source file, all configured to execute in parallel with no sequential dependency between them. Parallel execution was deliberately chosen to minimise total ingestion time and to reflect the independence of each source entity. Each Copy Data activity references a dedicated source dataset pointing to the raw subdirectory, and a corresponding sink dataset pointing to the ingested subdirectory. All datasets were configured with the DelimitedText format, First row as header enabled, and schema imported from the connection store. The pipeline canvas is shown in Figure 5.3.1.2.1.

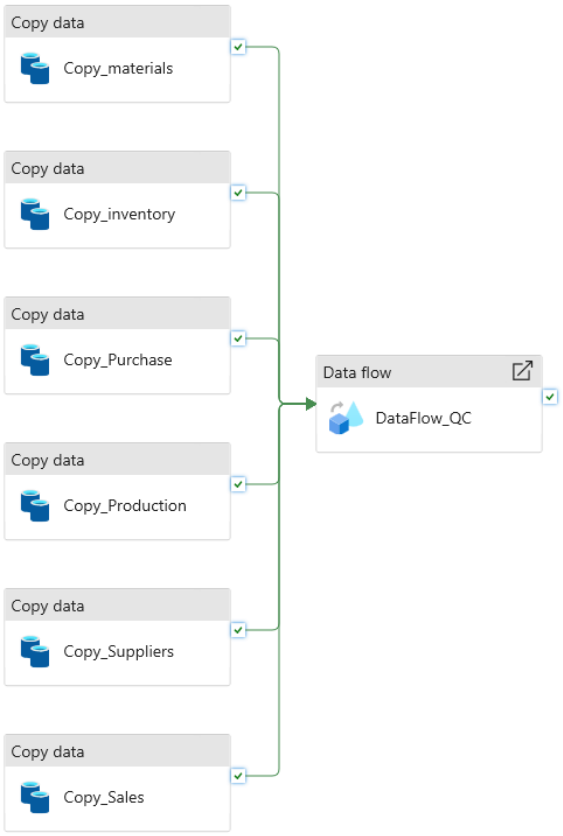


Figure 5.3.1.2.1: Bronze Ingestion Pipeline

Upon triggering the pipeline, all six CSV files were successfully copied from bronze/raw/ to bronze/ingested/, as confirmed in Figure 5.3.1.2.2.

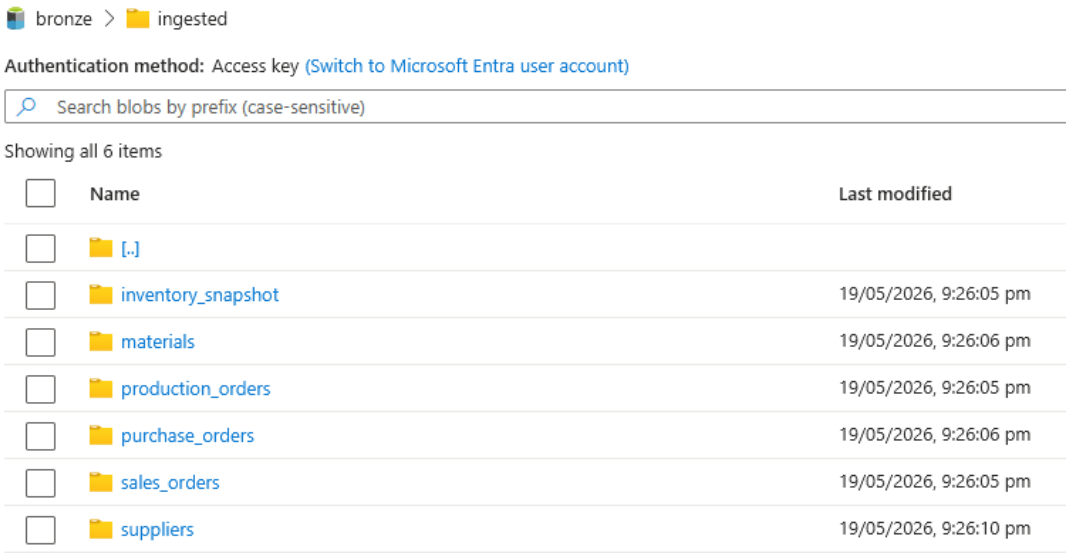


Figure 5.3.1.2.2: Six Source CSV Files in Ingested Directory

5.3.1.3 Data Quality Checks

Following ingestion, data quality validation was implemented using an ADF Data Flow named df_bronze_quality. A single Data Flow canvas was constructed containing twelve independent transformation chains, six for null and quality flag checking and six for duplicate detection, designed one pair per source entity.

Quality Flag Chains

Each quality chain begins with a dedicated Source transformation referencing the corresponding ingested dataset. A Derived Column transformation named QualityFlags then evaluates each critical field for null values using the ADF expression iif(isNull(column), 1, 0). For example, the expressions applied for materials dataset are as follows:

```
is_null_material_id = iif(isNull(material_id), 1, 0)
  is_null_unit_cost  = iif(isNull(unit_cost), 1, 0)
is_null_reorder     = iif(isNull(reorder_level), 1, 0)
  is_null_category   = iif(isNull(category), 1, 0)
 is_null_lead  = iif(isNull(lead_time_days), 1, 0)
```

A Conditional Split transformation then routes rows into two output streams based on these flags. Rows where any mandatory field is null are directed to the BadRows stream and sinked to bronze/quarantine/<entity>/. Rows where all mandatory fields are present are directed to the GoodRows stream and sinked to bronze/validated/<entity>/. The six quality flag chains on the Data Flow canvas are shown in Figure 5.3.1.3.1.

Figure 5.3.1.3.1: Data Flow Canvas for Quality Flag

Duplicate Detection

Each duplicate detection chain branches independently from a separate Source transformation pointing to the same ingested dataset. This architectural separation is necessary because chaining the Aggregate after the QualityFlags Derived Column would cause it to operate on an annotated schema and produce incorrect grouping results. An Aggregate transformation groups rows by the primary key of each entity, with row_count = count(1) as the computed column. A Filter transformation then retains only rows where row_count > 1, representing genuine duplicates. Table 5.3.1.3.1 lists the grouping keys applied per entity.

Table 5.3.1.3.1 Primary Keys Used for Duplicate Detection

Source File	Grouping Key

materials.csv	material_id
inventory_snapshot.csv	snapshot_date + material_id
purchase_orders.csv	po_id
production_orders.csv	production_order_id
suppliers.csv	supplier_id
sales_orders.csv	sales_order_id

Detected duplicates are sinked to bronze/quarantine/duplicates/<entity>/. The six duplicate detection chains are shown in Figure 5.3.1.3.2.

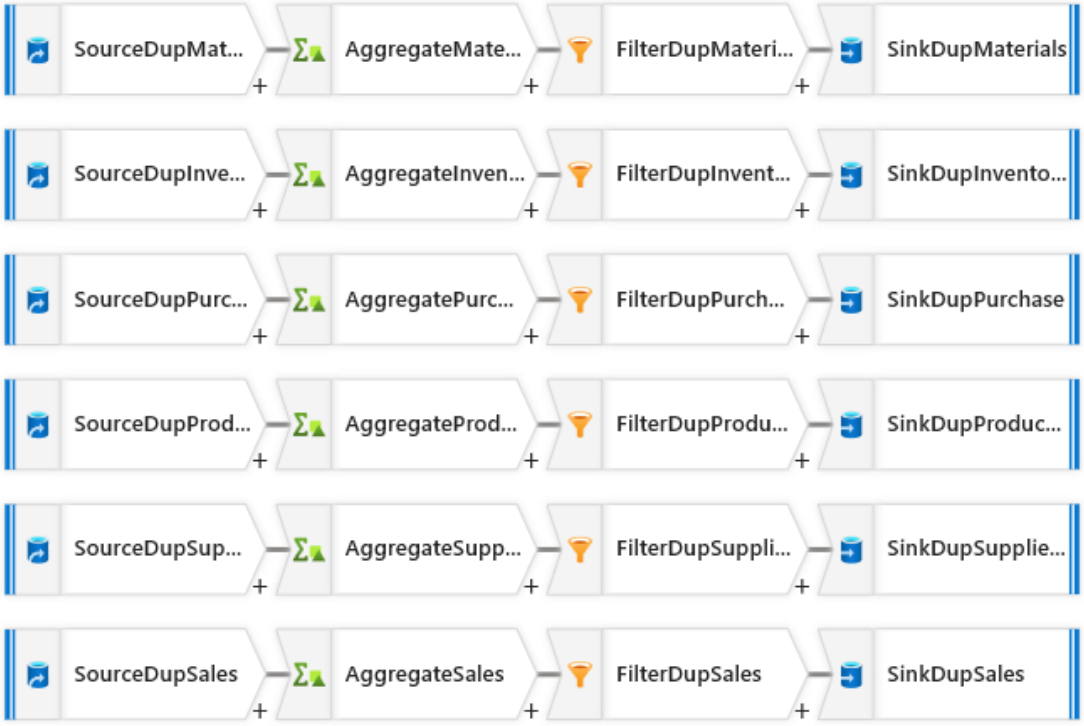


Figure 5.3.1.3.2: Data Flow Canvas for Duplicate Detection

The Data Flow activity df_bronze_quality was linked to pl_bronze_ingestion via a success dependency arrow, ensuring that quality checks execute only after all six Copy Data activities have completed successfully. Upon execution, the bronze/validated/ and bronze/quarantine/ output directories were populated as shown in Figures 5.3.1.3.3 and 5.3.1.3.4.

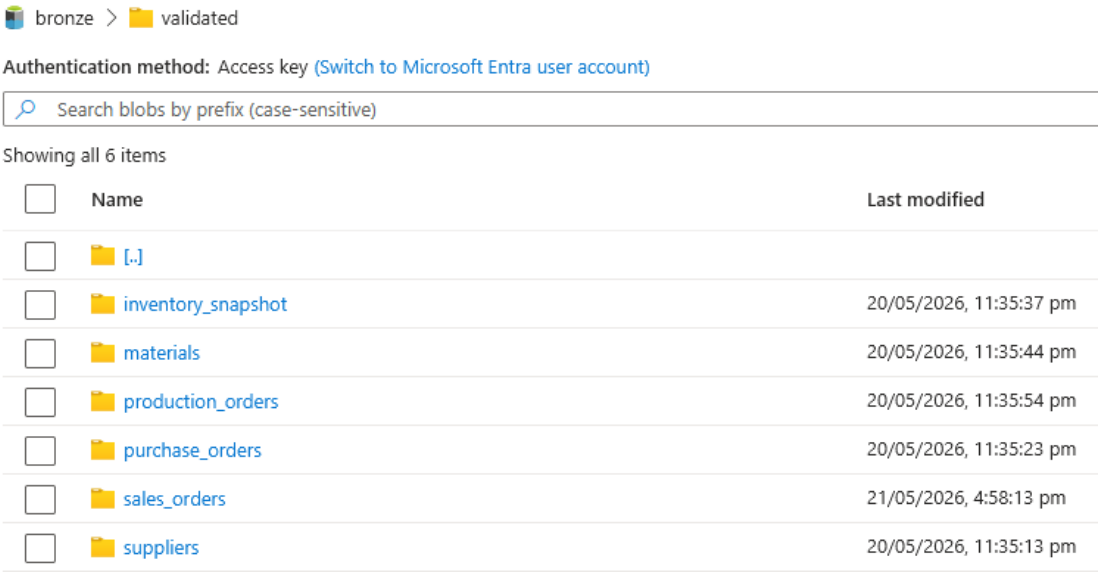


Figure 5.3.1.3.3: Validated Ouput Files in Bronze Container

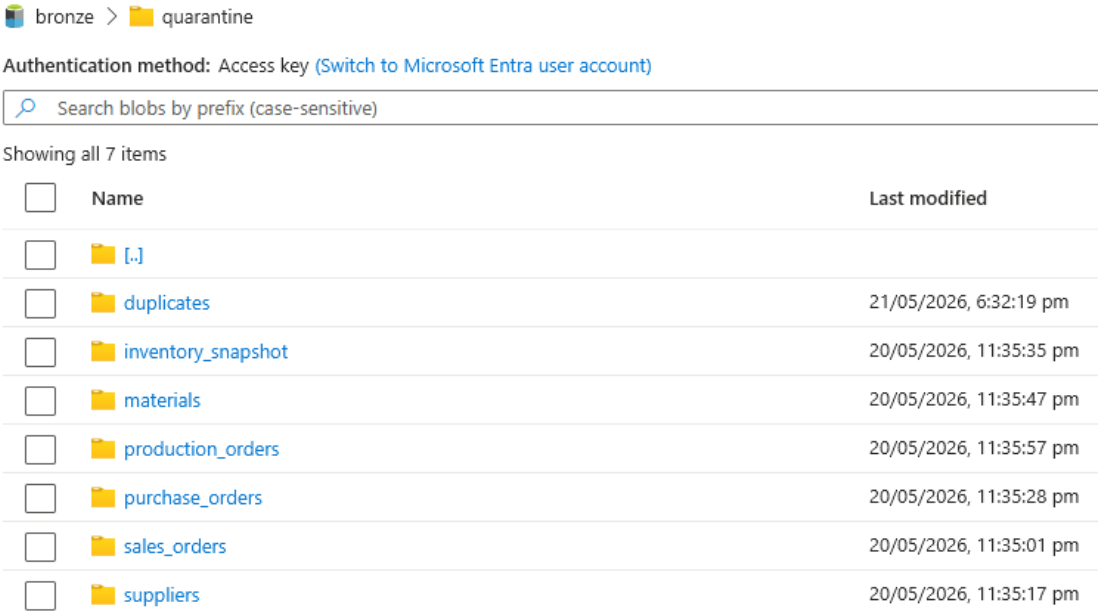


Figure 5.3.1.3.4: Quarantined Output Files in Bronze Container

5.3.2 RA & MC Business Logic

The main function of this component was divided into four parts that including 12-month consumption calculation, Recoverable Asset detection, Magna Carta Dead Stock detection, Magna Carta Expired detection, and financial provision calculation. These functions were implemented using SQL logic within the vw_silver_ra_mc_logic view in Azure Synapse Serverless SQL. A fixed reference date of 2026-03-31 was applied consistently across all calculations that includes date instead of the current system date, as the dataset is anchored to March 2026. This ensures that the 12-month consumption window, idle period calculation, and expiry date comparisons all produce consistent results that are aligned with the dataset rather than shifting based on when the query is executed.

```
latest_inventory AS (
    SELECT
        i.*,
        ROW_NUMBER() OVER (
            PARTITION BY i.material_id
            ORDER BY i.snapshot_date DESC
        ) AS rn
    FROM inventory i
),
```

Figure 5.3.2.1: Latest inventory selection SQL logic

Since the inventory snapshot dataset contains 12 monthly records per material, it was necessary to identify the most recent snapshot for each material before any calculations could be performed. As shown in Figure 5.3.2.1, the latest_inventory CTE applied a ROW_NUMBER() window function partitioned by material_id and ordered by snapshot_date descending. This assigned rank 1 to the most recent snapshot row for each material. Only rows where rn = 1 were carried forward into the final calculation, ensuring all flags and quantities were based on the current stock position rather than historical records.

```
consumption_12m AS (
    SELECT
        material_id,
        SUM(quantity_consumed) AS total_consumed_12m,
        MAX(production_date) AS last_consumed_date
    FROM production
    WHERE production_date >= DATEADD(MONTH, -12, CAST('2026-03-31' AS DATE))
    GROUP BY material_id
),
```

Figure 5.3.3.2 12-month consumption calculation SQL logic

As shown in Figure 5.3.2.2, the consumption calculation step calculated the total quantity consumed per material over the past 12 months from production order records.

12-Month Consumption = Total quantity Consumed

WHERE production_date >= 2026-03-31 minus 12 months

Only production orders where production_date >= DATEADD(MONTH, -12, CAST('2026-03-31' AS DATE)) were included, meaning only production records from 2025-03-31 onwards were considered. A fixed reference date of 2026-03-31 was used instead of the current system date to ensure consistency across all team members as the dataset is anchored to Marh 2026. SUM(quantity_consumed) accumulated the total units consumed per material, and MAX(production_date) captured the most recent date each material was used in production, which is required for dead stock detection in Q3.

```
idle_stats AS (
    SELECT
        m.material_id,
        c12m.last_consumed_date,
        CASE
            WHEN c12m.last_consumed_date IS NULL THEN 999.0
            ELSE CAST(DATEDIFF(MONTH, c12m.last_consumed_date, CAST('2026-03-31' AS DATE)) AS FLOAT)
        END AS months_since_consumed
    FROM materials m
    LEFT JOIN consumption_12m c12m ON m.material_id = c12m.material_id
)
```

Figure 5.3.2.3 Months since consumed calculation SQL logic

As shown in Figure 5.3.2.3, the idle calculation step calculated how many months had passed since each material was last consumed. The calculation logic

Months Since Consumed = DATEDIFF between last consumed date and 2026-03-31

If never consumed → Months Since Consumed = 999.0

A LEFT JOIN was applied between the materials and consumption calculation results to ensure all 20 materials were included even if they had no production records. The number of months was calculated using DATEDIFF(MONTH, last_consumed_date, CAST('2026-03-31' AS DATE)). The same fixed reference date of 2026-03-31 was applied here to maintain consistency with the consumption calculation window. Materials where last_consumed_date was NULL, meaning they had never appeared in any production order.

They were assigned a placeholder value of 999.0 to ensure they would always exceed the 9-month dead stock threshold.

```
-- Q2: Recoverable Asset Calculations
ISNULL(c12m.total_consumed_12m, 0) AS consumption_12m,
CASE
    WHEN li.quantity_on_hand > ISNULL(c12m.total_consumed_12m, 0)
        THEN li.quantity_on_hand - ISNULL(c12m.total_consumed_12m, 0)
    ELSE 0
END AS excess_quantity,
CASE
    WHEN li.quantity_on_hand > ISNULL(c12m.total_consumed_12m, 0) THEN 1
    ELSE 0
END AS is_recoverable_asset,
```

Figure 5.3.2.4 Recoverable Asset flag SQL logic

As shown in Figure 5.3.2.4, the Q2 logic compared each material's current quantity_on_hand from the latest inventory snapshot against its total 12-month consumption.

$$\text{Excess Quantity} = \text{Quantity on Hand} - \text{12-Month Consumption}$$

$$\text{Recoverable Asset} = 1 \text{ if Quantity on Hand} > \text{12-Month Consumption, else } 0$$

ISNULL(c12m.total_consumed_12m, 0) ensured that materials with no production records were treated as zero consumed rather than NULL, preventing calculation errors. A material was flagged as is_recoverable_asset = 1 when quantity on hand exceeded 12-month consumption, indicating PPG held more stock than it could realistically use within a year. The difference was stored as excess_quantity to show the exact surplus amount.

```
-- Q3: Magna Carta Dead Stock Calculations
ids.last_consumed_date,
ids.months_since_consumed,
CASE
    WHEN ids.months_since_consumed >= 9 AND li.lot_expiry_date IS NULL THEN 1
    ELSE 0
END AS is_mc_dead_stock,
```

Figure 5.3.2.5: Q3 MC Dead Stock flag SQL logic

As shown in Figure 5.3.2.5, the Q3 logic flagged a material as is_mc_dead_stock = 1 when two conditions were simultaneously true: months_since_consumed >= 9 and lot_expiry_date IS NULL.

$$\text{MC Dead Stock} = 1$$

if Months Since Consumed >= 9 AND lot_expiry_date is NULL, else 0

Both conditions had to be met together where a material idle for 9 or more months but having an expiry date would be handled separately under Q4, not Q3. The last consumed date and months since consumed values were also included in the output to provide supporting detail for each flagged material.

```
-- Q4: Magna Carta Expired Calculations
CASE
    WHEN li.lot_expiry_date IS NOT NULL AND li.lot_expiry_date < CAST('2026-03-31' AS DATE) THEN 1
    ELSE 0
END AS is_mc_expired,
```

Figure 5.3.2.6: Q4 MC Expired flag calculation

As shown in Figure 5.3.2.6, the Q4 logic flagged a material as is_mc_expired = 1 when its lot_expiry_date was not NULL and had already passed relative to the fixed reference date.

MC Expired = 1 if lot_expiry_date is not NULL AND

lot_expiry_date < 2026-03-31, else 0

CAST('2026-03-31' AS DATE) was used as the reference point for comparison instead of the current system date. A material with a lot expiry date before 2026-03-31 was classified as expired, while materials with future expiry dates or no expiry date were not flagged under Q4.

```
-- Q5: Composite Flags and Financial Provision Logic
CASE
    WHEN (ids.months_since_consumed >= 9 AND li.lot_expiry_date IS NULL)
        OR (li.lot_expiry_date IS NOT NULL AND li.lot_expiry_date < CAST('2026-03-31' AS DATE)) THEN 1
    ELSE 0
END AS is_mc_material,
CASE
    WHEN (ids.months_since_consumed >= 9 AND li.lot_expiry_date IS NULL)
        OR (li.lot_expiry_date IS NOT NULL AND li.lot_expiry_date < CAST('2026-03-31' AS DATE))
        THEN CAST(li.quantity_on_hand AS FLOAT) * m.unit_cost
    ELSE 0.0
END AS mc_provision_amount
```

Figure 5.3.2.7: Q5 composite MC flag and provision amount calculation

As shown in Figure 5.3.2.7, the Q5 logic applied a composite flag is_mc_material that equals 1 when a material qualifies as either MC Dead Stock or MC Expired.

MC Provision Amount = Quantity on Hand × Unit Cost

Total MC Provision = SUM of all mc_provision_amount where is_mc_material = 1

Either condition alone was sufficient to trigger the flag. The financial provision amount was then calculated as a 100% write-down by multiplying quantity_on_hand by unit_cost. CAST(li.quantity_on_hand AS FLOAT) was applied to ensure decimal precision in the multiplication. Materials that did not meet either MC condition received a provision amount of 0.0.

```
SELECT ···
FROM latest_inventory li
JOIN materials m ON li.material_id = m.material_id
LEFT JOIN consumption_12m c12m ON li.material_id = c12m.material_id
LEFT JOIN idle_stats ids ON li.material_id = ids.material_id
WHERE li.rn = 1;
GO
```

Figure 5.3.2.8 Final SELECT JOIN structure and filter

As shown in Figure 5.3.2.8, the final SELECT joined all calculation steps together to produce one output row per material. An inner JOIN was applied between the latest inventory and materials to ensure only materials existing in the master list were included. LEFT JOIN was applied for both the consumption and idle calculation results to preserve all materials even if they had no production history. The WHERE li.rn = 1 filter was applied to retain only the most recent inventory snapshot per material, producing a clean single-row output per material containing all computed flags and financial values ready for the Gold layer.

5.3.3 Stockout & Sales Order Impact Logic

The main function of this component was divided into three parts: below reorder level detection, stockout risk calculation, and impacted sales order tracing. These functions were implemented using SQL logic in Azure Synapse Serverless SQL.

First, below reorder level detection was performed by comparing the latest quantity on hand with the reorder level in the material dataset. The latest inventory snapshot for each material was selected using the ROW_NUMBER() function. After the dataset update, the latest inventory snapshot used by the system was 31 March 2026 for all 20 materials. This confirmed that the newly added January, February, and March 2026 inventory records were successfully included in the Silver Layer logic. A material was flagged as below reorder level when its quantity on hand was lower than the reorder level.

Second, stockout risk was calculated using average daily consumption and stock cover days. The total quantity consumed for each material was calculated from production order records within the 12-month window based on the fixed reference date of 31 March 2026. The average daily consumption was then derived by dividing the total consumed quantity by the active production date range. After that, stock cover days were calculated using the available stock quantity and average daily consumption.

Stock Cover Days = Quantity on Hand / Average Daily Consumption

A material was classified as having stockout risk when its stock cover days were lower than its lead time days. This means that the current available stock may not be sufficient to cover material usage before replenishment can arrive.

Stockout Risk = Stock Cover Days < Lead Time Days

```
ISNULL(cs.total_consumed, 0) AS total_consumed,
ISNULL(cs.avg_daily_consumption, 0.001) AS avg_daily_consumption,

CASE
    WHEN ISNULL(cs.avg_daily_consumption, 0.001) > 0
        THEN CAST(li.quantity_on_hand AS FLOAT) / ISNULL(cs.avg_daily_consumption, 0.001)
    ELSE 9999
END AS stock_cover_days,

CASE
    WHEN
        CASE
            WHEN ISNULL(cs.avg_daily_consumption, 0.001) > 0
                THEN CAST(li.quantity_on_hand AS FLOAT) / ISNULL(cs.avg_daily_consumption, 0.001)
            ELSE 9999
        END < m.lead_time_days
    THEN 1
    ELSE 0
END AS is_stockout_risk
```

Figure 5.3.3.1: Stockout risk calculation SQL logic

Figure 5.3.3.1 shows the SQL logic used to calculate average daily consumption, stock cover days, and stockout risk. The calculation compares stock cover days with lead time days to determine whether a material is at risk of stockout.

Third, impacted sales order tracing was performed by linking stockout-risk materials to finished products through the production order dataset. After the finished products were identified, they were linked to open sales orders from the sales order dataset. This allowed

the system to identify which customer orders could be affected by materials with stockout risk.

Stockout-risk material:

- Finished product
- Open sales order
- Affected customer

```
SELECT
    os.sales_order_id,
    os.order_date,
    os.customer_name,
    os.finished_product,
    os.quantity_ordered,
    os.required_delivery_date,
    os.status,
    pml.material_id,
    arm.material_name,
    arm.category,
    arm.quantity_on_hand,
    arm.reorder_level,
    arm.avg_daily_consumption,
    arm.stock_cover_days,
    arm.lead_time_days,
    arm.is_stockout_risk
FROM open_sales os
JOIN product_material_link pml
    ON os.finished_product = pml.finished_product
JOIN at_risk_materials arm
    ON pml.material_id = arm.material_id;
```

Figure 5.3.3.2: Impacted sales order tracing SQL logic

Figure 5.3.3.2 shows the SQL join logic used to trace impacted sales orders. Stockout-risk materials were linked to finished products, and the finished products were then linked to open customer sales orders.

Several data preparation steps were included in the SQL script to improve reliability. Duplicate records were reduced using SELECT DISTINCT. Date fields were read as text and converted using TRY_CONVERT because different date formats were found in the

source datasets. The production order date used the yyyy-mm-dd format, while other date fields used different formats. In the updated implementation, the SQL logic was also adjusted to use a fixed reference date of 31 March 2026. This was required to ensure that the 12-month consumption window and the latest inventory snapshot were aligned with the updated mock dataset. After the date conversion and reference date logic were updated, the consumption, stock cover, and stockout risk calculations were generated correctly.

5.3.4 Galaxy Schema & Fact Table Loading

This section documents the key SQL logic implemented to build the Gold Layer galaxy schema. All scripts were developed and executed using Azure Synapse Analytics SQL , connected to the ppg_gold database.

All views use CREATE OR ALTER VIEW, which is the correct approach for Synapse SQL. Plain CREATE TABLE is not supported on SQL Pool as it does not allow persistent physical tables. All Gold Layer objects are non-persistent views that compute results on-demand from the Bronze and Silver Layers.

The reference date is fixed at 2026-03-31 across all views to align with the dataset anchor date used to ensure consistent results regardless of when the scripts are executed.

Dimension View: dim_material

The dim_material view reads directly from the Bronze validated materials CSV files using OPENROWSET as shown in Figure 5.3.4.1. It provides descriptive attributes for each material including its category, unit of measure of reorder level, lead time and unit cost. These attributes are used as reference data by the fact view and by downstream Power BI visuals.

```sql
-- 1. Material Dimension (shared by fact_inventory_status + fact_sales_impact)
CREATE OR ALTER VIEW dim_material AS
SELECT
    material_id,
    material_name,
    category,
    unit_of_measure,
    reorder_level,
    lead_time_days,
    unit_cost
FROM OPENROWSET(
    BULK 'https://ppgdatalake.dfs.core.windows.net/bronze/validated/materials/*.csv',
    FORMAT = 'CSV',
    PARSER_VERSION = '2.0',
    FIRSTROW = 2
) WITH (
    material_id       VARCHAR(20)  COLLATE Latin1_General_100_BIN2_UTF8,
    material_name     VARCHAR(100) COLLATE Latin1_General_100_BIN2_UTF8,
    category          VARCHAR(50)  COLLATE Latin1_General_100_BIN2_UTF8,
    unit_of_measure   VARCHAR(20)  COLLATE Latin1_General_100_BIN2_UTF8,
    reorder_level     INT,
    lead_time_days    INT,
    unit_cost         FLOAT
) AS r;
GO
```

Figure 5.3.4.1: Material Dimension View SQL Scripts

Dimension View: dim_supplier

The dim_supplier view reads from the Bronze validated suppliers for CSV files. It provides supplier-level context including supplier name, country of origin, and reliability rating. This view supports potential future analysis of supplier performance and its relationship to stockout risk. The creation of supplier dimension view is shown in Figure 5.3.4.2.

```sql
-- 2. Supplier Dimension (standalone — supports future supplier risk analysis)
CREATE OR ALTER VIEW dim_supplier AS
SELECT
    supplier_id,
    supplier_name,
    country,
    reliability_rating
FROM OPENROWSET(
    BULK 'https://ppgdatalake.dfs.core.windows.net/bronze/validated/suppliers/*.csv',
    FORMAT = 'CSV',
    PARSER_VERSION = '2.0',
    FIRSTROW = 2
) WITH (
    supplier_id        VARCHAR(20)  COLLATE Latin1_General_100_BIN2_UTF8,
    supplier_name      VARCHAR(100) COLLATE Latin1_General_100_BIN2_UTF8,
    country            VARCHAR(50)  COLLATE Latin1_General_100_BIN2_UTF8,
    reliability_rating VARCHAR(10)  COLLATE Latin1_General_100_BIN2_UTF8
) AS r;
GO
```

Figure 5.3.4.2: Supplier Dimension View SQL Scripts

Dimension View: dim_customer

The dim_customer view extracts distinct customer names from the sales orders Bronze CSV. A DISTINCT as in Figure 5.3.4.3 is used to ensure that each customer appears only once in the dimension. All seven columns in the sales orders of CSV are declared in the WITH clause to satisfy the Synapse Serverless column mapping requirements, even though only customer_name is selected in the output.

```sql
-- 3. Customer Dimension (shared by fact_sales_impact for Q7 analysis)
CREATE OR ALTER VIEW dim_customer AS
SELECT DISTINCT
    customer_name
FROM OPENROWSET(
    BULK 'https://ppgdatalake.dfs.core.windows.net/bronze/validated/sales_orders/*.csv',
    FORMAT = 'CSV',
    PARSER_VERSION = '2.0',
    FIRSTROW = 2
) WITH (
    sales_order_id         VARCHAR(20)  COLLATE Latin1_General_100_BIN2_UTF8,
    order_date             VARCHAR(50)  COLLATE Latin1_General_100_BIN2_UTF8,
    customer_name          VARCHAR(100) COLLATE Latin1_General_100_BIN2_UTF8,
    finished_product       VARCHAR(100) COLLATE Latin1_General_100_BIN2_UTF8,
    quantity_ordered       VARCHAR(50)  COLLATE Latin1_General_100_BIN2_UTF8,
    required_delivery_date VARCHAR(50)  COLLATE Latin1_General_100_BIN2_UTF8,
    status                 VARCHAR(50)  COLLATE Latin1_General_100_BIN2_UTF8
) AS r;
GO
```

Figure 5.3.4.3: Customer Dimension View SQL Scripts

Dimension View: dim_warehouse

The dim_warehourse view is a new dimension introduced as part of the Galaxy Schema upgrade. It extracts the distinct warehouse location values from the inventory snapshot Bronze Layer CSV. This dimension is shared across both fact views, enabling the Power BI to filter and slice inventory risk metrics and sales order impact by warehouse location. In Figure 5.3.4.4, a WHERE clause filters out the column header row and NULL values to ensure only valid warehouse codes are included.

```
-- 4. Warehouse Dimension (NEW — extracted from inventory snapshot)
--    Shared by both fact views to support warehouse-level filtering in Power BI
CREATE OR ALTER VIEW dim_warehouse AS
SELECT DISTINCT
    warehouse_location
FROM OPENROWSET(
    BULK 'https://ppgdatalake.dfs.core.windows.net/bronze/validated/inventory_snapshot/*.csv'
    FORMAT = 'CSV',
    PARSER_VERSION = '2.0',
    FIRSTROW = 2
) WITH (
    snapshot_date_raw      VARCHAR(20)  COLLATE Latin1_General_100_BIN2_UTF8,
    material_id            VARCHAR(20)  COLLATE Latin1_General_100_BIN2_UTF8,
    warehouse_location     VARCHAR(50)  COLLATE Latin1_General_100_BIN2_UTF8,
    quantity_on_hand       INT,
    lot_expiry_date_raw    VARCHAR(30)  COLLATE Latin1_General_100_BIN2_UTF8
) AS r
WHERE warehouse_location IS NOT NULL
  AND warehouse_location <> 'warehouse_location';
GO
```

Figure 5.3.4.4: Warehouse Dimension View SQL Script

Dimension View: dim_date

The dim_date view was expanded as part of the Galaxy Schema upgrade to collect unique dates from four sources. The addition of sales order dates and required delivery dates from Q7 silver view enable Power BI to perform time-based filtering on both fact views using the same shared date dimension. A UNION of all four sources as in Figure 5.3.4.5 ensures all relevant dates are captured without duplicates.

```sql
-- 5. Dynamic Date Dimension (shared by both fact views)
--    Collects unique dates from both Silver views + sales orders
--    Reference date fixed at 2026-03-31 to match M2 and M3 scripts
CREATE OR ALTER VIEW dim_date AS
WITH unique_dates AS (
    -- Inventory snapshot dates from M2 silver view
    SELECT DISTINCT CAST(snapshot_date AS DATE) AS date_id
    FROM vw_silver_ra_mc_logic

    UNION

    -- Inventory snapshot dates from M3 silver view
    SELECT DISTINCT CAST(snapshot_date AS DATE) AS date_id
    FROM vw_silver_stockout_risk

    UNION

    -- Sales order dates from M3 Q7 silver view (for fact_sales_impact)
    SELECT DISTINCT CAST(order_date AS DATE) AS date_id
    FROM vw_silver_impacted_sales_orders
    WHERE order_date IS NOT NULL

    UNION

    -- Required delivery dates from M3 Q7 silver view
    SELECT DISTINCT CAST(required_delivery_date AS DATE) AS date_id
    FROM vw_silver_impacted_sales_orders
    WHERE required_delivery_date IS NOT NULL
)
SELECT
    date_id,
    DATEPART(YEAR,    date_id) AS year,
    DATEPART(QUARTER, date_id) AS quarter,
    DATEPART(MONTH,   date_id) AS month,
    DATENAME(MONTH,   date_id) AS month_name,
    DATEPART(WEEK,    date_id) AS week_of_year,
    DATEPART(WEEKDAY, date_id) AS day_of_week,
    DATENAME(WEEKDAY, date_id) AS day_name
FROM unique_dates;
GO
```

Figure 5.3.4.5: Date Dimension View SQL Scripts

Central Fact View: fact_inventory_status

The fact_inventory_status view is the central object of the Gold Layer. It merges outputs from two Silver Layer views: vw_silver_ra_mc_logic and vw_silver_stockout_risk. A FULL OUTER JOIN is used to ensure all 20 materials are represented regardless of whether they appear in one or both Silver views as shown in Figure 5.3.4.6.

```
CREATE OR ALTER VIEW fact_inventory_status AS
SELECT
    -- Shared keys (COALESCE handles materials only in one silver view)
    COALESCE(mc.material_id, st.material_id)
        COLLATE Latin1_General_100_BIN2_UTF8          AS material_id,
    COALESCE(mc.snapshot_date, st.snapshot_date)       AS snapshot_date,
    COALESCE(mc.warehouse_location, st.warehouse_location)
        COLLATE Latin1_General_100_BIN2_UTF8          AS warehouse_location,
    COALESCE(mc.quantity_on_hand, st.quantity_on_hand)  AS quantity_on_hand,

    -- Foreign keys for dimension joins
    -- JOIN dim_material ON fact_inventory_status.material_id = dim_material.material_id
    -- JOIN dim_date ON fact_inventory_status.snapshot_date = dim_date.date_id
    -- JOIN dim_warehouse ON fact_inventory_status.warehouse_location = dim_warehouse.warehouse_location

    -- M2 metrics: Recoverable Assets & Magna Carta Provision (Q2, Q3, Q4, Q5)
    CAST(ISNULL(mc.consumption_12m, 0)       AS FLOAT) AS consumption_12m,
    CAST(ISNULL(mc.excess_quantity, 0)        AS FLOAT) AS excess_quantity,
    CAST(ISNULL(mc.is_recoverable_asset, 0)   AS BIT)   AS is_recoverable_asset,
    CAST(ISNULL(mc.is_mc_dead_stock, 0)       AS BIT)   AS is_mc_dead_stock,
    CAST(ISNULL(mc.is_mc_expired, 0)          AS BIT)   AS is_mc_expired,
    CAST(ISNULL(mc.is_mc_material, 0)         AS BIT)   AS is_mc_material,
    CAST(ISNULL(mc.mc_provision_amount, 0)    AS FLOAT) AS mc_provision_amount,
    CAST(ISNULL(mc.months_since_consumed, 0)  AS FLOAT) AS months_since_consumed,
    mc.lot_expiry_date,
    mc.last_consumed_date,

    -- M3 metrics: Stockout Risk & Reorder Flags (Q1, Q6)
    CAST(ISNULL(st.is_below_reorder, 0)       AS BIT)   AS is_below_reorder,
    CAST(ISNULL(st.is_stockout_risk, 0)       AS BIT)   AS is_stockout_risk,
    CAST(ISNULL(st.stock_cover_days, 0)       AS FLOAT) AS stock_cover_days,
    CAST(ISNULL(st.avg_daily_consumption, 0)  AS FLOAT) AS avg_daily_consumption,
    CAST(ISNULL(st.total_consumed, 0)         AS FLOAT) AS total_consumed,
    st.reorder_level,
    st.lead_time_days

FROM vw_silver_ra_mc_logic mc
FULL OUTER JOIN vw_silver_stockout_risk st
    ON  mc.material_id   COLLATE Latin1_General_100_BIN2_UTF8
      = st.material_id   COLLATE Latin1_General_100_BIN2_UTF8
    AND CAST(mc.snapshot_date AS DATE) = CAST(st.snapshot_date AS DATE);
GO
```

Figure 5.3.4.6: Central Fact View (fact_inventory_status) SQL Scripts

Several important technical decisions were made in this view:

- FULL OUTER JOIN is used to ensure materials that appear in only one silver view are not excluded from the fact view. This is important for materials with no production consumption history, which appear in M2's view but may not match M3's stockout calculations.

- COALESCE is used on material_id, snapshot_date, warehouse_location, and quantity_on_hand to safely handle cases where a material only exists in one Silver view.
- COLLATE Latin1_General_100_BIN2_UTF8 is applied on VARCHAR join keys to resolve potential collation mismatch errors between the two Silver views.
- ISNULL with default values is used for all flag and metric columns to prevent NULL propagation into the Power BI dashboard.
- CAST to BIT is applied on all Boolean flag columns (is_recoverable_asset, is_mc_dead_stock, is_mc_expired, is_mc_material, is_below_reorder, is_stockout_risk) for type safety.
- lot_exiry_date and last_consumed_date are preserved in the fact view to support Q3 and Q4 analysis.

Central Fact View: fact_sales_impact

The fact_sales_impact view is the second fact view introduced as part of the Galaxy Schema update. It elevates the Q7 Silver view (vw_silver_impacted_sales_orders) into a proper Gold Layer fact view that participates in the share dimension structure.

Figure 5.3.4.7 shows the SQL Scripts that identify the delivery days either positive or negative. A positive delivery_risk_days value indicates the order is at risk of missing its delivery date because stock will run out before the delivery is due. A negative value means there is sufficient stock to cover the delivery. This metric gives a ready-mad urgency indicator for Power BI visuals without requiring DAX calculations.

```sql
CREATE OR ALTER VIEW fact_sales_impact AS
SELECT
    -- Order identifiers
    iso.sales_order_id,
    iso.customer_name,
    iso.material_id             COLLATE Latin1_General_100_BIN2_UTF8 AS material_id,
    CAST(iso.order_date AS DATE)             AS order_date,
    CAST(iso.required_delivery_date AS DATE)  AS required_delivery_date,

    -- Order details
    iso.finished_product,
    iso.quantity_ordered,
    iso.status,

    -- Material risk context (denormalised from M3 for reporting convenience)
    iso.material_name,
    iso.category,
    iso.quantity_on_hand,
    iso.reorder_level,
    iso.avg_daily_consumption,
    iso.stock_cover_days,
    iso.lead_time_days,
    iso.is_stockout_risk,

    -- Derived urgency metric: days until delivery minus stock cover days
    -- Positive = order at risk of missing delivery
    -- Negative = stock sufficient to cover delivery
    CAST(
        DATEDIFF(DAY, CAST('2026-03-31' AS DATE), CAST(iso.required_delivery_date AS DATE))
        - iso.stock_cover_days
    AS FLOAT) AS delivery_risk_days

FROM vw_silver_impacted_sales_orders iso;
GO
```

Figure 5.3.4.7: Central Fact View (fact_sales_impact) SQL Scripts

5.3.5 Power BI Data Model and Dashboard Measures

This section documents the key data model configurations, DAX measures, and calculated columns implemented within Power BI Desktop to support the PPG inventory analytics dashboard.

5.3.5.1 Data Model Relationships and Filter Context Issue

The data model was established in Power BI Model view by importing all Gold Layer views. During development, a critical issue was identified: columns such as quantity_on_hand, consumption_12m, lead_time_days, and stock_cover_days were returning inflated repeated values across all rows in table visuals. Investigation confirmed that standard model relationships between dim_material and the analytical views were not propagating filter context correctly, causing every row to display the grand total instead of the per-material value.

This issue was diagnosed by testing a standalone table visual using only columns from fact_inventory_status, which returned correct per-material values. The problem was isolated to cross-table visuals that mixed dimension columns from dim_material with measure columns from the analytical views. The root cause was ambiguous or inactive relationship paths created by the presence of multiple views sharing the same material_id key field.

The fix was applied by replacing all raw column references with TREATAS-based DAX measures. TREATAS explicitly instructs Power BI to treat the filtered values of dim_material[material_id] as if they were filter values on the target view's material_id column, bypassing the broken relationship path entirely.

5.3.5.2 DAX Measures: TREATAS Pattern

All numeric measures that cross from dim_material into an analytical view follow this pattern:

```
Quantity On Hand Fixed =
CALCULATE(
    SUM(fact_inventory_status[quantity_on_hand]),
    TREATAS(
        VALUES(dim_material[material_id]),
        fact_inventory_status[material_id]
    )
)
```

The same pattern was applied to produce the following measures used across the dashboard:

- Consumption 12M Fixed - SUM of vw_silver_ra_mc_logic[consumption_12m]
- Excess Quantity Fixed - SUM of vw_silver_ra_mc_logic[excess_quantity]
- Lead Time Fixed - SUM of vw_silver_stockout_risk[lead_time_days]
- Stock Cover Fixed - SUM of vw_silver_stockout_risk[stock_cover_days]

5.3.5.3 DAX Measures: Dead Stock and Expired Materials

For Q3 and Q4, measures that aggregate across filtered subsets of vw_silver_ra_mc_logic use the FILTER iterator pattern instead of TREATAS, as these measures are used in standalone KPI cards with no row context from dim_material:

```
Dead Stock Count =
COUNTROWS(
  FILTER(
    vw_silver_ra_mc_logic,
    vw_silver_ra_mc_logic[is_mc_dead_stock] = 1
  )
)

Dead Stock Provision =
SUMX(
  FILTER(
    vw_silver_ra_mc_logic,
    vw_silver_ra_mc_logic[is_mc_dead_stock] = 1
  ),
  vw_silver_ra_mc_logic[mc_provision_amount]
)
```

Equivalent measures were defined for expired materials using is_mc_expired = 1 as the filter condition.

5.3.5.4 DAX Measures: Operational Risk

The following measures support the Q6 and Q7 visuals on the Operational Risk page:

```
Stockout Risk Count =
COUNTROWS(
  FILTER(
    vw_silver_stockout_risk,
    vw_silver_stockout_risk[is_stockout_risk] = 1
```

```
    )
)

Customers Affected =
COUNTROWS(
  DISTINCT(
    SELECTCOLUMNS(
      FILTER(
        vw_silver_impacted_sales,
        vw_silver_impacted_sales[is_stockout_risk] = 1
      ),
      "customer", vw_silver_impacted_sales[customer_name]
    )
  )
)
```

5.3.5.5 Calculated Column: Risk Status

A calculated column named Risk Status was created on the vw_silver_stockout_risk table to classify each at-risk material for display in the Q6 table visual:

```
Risk Status = IF(vw_silver_stockout_risk[stock_cover_days] = 0, "Critical", "At Risk")
```

This column was created as a calculated column rather than a measure because it operates row by row on the table, classifying each material individually based on its stock cover days value.

5.3.6 Master Pipeline Orchestration

To orchestrate the complete end-to-end execution of the data pipeline across all three Medallion layers, a master pipeline named pl_master_ppg was constructed in ADF Studio. This pipeline coordinates the sequential execution of Bronze ingestion, Silver

transformation, and Gold loading through four activities connected via **success** dependency arrows, ensuring that each stage begins execution only after its required predecessor(s) have completed without error.

The pipeline begins with an Execute Pipeline activity named Run_Bronze, which invokes pl_bronze_ingestion, triggering the six parallel Copy Data activities and the df_bronze_quality Data Flow described in Section 5.3.1. Upon successful completion of Run_Bronze, two Script activities which are Script_M2_Views and Script_M3_Views were executed in parallel. Both activities run direct SQL statements against the Synapse Serverless SQL Pool via the ls_synapse_ppg linked service, applying the Silver-layer transformation logic developed by Member 2 (RA/MC business rules) and Member 3 (stockout and sales order impact logic) respectively. Parallel execution was used here because the two Silver transformations operate on independent logic paths and have no dependency on each other, allowing both to run concurrently once Bronze validated data is available.

The final activity, a Script activity named Script_M4_Views, executes only after both Script_M2_Views and Script_M3_Views have been completed successfully. This activity contains seven separate query blocks, each responsible for creating and loading one Gold layer dimension or fact table including dim_material, dim_supplier, dim_customer, dim_date, dim_warehouse, fact_inventory_status, and fact_sales_impact. The complete master pipeline canvas, showing all four activities with Succeeded status, is shown in Figure 5.3.6.1.

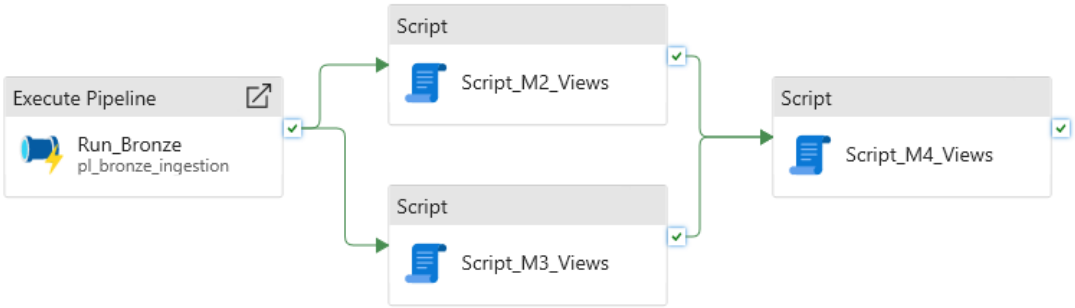


Figure 5.3.6.1 Master Pipeline of PPG RA and IRM System

This design centralises end-to-end pipeline execution behind a single trigger, fulfilling the automation requirement for the Pipeline Design & Architecture criterion of the project rubric. Running Script_M2_Views and Script_M3_Views in parallel reduces overall pipeline runtime compared to sequential execution, while the success dependency into Script_M4_Views ensures the Gold layer is only populated once both sets of Silver-layer SQL objects exist.

5.4 Essential Interfaces

This section showcases the primary visual interfaces and query outputs generated at each stage of the completed pipeline. By reviewing these system interfaces, the data can be visually confirmed that they were successfully flows from the initial data lake ingestion status to the structured data warehouse views, and finally into the reporting layer. The essential interfaces included the execution logs, intermediate database query tables, and interactive dashboard screens that users interact with to monitor inventory risks and financial write-downs.

5.4.1 Bronze Pipeline Results

This section presents the essential interfaces and execution results produced by the Bronze Layer pipeline. The outputs demonstrated here confirm that the ingestion, data quality, and duplicate detection processes were executed successfully within Azure Data Factory.

5.4.1.1 Pipeline Execution Monitor

Upon triggering pl_bronze_ingestion via the Trigger Now function in ADF Studio, all pipeline activities were monitored through the ADF Monitor tab. The monitor view provides a real-time status update for each activity, displaying the execution state, duration, and result for both the six parallel Copy Data activities and the subsequent df_bronze_quality Data Flow activity. A successful run is indicated when all activities display the Succeeded status, as shown in Figure 5.4.1.1.1.

Activity runs

Pipeline run ID a7a800da-5e6b-4e30-a4d0-98dbc276e809

All status

Showing 1 - 7 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
DataFlow_QC	Succeeded	Data flow	6/6/2026, 5:29:16 PM	4m 41s	AutoResolveIntegrationRuntime (Southeast Asia)
Copy_Production	Succeeded	Copy data	6/6/2026, 5:28:55 PM	20s	AutoResolveIntegrationRuntime (Southeast Asia)
Copy_Sales	Succeeded	Copy data	6/6/2026, 5:28:55 PM	18s	AutoResolveIntegrationRuntime (Southeast Asia)
Copy_materials	Succeeded	Copy data	6/6/2026, 5:28:55 PM	18s	AutoResolveIntegrationRuntime (Southeast Asia)
Copy_Purchase	Succeeded	Copy data	6/6/2026, 5:28:55 PM	20s	AutoResolveIntegrationRuntime (Southeast Asia)
Copy_Suppliers	Succeeded	Copy data	6/6/2026, 5:28:55 PM	17s	AutoResolveIntegrationRuntime (Southeast Asia)
Copy_inventory	Succeeded	Copy data	6/6/2026, 5:28:55 PM	18s	AutoResolveIntegrationRuntime (Southeast Asia)

Figure 5.4.1.1.1: Execution Result of the Pipeline

5.4.1.2 Data Flow Execution

Beyond the pipeline-level monitor, the ADF Monitor tab also provides activity-level execution details for the Data Flow. Clicking into the df_bronze_quality activity reveals a breakdown of each transformation stage, including the number of rows read from each source, the number of rows passed to GoodRows, and the number of rows routed to BadRows for each of the six entities. This row-level traceability provides direct evidence that the quality logic executed as intended, as shown in Figure 5.4.1.2.1.

Sink	Status	Processing time	Highest processing time	Rows written
SinkDupPurchase	Succeeded	5s	1s 387ms	0
SinkBadProduction	Succeeded	3s 383ms	1s 151ms	0
SinkGoodProduction	Succeeded	4s 63ms	1s 151ms	336
SinkDupInventory	Succeeded	3s 165ms	638ms	0
SinkBadPurchase	Succeeded	5s	941ms	0
SinkGoodPurchase	Succeeded	3s 56ms	941ms	40
SinkBadSuppliers	Succeeded	3s 803ms	717ms	0
SinkGoodSuppliers	Succeeded	3s 101ms	717ms	10
SinkDupSuppliers	Succeeded	3s 918ms	835ms	0
SinkBadMaterials	Succeeded	6s	3s 995ms	0
SinkGoodMaterials	Succeeded	2s 758ms	3s 995ms	20
SinkBadSales	Succeeded	4s 30ms	685ms	0
SinkGoodSales	Succeeded	2s 746ms	685ms	48
SinkDupProduction	Succeeded	3s 942ms	652ms	84
SinkBadInventory	Succeeded	3s 692ms	631ms	0
SinkGoodInventory	Succeeded	2s 807ms	631ms	240
SinkDupMaterials	Succeeded	2s 824ms	563ms	0
SinkDupSales	Succeeded	2s 801ms	530ms	0

Figure 5.4.1.2.1: Data Flow Execution Details

5.4.2 Silver RA/MC Query Results

The results of the silver layer RA and MC logic were verified by running four test queries against the vw_silver_ra_mc_logic view in Azure Synapse Studio.

```
191  -- Q2: Recoverable Assets
192  SELECT material_id, material_name, category, quantity_on_hand, consumption_12m, excess_quantity, is_recoverable_asset
193  FROM vw_silver_ra_mc_logic WHERE is_recoverable_asset = 1;
194
```

Results Messages

Select Query 6 View Table Chart Export results

Search

material_id	material_name	category	quantity_on_h...	consumption_1...	excess_quantity	is_recoverable...
MAT016	Metal Can 1L	Packaging	19410	9140	10270	1
MAT019	Cardboard Box	Packaging	11300	0	11300	1
MAT020	Plastic Pail 5L	Packaging	4760	3100	1660	1
MAT014	Acetone	Solvent	5200	0	5200	1
MAT005	Chrome Yellow	Pigment	120	0	120	1
MAT009	Polyurethane Resin	Resin	210	0	210	1
MAT012	Toluene	Solvent	6450	1030	5420	1
MAT010	Vinyl Resin	Resin	2220	0	2220	1
MAT015	Butanol	Solvent	1700	900	800	1
MAT017	Metal Can 4L	Packaging	10500	4780	5720	1

Figure 5.4.2.1: Recoverable Assets query result

As shown in Figure 5.4.2.1, the query returned 10 materials classified as Recoverable Assets. These materials had a current quantity on hand that exceeded their total 12-month consumption. The largest excess quantities were identified in MAT016 Metal Can 1L with an excess of 16,310 units, MAT019 Cardboard Box with 14,000 units, and MAT017 Metal Can 4L with 8,900 units. Materials such as MAT005, MAT009, MAT014, MAT10 and MAT019 showed zero 12-month consumption, indicating they were not used in any production activity over the past year despite having stock available.

```
183  -- Q3: Magna Carta Dead Stock
184  SELECT material_id, material_name, category, quantity_on_hand, last_consumed_date, months_since_consumed, is_mc_dead_stock
185  FROM vw_silver_ra_mc_logic WHERE is_mc_dead_stock = 1;
```

Results Messages

Select Query 5 View Table Chart Export results

Search

material_id	material_name	category	quantity_on_hand	last_consumed_date	months_since_consumed	is_mc_dead_stock
MAT010	Vinyl Resin	Resin	2800	(NULL)	999	1

Figure 5.4.2.2: MC Dead Stock query result

As shown in Figure 5.4.2.2, the query returned one material classified as MC Dead Stock. MAT010 Vinyl Resin, a Resin category material, had a quantity on hand of 2,800 units with a last_consumed_date of NULL and months_since_consumed of 999, confirming it

had never been consumed in any production order throughout the entire dataset. As it also had no lot expiry date recorded, it satisfied both conditions required for the MC Dead Stock classification.

```
183   -- Q4: Magna Carta Expired Stock
184   SELECT material_id, material_name, category, quantity_on_hand, lot_expiry_date, is_mc_expired
185   FROM vw_silver_ra_mc_logic WHERE is_mc_expired = 1;
186
```

Results Messages

Select Query 4 View Table Chart Export results

Search

material_id	material_name	category	quantity_on_h...	lot_expiry_date	is_mc_expired
MAT014	Acetone	Solvent	5200	2025-03-15	1
MAT009	Polyurethane Resin	Resin	210	2025-05-15	1

Figure 5.4.2.3: MC Expired query result

As shown in Figure 5.4.2.3, the query returned two materials classified as MC Expired. These materials had a lot expiry date that had already passed relative to the current system date. The expired materials were MAT014 Acetone with expiry date 2025-03-15 and MAT009 Polyurethane Resin with expiry date 2025-05-15. All two materials had is_mc_expired = 1 confirming the flag was correctly applied.

```
179   -- Q5: Total Financial Provision
180   SELECT SUM(mc_provision_amount) AS grand_total_mc_provision_USD
181   FROM vw_silver_ra_mc_logic WHERE is_mc_material = 1;
182
```

Results Messages

Select Query 3 View Table Chart Export results

Search

grand_total_mc_provision_USD
30144

Figure 5.4.2.4: Total MC provision query result

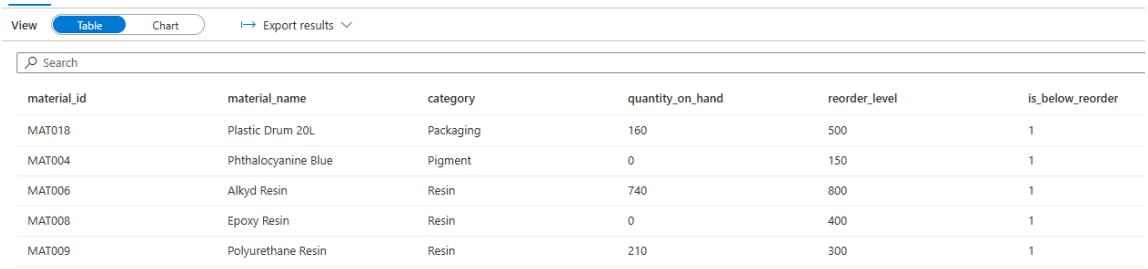
As shown in Figure 5.4.2.4, the query returned a total MC provision amount of USD 30144. This figure represents the 100% financial write-down value across all six materials classified as either MC Dead Stock or MC Expired, calculated as the sum of each material's quantity on hand multiplied by its unit cost. This total provision amount is the key financial output of the Silver layer and will be surfaced in the Power BI dashboard for management reporting.

5.4.3 Silver Stockout Query Results

The query results were produced in Azure Synapse Studio after the updated mock datasets were uploaded into the pipeline. The updated datasets increased the inventory snapshot records, production order records, sales order records, and purchase order records. For the stockout and sales impact component, the most important changes were the additional January, February, and March 2026 inventory snapshots, additional production orders, and additional sales orders.

The first output was generated from the vw_silver_stockout_risk view. This view was used to answer Q1 and Q6. Before the final result was checked, the latest inventory snapshot was validated. The query result showed that all 20 materials were using the latest snapshot date of 31 March 2026. This confirmed that the Silver Layer logic had successfully picked up the newly added inventory records.

Based on the final quick check, 5 materials were identified as below reorder level, while 2 materials were identified as having stockout risk. The result was lower than the previous output because the newly added March 2026 inventory records reflected updated stock quantities. Some materials that were previously below reorder level or at stockout risk were no longer classified as risky after the latest stock data was included.



material_id	material_name	category	quantity_on_hand	reorder_level	is_below_reorder
MAT018	Plastic Drum 20L	Packaging	160	500	1
MAT004	Phthalocyanine Blue	Pigment	0	150	1
MAT006	Alkyd Resin	Resin	740	800	1
MAT008	Epoxy Resin	Resin	0	400	1
MAT009	Polyurethane Resin	Resin	210	300	1

Figure 5.4.3.1: Query result for materials below reorder level

Figure 5.4.3.1 shows the query result for materials below reorder level. The result shows that five materials had a quantity on hand lower than their reorder level. These materials need attention because their stock level has dropped below the minimum required level.

Results Messages

View Table Chart Export results

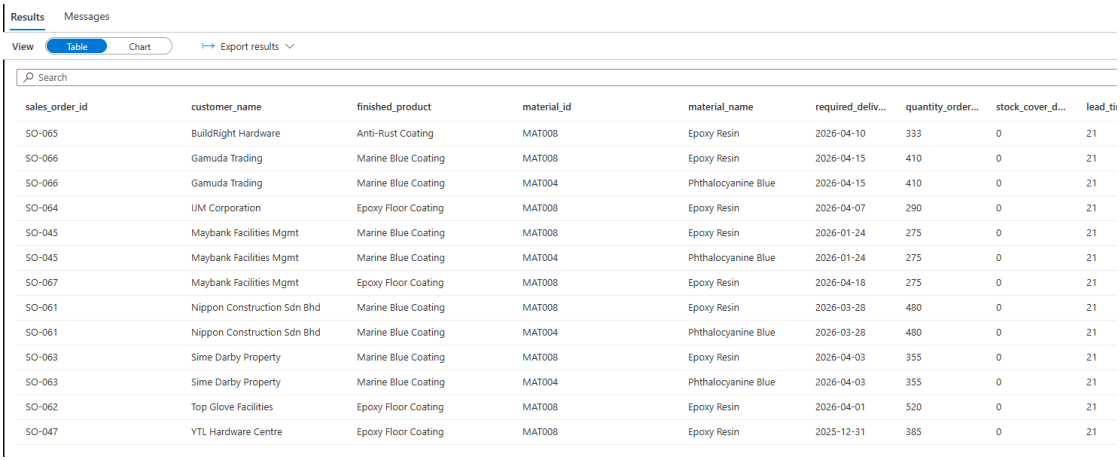
Search

material_id	material_name	category	quantity_on_hand	avg_daily_consumption	stock_cover_days	lead_time_days	is_stockout_risk
MAT004	Phthalocyanine Blue	Pigment	0	7.071428571428571	0	21	1
MAT008	Epoxy Resin	Resin	0	24.651162790697676	0	21	1

Figure 5.4.3.2: Query result for materials at stockout risk

Figure 5.4.3.2 shows the query result for materials at stockout risk. The result shows that two materials, Phthalocyanine Blue and Epoxy Resin, were at stockout risk. Both materials had zero quantity on hand, causing their stock cover days to be lower than their lead time days.

The second output was generated from the vw_silver_impacted_sales_orders view. This view was used to answer Q7. The result showed that 13 impacted sales order-material records were identified. These records involved 8 affected customers. This confirmed that the system was able to trace the relationship between material shortage risk and customer order fulfilment using the updated mock dataset.



Results Messages

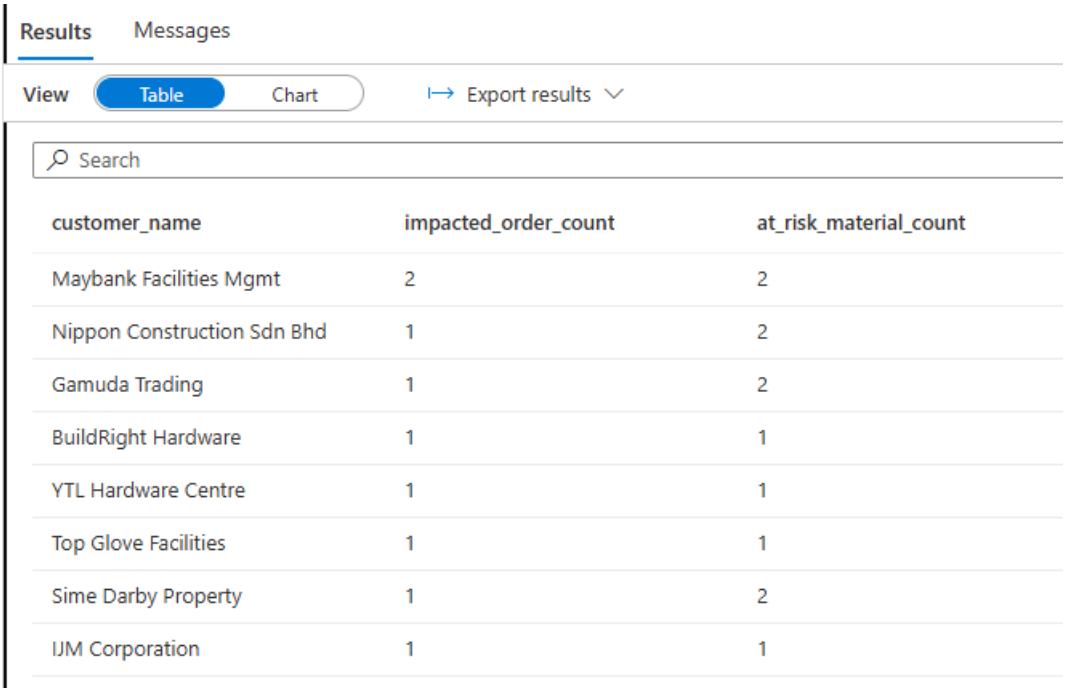
View Table Chart Export results

Search

sales_order_id	customer_name	finished_product	material_id	material_name	required_deliv...	quantity_order...	stock_cover_d...	lead_ti
SO-065	BuildRight Hardware	Anti-Rust Coating	MAT008	Epoxy Resin	2026-04-10	333	0	21
SO-066	Gamuda Trading	Marine Blue Coating	MAT008	Epoxy Resin	2026-04-15	410	0	21
SO-066	Gamuda Trading	Marine Blue Coating	MAT004	Phthalocyanine Blue	2026-04-15	410	0	21
SO-064	UM Corporation	Epoxy Floor Coating	MAT008	Epoxy Resin	2026-04-07	290	0	21
SO-045	Maybank Facilities Mgmt	Marine Blue Coating	MAT008	Epoxy Resin	2026-01-24	275	0	21
SO-045	Maybank Facilities Mgmt	Marine Blue Coating	MAT004	Phthalocyanine Blue	2026-01-24	275	0	21
SO-067	Maybank Facilities Mgmt	Epoxy Floor Coating	MAT008	Epoxy Resin	2026-04-18	275	0	21
SO-061	Nippon Construction Sdn Bhd	Marine Blue Coating	MAT008	Epoxy Resin	2026-03-28	480	0	21
SO-061	Nippon Construction Sdn Bhd	Marine Blue Coating	MAT004	Phthalocyanine Blue	2026-03-28	480	0	21
SO-063	Sime Darby Property	Marine Blue Coating	MAT008	Epoxy Resin	2026-04-03	355	0	21
SO-063	Sime Darby Property	Marine Blue Coating	MAT004	Phthalocyanine Blue	2026-04-03	355	0	21
SO-062	Top Glove Facilities	Epoxy Floor Coating	MAT008	Epoxy Resin	2026-04-01	520	0	21
SO-047	YTL Hardware Centre	Epoxy Floor Coating	MAT008	Epoxy Resin	2025-12-31	385	0	21

Figure 5.4.3.3: Query Result for Impacted Sales Orders

Figure 5.4.3.3 shows the impacted sales order result. The output lists open customer sales orders that may be affected by stockout-risk materials. The result shows 13 impacted sales order-material records.



customer_name	impacted_order_count	at_risk_material_count
Maybank Facilities Mgmt	2	2
Nippon Construction Sdn Bhd	1	2
Gamuda Trading	1	2
BuildRight Hardware	1	1
YTL Hardware Centre	1	1
Top Glove Facilities	1	1
Sime Darby Property	1	2
IJM Corporation	1	1

Figure 5.4.3.4 Query Result for Affected Customer Summary

Figure 5.4.3.4 shows the affected customer summary result. The output summarizes the number of impacted orders and at-risk materials for each customer. The result confirms that 8 customers were affected by stockout-risk materials.

5.4.4 Gold Layer Tables

The Gold Layer exposes its data through a set of SQL views that serve as the primary interface between the data pipeline and the Power BI dashboard. All views reside in the ppg_gold database on the Azure Synapse Analytics SQL endpoint. Table 5.4.4.1 show the summary of the Gold Layer data and Table 5.4.4.2 and Table 5.4.4.3 shows the fact table column references.

Table 5.4.4.1 Gold Layer View Summary

View	Columns	Row Count
dim_material	material_id, material_name, category, unit_of_measure, reorder_level, lead_time_days, unit_cost	20
dim_supplier	supplier_id, supplier_name, country, realibility_rating	10
dim_customer	customer_name	15
dim_warehouse	warehouse_location	5

dim_date	date_id, year, quarter, month, month_name, week_of_year, day_of_week, day_name	19
fact_inventory_status	All RA, MC, and stockout flags	20
fact_sales_impact	All RA, MC and sales impact	13

Table 5.4.4.2 fact_inventory_status Column References

Column Name	Data Type	Description
material_id	VARCHAR	Primary key as unique material identifier
snapshot_date	DATE	Date of inventory snapshot
warehouse_location	VARCHAR	Storage loaction of material
quantity_on_hand	INT	Current stock quantity
consumption_12m	FLOAT	Total quantity consumed in past 12 months
excess_quantity	FLOAT	Quantity above 12-month consumption
is_recoverable_asset	BIT	1 = excess stock qualifies as recoverable asset
is_mc_dead_stock	BIT	1 = no consumption for 9+ months, no expiry
is_mc_expired	BIT	1 = lot expiry date has passed
is_mc_material	BIT	1 = material qualifies for MC financial provision
mc_provision_amount	FLOAT	Financial write-down value (quantity x unit cost)
months_since_consumed	FLOAT	Months elapsed since last consumption (999 = never)
lot_expiry_date	DATE	Lot expiry date (NULL if no expiry)
last_consumed_date	DATE	Most recent production consumption date
is_below_reorder	BIT	1 = quantity on hand is below reorder level
is_stockout_risk	BIT	1 = stock cover days less than lead time days
stock_cover_days	FLOAT	Estimated days of stock remaining based on consumption rate
avg_daily_consumption	FLOAT	Average units consumed per day
total_consumed	FLOAT	Total units consumed across all production history
reorder_level	INT	Minimum stock threshold before reorder
lead_time_days	INT	Supplier lead time in days

Table 5.4.4.3 fact_sales_impact Column References

Column Name	Data Type	Description
sales_order_id	VARCHAR	Sales order identifier
customer_name	VARCHAR	Customer name (FK to dim_customer)
material_id	VARCHAR	At-risk material (FK to dim_material)
order_date	DATE	Sales order date (FK to dim_date)
required_delivery_date	DATE	Required delivery date (FK to dim_date)
finished_product	VARCHAR	Finished product affected by the at-risk material
quantity_ordered	INT	Ordered quantity
status	VARCHAR	Order status (OPEN/PENDING/BACKORDER)
material_name	VARCHAR	At-risk material name
category	VARCHAR	Material category
quantity_on_hand	INT	Current stock of at-risk material
reorder_level	INT	Reorder threshold for the material
avg_daily_consumption	FLOAT	Avergae daily consumption rate
stock_cover_days	FLOAT	Days of stock remaining
lead_time_days	INT	Suuplier lead time in days
is_stockout_risk	BIT	1 = material confirmed at stockout risk
delivery_risk_days	FLOAT	Days until delivery minus stock cover days; positive = at risk

5.4.5 Power BI Dashboard

The Power BI dashboard was developed in Power BI Desktop connected to the ppg_gold database on Azure Synapse Analytics via Import connection mode. The dashboard is organised into two pages, each addressing a distinct category of inventory risk.

5.4.5.1 Page 1: Inventory Health (Q1 to Q5)

The Inventory Health page addresses Q1 through Q5 across four quadrants.

For Q1, a clustered column chart was constructed using material_name on the axis and both Quantity On Hand Fixed and Reorder Level Fixed measures as the two value series. The chart was filtered to rows where is_below_reorder equals 1. A dashed reference line was added at zero to visually anchor the comparison. The result identified 5 materials below their reorder level, with Alkyd Resin, Polyurethane Resin, Plastic Drum 20L, Epoxy Resin, and Phthalocyanine Blue shown as the visible at-risk materials in the chart.

For Q2, a table visual was constructed using material_name, Quantity On Hand Fixed, Consumption 12M Fixed, and Excess Quantity Fixed measures, filtered to rows where is_recoverable_asset equals 1 and sorted descending by excess quantity. The result identified 11 materials classified as Recoverable Assets with a total excess quantity of 42,920 units.

For Q3, two KPI cards display Dead Stock Count and Dead Stock Provision, showing 1 material and $16.87K respectively. A supporting table below the cards shows the detail row for Vinyl Resin with its category, quantity on hand, and provision amount.

For Q4, a donut chart was constructed using material_name as the legend and mc_provision_amount as the value, filtered to rows where is_mc_expired equals 1. The chart shows the proportion of provision amount contributed by each of the 2 expired materials. Acetone contributes the largest share at 82.28% ($10.92K), followed by Polyurethane Resin at 17.72% ($2.35K). Two KPI cards alongside the chart display the expired material count and total expired provision amount.

For Q5, a prominent KPI card displays the total MC provision amount of $30.14K, combining dead stock and expired material provisions.

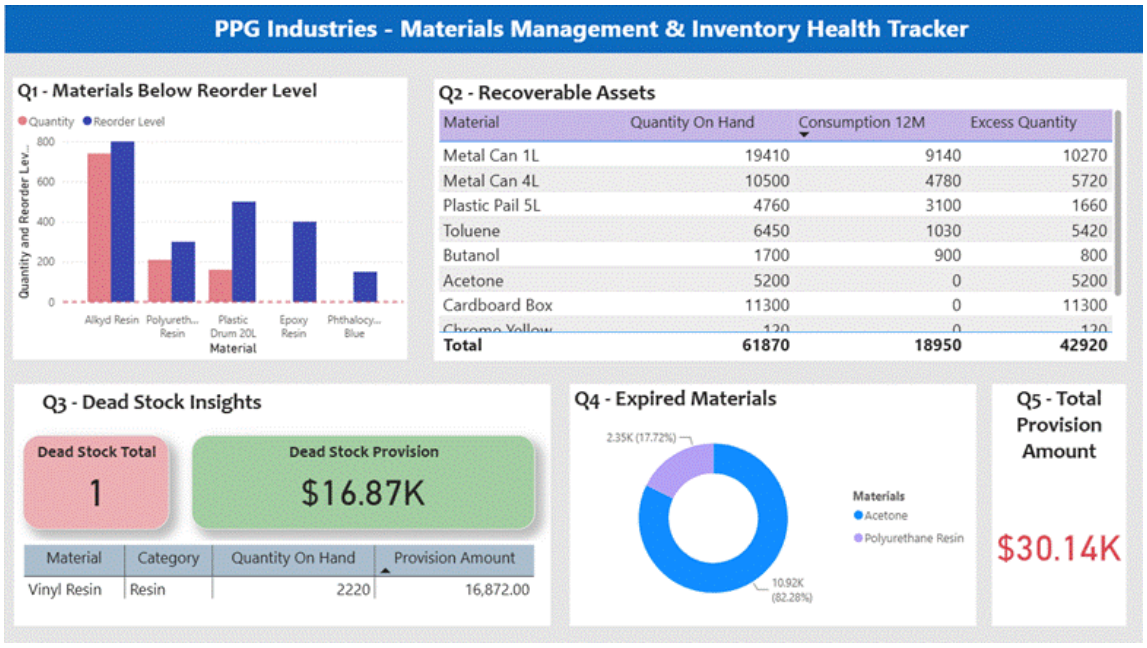


Figure 5.4.5.1: Power BI Dashboard, Inventory Health Page (Q1 to Q5)

5.4.5.2 Page 2: Operational Risk (Q6 and Q7)

The Operational Risk page addresses Q6 and Q7. Three KPI cards at the top of the page display Stockout Risk (2 materials), Affected Customers (8), and Orders at Risk (13).

For Q6, a table visual was constructed using material_name, Lead Time Fixed, Stock Cover Fixed, Quantity On Hand Fixed measures, and the Risk Status calculated column, filtered to rows where is_stockout_risk equals 1. Conditional formatting was applied to the Quantity On Hand Fixed column with a red background to highlight critically low stock levels. The Risk Status column classifies both visible materials as Critical, as both Epoxy Resin and Phthalocyanine Blue have zero stock cover days against a supplier lead time of 21 days.

For Q7, a matrix visual was constructed with customer_name on rows, material_name on columns, and the count of orders as the value. Conditional formatting was applied to the matrix cells using a gradient colour scale from white to red, creating a heatmap effect that immediately highlights the customers with the highest order exposure. Maybank Facilities Mgmt appears as the most impacted customer with 3 orders across both at-risk materials. The matrix shows a total of 13 impacted orders across 8 customers.

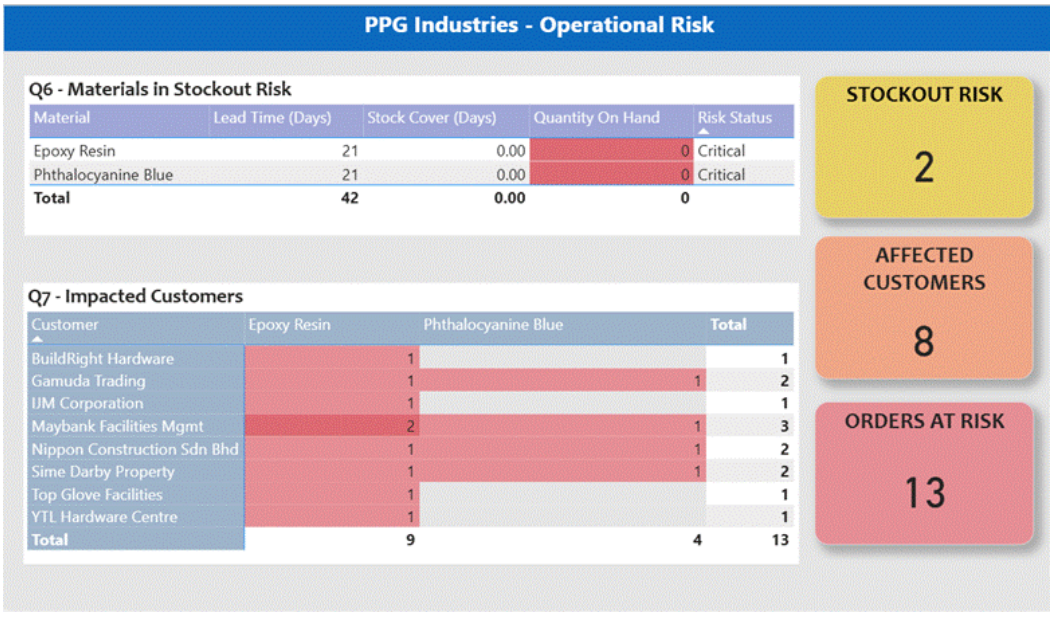


Figure 5.4.5.2: Power BI Dashboard, Operational Risk Page (Q6 and Q7)

5.4.6 Master Pipeline Execution Results

Upon triggering pl_master_ppg via Trigger Now, the ADF Monitor tab displayed the execution status of all four activities: Run_Bronze, Script_M2_Views, Script_M3_Views, and Script_M4_Views. A successful end-to-end run is confirmed when all four activities display the Succeeded status, with Script_M2_Views and Script_M3_Views beginning

only after Run_Bronze completes, executing concurrently, and Script_M4_Views beginning only after both have finished. This sequence is shown in Figure 5.4.6.1.

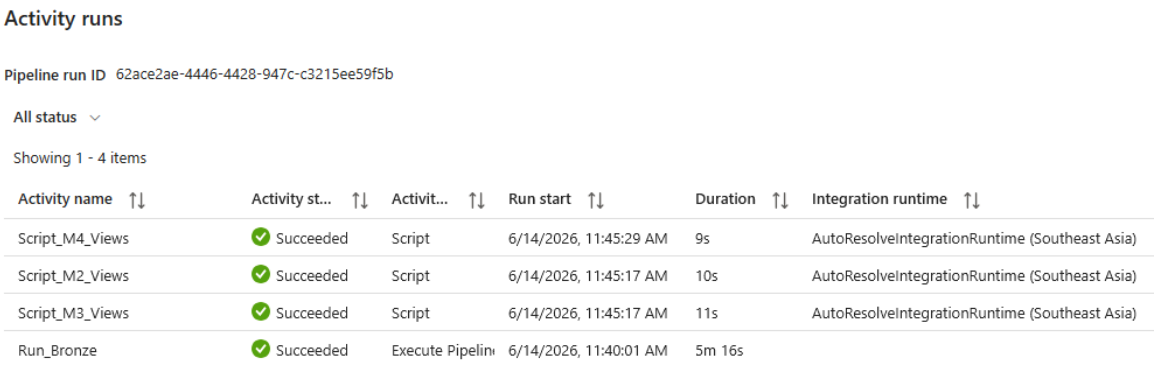


Figure 5.4.6.1: Master Pipeline Execution Result

5.5 Testing

To ensure the pipeline is both accurate and reliable, the entire system is subjected to evaluation process. This section documents the system testing strategy, which examines the platform from multiple technical perspectives to verify overall stability. The evaluation framework is divided into three main stages, black box testing to validate that each layer outputs the correct operational results, white box testing to verify the internal logic of the underlying code and schemas, and user acceptance testing to confirm that stakeholders can easily interpret the final dashboard views. Together, these procedures provide objective proof that the data pipeline performs exactly as required.

5.5.1 Black Box Testing

This section evaluates the functional behavior of the system by focusing purely on inputs and outputs without examining the internal code structure. The primary objective of black box testing is to verify that the end-to-end data pipeline correctly handles file ingestion, applies data quality checks, and executes business logic accurately.

5.5.1.1 Bronze Ingestion and Quality Check Output Testing

Black box testing conducted in Bronze layer was to verify that given a set of known input CSV files, the pipeline produces the correct and expected output directories within ppgdatalake storage. The tester interacts only with the pipeline trigger and the resulting storage outputs, treating the internal ADF pipeline and Data Flow logic as opaque.

Table 5.5.1.1 presents the black box test cases designed and executed for the Bronze ingestion and quality check components.

Table 5.5.1.1: Black Box Test Cases for Bronze Ingestion and Quality Check

Test Case	Expected Output	Actual Output	Status
All six CSV files are successfully ingested from raw to ingested directory	Six corresponding files appear in bronze/ingested/	Six files found in bronze/ingested/	Pass
Clean rows are routed to the validated directory	Files appear in bronze/validated/<entity>/ for all six entities	All six validated subfolders populated with output files	Pass
Null rows are routed to the quarantine directory	Affected rows appear in bronze/quarantine/<entity>/	Quarantine folder exists and check executed, no nulls found in synthetic data	Pass
Duplicate rows are detected and routed to quarantine	Duplicate rows appear in bronze/quarantine/duplicates/<entity>/	Duplicates folder exists and check executed, no duplicates found in synthetic data	Pass

The results of the black box tests confirm that the Bronze Layer pipeline behaves correctly from an end-user perspective. All six source files were successfully ingested into bronze/ingested/, and the quality check routing logic correctly directed clean rows to bronze/validated/ and flagged rows to bronze/quarantine/. The empty quarantine and duplicates directories for the synthetic dataset are an expected and valid outcome, as the source data was deliberately designed to be free of nulls and duplicates. Their existence nonetheless confirms that the detection logic was triggered and evaluated every row.

5.5.1.2 RA and MC Logic Output Testing

Black box testing was conducted to verify whether the RA and MC provision component produced the expected outputs based on the defined business requirements. The testing was focused on the final results generated by the SQL view without examining the internal SQL statements. The output of vw_silver_ra_mc_logic was used to validate Q2, Q3, Q4, and Q5.

For Q2, the expected output was a list of materials where the current quantity on hand exceeded their total 12-month consumption. The test result showed that 11 materials were identified as Recoverable Assets. For Q3, the expected output was a list of materials that had not been consumed for nine or more months and had no lot expiry date recorded. The test result showed that 1 material was identified as MC Dead Stock. For Q4, the expected output was a list of materials whose lot expiry date had already passed relative to the current date. The test result showed that 5 materials were identified as MC Expired. For Q5, the expected output was the total financial provision amount calculated as a 100% write-down across all MC materials. The test result returned a total of 6 MC materials with a combined provision of USD 76,586.

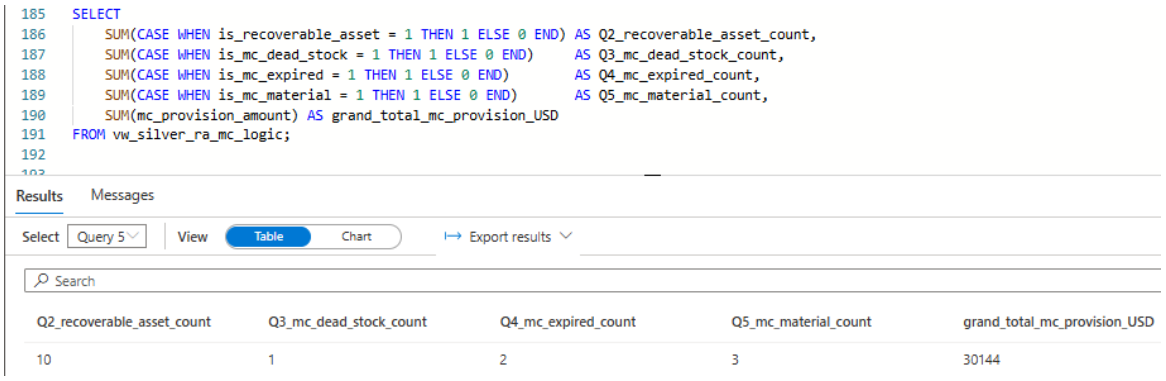


Figure 5.5.1.2.1: Final quick check count results for Q2, Q3, Q4 and Q5

Figure 5.5.1.2.1 shows the quick count check results for the RA and MC provision components. The results confirm that the expected output was successfully produced for all four business questions.

Table 5.5.1.2.1 shows the quick count check results for the RA and MC provision component. The results confirm that the expected output was successfully produced for all four business questions.

Table 5.5.1.2.1 Black Box Testing Results for RA and MC Provision Component

Test Case	Expected Output	Actual Output	Status
Q2 Recoverable Asset	Materials where stock exceeds 12-month consumption are identified	10 materials identified	Passed
Q3 MC Dead Stock	Materials with 9+ months no consumption and no expiry date are identified	1 material identified	Passed
Q4 MC Expired	Materials with passed lot expiry date are identified	2 materials identified	Passed
Q5 Total Provision	Total financial write-down amount is calculated	3 MC materials, USD 30144 returned	Passed

5.5.1.3 Stockout and Sales Impact Output Testing

Black box testing was carried out to verify whether the SQL views produced the expected outputs based on the business questions. The testing focused on the final query results without examining the internal SQL code.

For Q1, the expected output was a list of materials below reorder level. The updated query result showed that 5 materials were below reorder level. For Q6, the expected output was a list of materials at stockout risk. The updated query result showed that 2 materials were at stockout risk. For Q7, the expected output was a list of open sales orders and customers affected by stockout-risk materials. The updated query result showed 13 impacted sales order-material records involving 8 customers.

Table 5.5.1.3.1 Black Box Testing Results for Stockout and Sales Impact

Test Case	Expected Output	Actual Output	Status
Q1 Below Reorder Level	Materials below reorder level are identified	9 materials identified	Passed
Q6 Stockout Risk	Materials at stockout risk are identified	8 materials identified	Passed
Q7 Sales Order Impact	Impacted sales orders and affected customers are identified	23 records and 10 customers identified	Passed

Table 5.5.1.3.1 shows the black box testing results for the stockout and sales impact component. The test was conducted to verify whether the final query outputs answered Q1, Q6, and Q7 correctly.

5.5.1.4 Gold Layer Table Output Testing

Black box testing is used to ensure that the output of each view and fact table is accurate according to the business without inspecting the internal SQL logic. The testing approach as shown in Figure 5.5.1.4.1 by using SQL scripts treats Gold Layer objects and gives the results as in Figure 5.5.1.4.2.

```
SELECT 'dim_material'          AS object_name, COUNT(*) AS record_count FROM dim_material
UNION ALL
SELECT 'dim_supplier',                         COUNT(*) FROM dim_supplier
UNION ALL
SELECT 'dim_customer',                         COUNT(*) FROM dim_customer
UNION ALL
SELECT 'dim_warehouse',                        COUNT(*) FROM dim_warehouse
UNION ALL
SELECT 'dim_date',                             COUNT(*) FROM dim_date
UNION ALL
SELECT 'fact_inventory_status',                COUNT(*) FROM fact_inventory_status
UNION ALL
SELECT 'fact_sales_impact',                    COUNT(*) FROM fact_sales_impact;
GO
```

Figure 5.5.1.4.1: SQL Scripts of Galaxy Schema Validation

object_name	record_count
dim_material	20
dim_supplier	10
dim_customer	15
dim_warehouse	5
dim_date	19
fact_inventory_status	20
fact_sales_impact	13

Figure 5.5.1.4.2: Result of Galaxy Schema Validation

All the views created were passed showing that the black box test was a success. The Gold Layer outputs are verified to be consistent with the expected business questions answers. Table 5.5.1.4.3 shows the dimension and fact record count tests.

Table 5.5.1.4.3: Dimensions and Fact Record Count Tests

Object	Expected Rows	Actual Rows	Status
dim_material	20	20	Pass
dim_supplier	10	10	Pass
dim_customer	15	15	Pass
dim_warehouse	5	5	Pass
dim_date	19	19	Pass
fact_inventory_status	20	20	Pass
fact_sales_impact	13	13	Pass

5.5.1.5 Power BI Dashboard Output Testing

Black box testing was conducted to verify whether the Power BI dashboard correctly displayed the expected outputs for all seven business questions based on the data loaded from the Gold Layer views. Testing focused on the final visual outputs presented on each dashboard page without examining internal DAX logic or data model configuration.

For Q1, the expected output was a clustered bar chart showing only materials where quantity on hand was lower than the reorder level. The test result showed 5 materials in the Q1 chart, consistent with the Silver Layer output. For Q2, the expected output was a table showing only materials classified as Recoverable Assets sorted by excess quantity. The test result showed 10 materials with correct Quantity On Hand Fixed, Consumption 12M Fixed, and Excess Quantity Fixed values per material. For Q3, the expected output was two KPI cards showing dead stock count and provision amount, with a supporting detail table. The test result showed 1 material (Vinyl Resin) with a provision of $16,872, and the KPI cards displayed correctly. For Q4, the expected output was a donut chart showing the proportion of provision amount per expired material alongside KPI cards. The test result showed 2 expired materials with Acetone contributing the largest share, and the donut chart rendered correctly.

For Q5, the expected output was a KPI card showing the combined total MC provision. The test result showed $30.14K. For Q6, the expected output was a table showing only materials at stockout risk with red conditional formatting and a Risk Status label. The test result showed 2 materials, Epoxy Resin and Phthalocyanine Blue, both classified as Critical with

red formatting applied. For Q7, the expected output was a heatmap matrix showing impacted orders per customer per material. The test result showed 13 impacted orders across 8 customers with gradient red colouring applied correctly.

Table 5.5.1.5: Black Box Testing Results for Power BI Dashboard Output

Test Case	Expected Output	Actual Output	Status
Q1 Below Reorder Level	Materials below reorder level shown in bar chart	5 materials displayed	Passed
Q2 Recoverable Assets	Materials where stock exceeds 12-month consumption shown in table	10 materials displayed with correct per-material values	Passed
Q3 MC Dead Stock	KPI cards show dead stock count and provision; detail table shows Vinyl Resin	1 material, $16,872 provision displayed	Passed
Q4 MC Expired	Donut chart shows provision proportion per expired material	5 materials in donut chart; Acetone largest at 82.28%	Passed
Q5 Total MC Provision	KPI card displays combined total provision amount	$30.14K displayed	Passed
Q6 Stockout Risk	Table shows at-risk materials with Risk Status and red formatting	2 materials displayed, both Critical with red cells	Passed
Q7 Customer Impact	Heatmap matrix shows impacted orders per customer per material	13 orders across 8 customers with gradient colouring	Passed

5.5.2 White Box Testing

Unlike the previous functional checks, this section examines the internal logic, script execution, and structural integrity of the pipeline's codebase. White box testing focuses on verifying the backend processes, including the conditional loops within pipeline, the data

transformation integrity, and the calculation. This internal review ensures that the data schemas are highly optimized, scripts run without structural errors, and data transitions smoothly between the Medallion layers.

5.5.2.1 Bronze Ingestion and Quality Check Logic Testing

White box testing was conducted by inspecting the internal configuration of each transformation stage within pl_bronze_ingestion and df_bronze_quality. This includes verifying the null detection expressions within the QualityFlags Derived Column, the routing conditions of the Conditional Split, the grouping keys applied in the Aggregate transformation for duplicate detection, the success dependency between the Copy Data activities and the Data Flow activity, and the authentication configuration of both linked services. Table 5.5.2.1.1 presents the white box test cases designed and executed for these components.

Table 5.5.2.1.1: White Box Test Cases for Bronze Ingestion and Quality Check

Test Case	Component Tested	Expected Outcome	Actual Outcome	Status
Copy Data source dataset correctly references the raw CSV path	ds_src_materials dataset configuration	File path set to bronze/raw/materials/materials.csv with First row as header enabled	File path and header setting confirmed correct in dataset properties	Pass
Copy Data sink dataset correctly references the ingested path	ds_sink_materials dataset configuration	File path set to bronze/ingested/materials/ with linked service ls_adls_ppg	Sink path and linked service confirmed correct	Pass
Null detection expression correctly evaluates null fields	QualityFlags Derived Column in df_bronze_quality	iif(isNull(material_id), 1, 0) returns 1 for null values and 0 for non-null values	Expression validated in Data Flow expression builder and correct return	Pass

			values confirmed	
Conditional Split correctly routes BadRows on null condition	Conditional Split transformation in df_bronze_quality	Stream BadRows condition: is_null_material_id == true() || is_null_unit_cost == true() routes null rows away from GoodRows	Routing logic verified in Conditional Split configuration, correct stream assignment confirmed	Pass
Duplicate detection Aggregate groups by correct primary key	Aggregate transformation in duplicate detection chain	materials entity groups by material_id, inventory_snapshot groups by snapshot_date + material_id	Group-by keys verified in Aggregate transformation settings for all six entities	Pass
Duplicate detection Filter retains only rows with row_count greater than 1	Filter transformation in duplicate detection chain	Expression row_count > 1 retains only genuine duplicate rows	Filter expression verified in transformation configuration	Pass
Duplicate detection chain branches from an independent Source transformation	Data Flow canvas structure	Aggregate transformation connects to a separate Source, not to the QualityFlags Derived Column	Verified on canvas that each duplicate chain has its own Source transformation independent of the quality chain	Pass
Data Flow activity is connected to Copy Data	pl_bronze_ingestion pipeline canvas	df_bronze_quality activity executes only on success of all six Copy Data activities	Success dependency arrow verified	Pass

activities via a success dependency			between Copy Data activities and Data Flow activity in pipeline canvas	
Linked service uses System-Assigned Managed Identity for ADLS Gen2 authentication	ls_adls_ppg linked service configuration	Authentication method set to System Managed Identity targeting storage account ppgdatalake	Linked service configuration verified, Test Connection returned Succeeded	Pass
Linked service uses Azure SQL Database connector for Synapse Serverless endpoint	ls_synapse_ppg linked service configuration	Connector type is Azure SQL Database targeting synapse-ppg-ondemand.sql.azuresynapse.net	Connector type and endpoint verified in linked service properties	Pass

The white box test results confirm that all internal logic components of the Bronze Layer are correctly configured. Attention was given to two architectural decisions that required verification at the configuration level. First, the duplicate detection chains were confirmed to branch from independent Source transformations rather than from the QualityFlags Derived Column, which is necessary to prevent schema contamination between the two transformation chains. Second, the null detection expressions within the Derived Column transformation were verified to use the correct iif(isNull(column), 1, 0) syntax, as the isNull() function alone would produce a type mismatch error within the ADF Data Flow expression engine. An example of evidence for the null detection expression configuration for materials source dataset is shown in Figure 5.5.2.1.1.

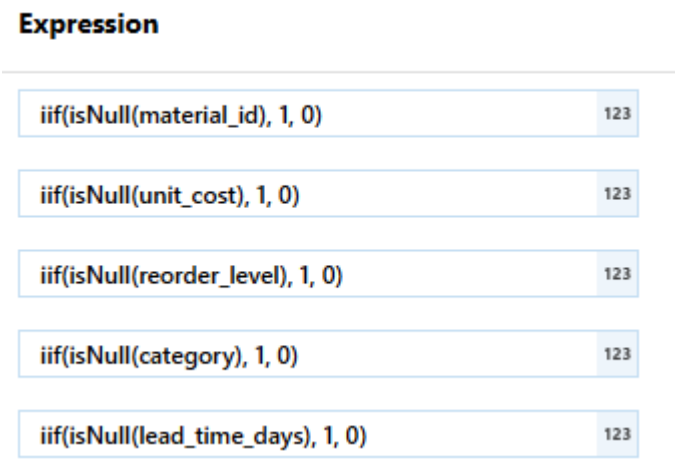


Figure 5.5.2.1.1: Null Detection Expression Syntax in the QualityFlagMaterials Derived Column

5.5.2.2 RA and MC Business Logic Testing

White box testing was performed to verify the internal SQL logic used in the RA and MC provision process. The testing covered five internal logic areas that includes latest inventory selection, never consumed material handling, MC Dead Stock condition logic, MC Expired date comparison logic, and financial provision calculation accuracy. A single verification query was constructed to test all five logic areas simultaneously and produce a consolidated result. Each column in the output represented one internal logic test.

For the first test, the total count of unique materials returned by the view was checked to verify that the ROW_NUMBER() window function correctly partitioned records by material_id and selected only the most recent snapshot per material. The expected result was 20, representing one row per material.

For the second test, the maximum months_since_consumed value was checked to verify that the internal CASE WHEN last_consumed_date IS NULL THEN 999.0 logic correctly assigned 999.0 to materials that had never appeared in any production order. The expected result was 999.

For the third test, the count of materials where both months_since_consumed >= 9 and lot_expiry_date IS NULL were simultaneously true and is_mc_dead_stock = 1 was

checked to verify that the AND condition logic inside the Q3 flag was applied correctly. The expected result was 1.

For the fourth test, the count of materials where lot_expiry_date was earlier than today and is_mc_expired = 1 was checked to verify that the internal date comparison logic of the Q4 flag was functioning correctly. The expected result was 2.

For the fifth test, the count of MC materials where the calculated quantity_on_hand × unit_cost matched the stored mc_provision_amount exactly was checked to verify the internal provision formula was producing accurate results. The expected result was 3.

```
193  SELECT
194      -- Test 1: Latest inventory selection
195      COUNT(DISTINCT material_id)                                  AS T1_unique_material_count,
196
197      -- Test 2: Never consumed assigned 999
198      MAX(CASE WHEN last_consumed_date IS NULL
199          THEN months_since_consumed END)                          AS T2_max_months_never_consumed,
200
201      -- Test 3: MC Dead Stock both conditions true
202      SUM(CASE WHEN months_since_consumed >= 9
203          AND lot_expiry_date IS NULL
204          AND is_mc_dead_stock = 1 THEN 1 ELSE 0 END)              AS T3_dead_stock_logic_correct,
205
206      -- Test 4: MC Expired lot_expiry_date < today
207      SUM(CASE WHEN lot_expiry_date < CAST('2026-03-31' AS DATE)
208          AND is_mc_expired = 1 THEN 1 ELSE 0 END)                 AS T4_expired_logic_correct,
209
210      -- Test 5: Provision calculation matches
211      SUM(CASE WHEN ROUND(quantity_on_hand * unit_cost, 2)
212          = ROUND(mc_provision_amount, 2)
213          AND is_mc_material = 1 THEN 1 ELSE 0 END)                AS T5_provision_calc_correct
214  FROM vw_silver_ra_mc_logic;
```

Results Messages

Select Query 5 View Table Chart Export results

Search

T1_unique_material_count	T2_max_months_never_consum...	T3_dead_stock_logic_correct	T4_expired_logic_correct	T5_provision_calc_correct
20	999	1	2	3

Figure 5.5.2.2.1: White Box Testing Verification Output

Figure 5.5.2.2.1 shows the consolidated white box testing output. All five internal logic areas returned the expected values, confirming that each condition, calculation, and logic path within the view was functioning correctly.

Table 5.5.2.2.1: White Box Testing Results for RA and MC Provision Logic

Test Area	Internal Logic Tested	Expected	Actual	Status
Latest Inventory Selection	ROW_NUMBER() partitioned by material_id returns one row per material	20	20	Passed

Never Consumed Handling	CASE WHEN last_consumed_date IS NULL assigns 999.0	999	999	Passed
MC Dead Stock Condition	AND logic requires months >= 9 AND lot_expiry_date IS NULL simultaneously	1	1	Passed
MC Expired Date Comparison	lot_expiry_date < CAST('2026-03-31' AS DATE) correctly identifies expired materials	2	2	Passed
Provision Calculation	quantity_on_hand × unit_cost matches mc_provision_amount for all MC materials	3	3	Passed

Table 5.5.2.2.1 shows the white box testing results for the internal SQL logic used in the RA and MC provision component. All five internal logic areas passed, confirming that the view logic was implemented correctly and producing accurate results across all business rules.

5.5.2.3 Stockout and Sales Impact Logic Testing

White box testing was performed by checking the internal SQL logic used in the stockout and sales impact process. The first test focused on date conversion. During earlier testing, the average daily consumption initially returned fallback values of 0.001. This showed that the production order dates were not being converted correctly. After the production dataset was checked, it was found that the production_date column used the yyyy-mm-dd format.

The SQL view was then updated to support multiple date formats using TRY_CONVERT. The second test focused on the fixed reference date logic. After the updated mock data was added, the SQL script was adjusted to use 31 March 2026 as the reference date. This ensured that the latest inventory snapshot was selected correctly and that the 12-month production consumption window remained consistent with the updated dataset. The query result confirmed that all 20 materials used the 31 March 2026 inventory snapshot.

The third test focused on the consumption calculation. After the date conversion and fixed reference date logic were corrected, the total consumption and average daily consumption values were calculated correctly from the production order records. This confirmed that the stock cover days could be calculated using valid consumption values.

The fourth test focused on the join logic. The stockout-risk materials were checked to ensure that they could be linked to finished products through production orders. The finished products were then linked to open sales orders. The final output confirmed that impacted sales orders and affected customers could be traced correctly.

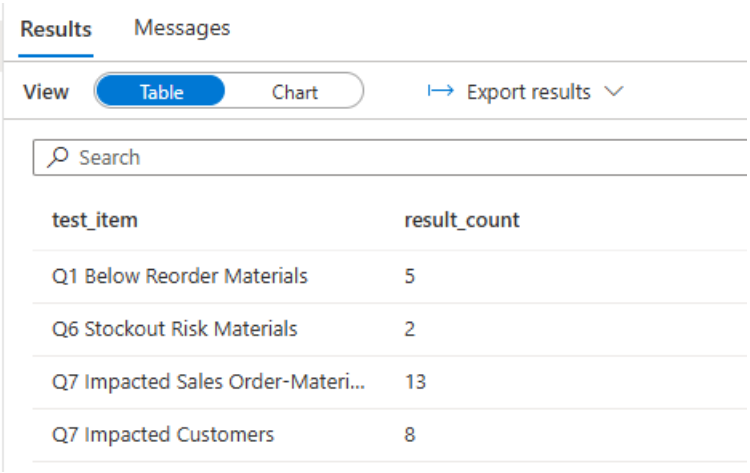


Figure 5.5.2.3.1: Final Quick Check Results for Stockout and Sales Impact Views

Figure 5.5.2.3.1 shows the final quick check results for the stockout and sales impact views. The results confirm that the SQL views were functioning correctly after the updated mock dataset was reloaded and the fixed reference date was applied.

Table 5.5.2.3.1: White Box Testing Results for Stockout Risk and Sales Impact Logic

Test Area	**Internal Logic Tested**	**Result**	**Status**
Fixed Reference Date	Reference date was set to 31 March 2026 to align with the updated mock dataset	Latest inventory snapshot and 12-month consumption window were applied correctly	Passed

Table 5.5.2.3.1 shows the white box testing results for the internal SQL logic used in the stockout and sales impact component. The test focused on date conversion, fixed reference date logic, consumption calculation, stockout risk logic, and impacted sales order tracing.

5.5.2.4 Gold Layer Galaxy Schema and Loading Logic Testing

White box testing for the Gold Layer examines the internal SQL logic of each view to verify that the transformation rules, join conditions, data type handling, and NULL management are correctly implemented. This level of testing ensures that the code is not only producing correct results but is also robust, maintainable, and free from logical defects.

Table 5.5.2.4.1: White Test Cases in Gold Layer Star Schema

Component	Logic Tested	Method	Result
fact_inventory_status	FULL OUTER JOIN captures all materials from both silver view	Verify row count = 20 (all materials represented)	PASS
fact_inventory_status	COALESCE on material_id returns non-NULL for all rows	SELECT COUNT(*) WHERE material_id IS NULL should return 0	PASS
fact_inventory_status	COLLATE fix prevents join collation mismatch error	View runs without error after COLLATE applied on join keys	PASS
fact_inventory_status	ISNULL defaults prevent NULL propagation into flags	SELECT COUNT (*) WHERE is_recoverable_assets IS NULL should return 0	PASS
fact_inventory_status	mc_provision_amount = quantity_on_hand x unit_cost for MC materials	Cross check each MC material: provision / qty = unit_cost from dim_material	PASS

fact_inventory_status	months_since_consumed = 999 for never consumed materials	Verify material with NULL last_consumed_date have months_since_consumed = 999	PASS
fact_sales_impact	Delivery_risk_days formula is correct	Manually verify: DATEDIFF(required_delivery_date, 2026-03-31) - stock_cover_days	PASS
fcct_sales_impact	Only OPEN/ PENDING/ BACKORDER orders included	Verify no CLOSED or COMPLETED status rows appear in fact_sales_impact	PASS
dim_customer	All 7 CSV columns declared in WITH clause	View runs without column mapping error	PASS
dim_date	4 UNION sources collect all unique dates without duplicate	SELECT COUNT (*) vs COUNT (DISTINCT date_id_ must be equal	PASS
dim_warehouse	Header row excluded by WHERE clause	No row with warehouse_location = “warehouse_location” appears	PASS
fact_inventory_status	avg_daily_consumption = 0.001 fallback for zero consumption materials	Verify MAT005, MAT009, MAT010, MAT014, MAT019 show 0.001	PASS
fact_inventory_status	High stock_cover_days for never consumed materials is expected	High values only occur for materials with avg_daily = 0.001	PASS

During white box testing, it was observed that five materials (MAT005, MAT009, MAT010, MAT014, MAT019) have no production consumption history. For these

materials, the avg_daily_consumption defaults to 0.001 as implemented by Member 3 to prevent division-by-zero errors. As a result, their stock_cover_days values appear extremely large (ranging from 210,000 to 14,000,000 days).

This behavior is intentional and correct. These materials effectively have unlimited stock cover since they are never consumed, making them strong candidates for Recoverable Asset or MC Dead Stock classification. The high stock_cover_days values do not represent data quality issues; they simply reflect the absence of consumption activity.

5.5.2.5 Power BI Data Model and DAX Logic Testing

White box testing was performed to verify the internal DAX logic, data model relationship handling, and measure definitions implemented within the Power BI data model. Testing focused on five internal logic areas: the TREATAS filter context fix, the FILTER-based KPI measures, the Customers Affected measure, the Risk Status calculated column, and the heatmap conditional formatting logic.

For the first test, the TREATAS fix was verified by comparing the output of raw SUM measures against the TREATAS-based measures in a table visual with material_name from dim_material. Raw SUM measures returned the same inflated value (71,745) across all rows, confirming the relationship path was broken. After replacing with TREATAS measures, each row correctly displayed its individual per-material value, confirming the fix resolved the filter context issue.

For the second test, the Dead Stock Count and Dead Stock Provision FILTER measures were verified by cross-referencing the KPI card values against the Gold Layer view data. The KPI cards displayed 1 material and $16,872 respectively, consistent with the single Vinyl Resin row where is_mc_dead_stock equals 1 in vw_silver_ra_mc_logic.

For the third test, the Expired Count and Expired Provision measures were verified by filtering vw_silver_ra_mc_logic to is_mc_expired equals 1 in Power Query and confirming the row count and provision sum matched the KPI card values of 2 materials and $13,263.

For the fourth test, the Customers Affected measure was verified by inspecting the distinct customer_name values in vw_silver_impacted_sales where is_stockout_risk equals 1. The measure returned 8 distinct customers, matching the number of unique customer rows visible in the Q7 matrix.

For the fifth test, the Risk Status calculated column was verified in Data view by confirming that all rows where stock_cover_days equals 0 displayed Critical, while rows with stock_cover_days greater than 0 displayed At Risk. The column was confirmed to be of text type and rendered correctly in the Q6 table visual.

Table 5.5.2.5.1: White Box Testing Results for Power BI Data Model and DAX Logic

Test Area	Internal Logic Tested	Expected	Actual	Status
TREATAS Filter Context	TREATAS maps dim_material filter into analytical views	Per-material values in table visual	Correct per-material values after fix	Passed
Dead Stock Measures	FILTER iterates vw_silver_ra_mc_logic for is_mc_dead_stock = 1	1 material, $16,872	1 and $16,872 on KPI cards	Passed
Expired Measures	FILTER iterates vw_silver_ra_mc_logic for is_mc_expired = 1	2 materials, $13,263	2 and $13,263 on KPI cards	Passed
Customers Affected	DISTINCT + SELECTCOLUMNS counts unique customers where is_stockout_risk = 1	8 distinct customers	8 displayed on KPI card	Passed
Risk Status Column	IF condition classifies stock_cover_days = 0 as Critical, else At Risk	Critical for 0 cover, At Risk otherwise	Labels display correctly in Q6 table	Passed

5.5.3 User Testing

5.5.3.1 Bronze Pipeline Usability Review

User testing for the Bronze Layer was conducted by Member 1 as the pipeline owner, with feedback gathered from Member 2 and Member 3 who depend on the Bronze Layer outputs as inputs to their respective Silver Layer components. Testing was structured around three

key interaction points which are triggering the pipeline, monitoring its execution, and locating the output directories in the storage account.

The pipeline trigger was initiated without difficulty through the Trigger Now function in ADF Studio. The ADF Monitor tab provided clear activity-level visibility into execution progress, displaying the status, duration, and result for all six Copy Data activities and the subsequent Data Flow activity. Member 2 and Member 3 were both able to independently locate the bronze/validated/ output directories in the Azure Portal and confirm the correct ABFSS path for use in their respective Synapse Notebooks without requiring assistance from Member 1.

One procedural note was established during testing, when re-triggering the pipeline after a source file update such as the addition of the lot_expiry_date column to inventory_snapshot.csv, existing files in bronze/validated/ should be deleted prior to re-execution to prevent stale data from persisting alongside the new output. The pipeline re-ran successfully following this procedure without any modification to the pipeline configuration itself. Overall, the Bronze Layer pipeline was found to be operationally clear and accessible to all relevant team members.

5.5.3.2 RA and MC Output Review

User testing was conducted by reviewing whether the generated RA and MC provision outputs were understandable and suitable for reporting purposes. The Silver Layer views were rerun using the fixed reference date of 31 March 2026. The output from vw_silver_ra_mc_logic was checked to ensure that all important fields were available and clearly presented for dashboard development.

For Q2, the output included material name, category, quantity on hand, 12-month consumption, and excess quantity, which provided sufficient detail to identify and communicate Recoverable Asset findings to business users. For Q3, the output included last consumed date and months since consumed alongside the dead stock flag, making it straightforward to understand why each material was classified as MC Dead Stock. For Q4, the output included the lot expiry date alongside the expired flag, allowing business users

to clearly see which materials had passed their expiry. For Q5, the output included the individual provision amount per material as well as the aggregated total, providing both detail and summary views suitable for financial reporting.

Based on the latest results, the system identified 10 recoverable assets, 1 material flag at dead stock stage, 2 materials were expired, and provision amount for dead stock and expired stock is USD 30144. The query outputs were found to be clear, relevant, and complete. All fields required for Power BI dashboard visualisation were present, and the results were considered suitable for use in the Gold layer and dashboard development stage.

5.5.3.3 Stockout and Sales Impact Output Review

User testing was conducted by reviewing whether the generated stockout and sales impact outputs were understandable and suitable for reporting purposes. After the updated mock dataset was reloaded, the Silver Layer views were rerun using the fixed reference date of 31 March 2026. The output from vw_silver_stockout_risk was checked to ensure that important fields such as material name, category, quantity on hand, reorder level, stock cover days, and lead time days were available for dashboard development.

The output from vw_silver_impacted_sales_orders was also reviewed to ensure that affected customer orders could be clearly identified. Fields such as sales order ID, customer name, finished product, material name, required delivery date, and quantity ordered were included in the result. These fields were considered suitable for Power BI dashboard visualisation because they support the analysis of stockout risk and customer order impact.

Based on the updated results, the system identified 5 materials below reorder level, 2 materials at stockout risk, 13 impacted sales order-material records, and 8 affected customers. Overall, the query outputs were found to be clear, relevant, and suitable for use in the Gold Layer and dashboard development stage.

5.5.3.4 Gold Layer Data Readiness Review

User testing for the Gold Layer was conducted as a data readiness review to confirm that the Gold Layer objects are accessible, understandable, and ready for consumption by

Member 5 (Power BI dashboard) and by the project team. The review involved cross-checking the Gold Layer outputs against the results reported by upstream members and verifying that all required data is available for dashboard development.

The Gold Layer data readiness review confirmed that all required data objects are in place, all business question answers are verified, and the layer is fully ready for Power BI dashboard development. The following guidance was provided to Member 5 to ensure a smooth integration:

- Use Import Mode in Power BI instead of DirectQuery, as the Synapse Analytics SQL Pool is optimized for batch queries rather than live interactive queries.
- Boolean flag columns such as is_stockout_risk and is_below_reorder are stored as BIT (0 or 1) values. In Power BI, filters should use = 1 rather than TRUE.
- For Q7 visuals, use fact_sales_impat directly, it contains full order and customer details joined with stockout risk context, and includes the derived delivery_risk_days urgency metric.
- Materials with months_since_consumed = 999 represent materials that have never been consumed. It is recommended that Member 5 applies a conditional column in Power BI to display these as 'Never Consumed' for better readability in dashboard visuals.
- Use delivery_risk_days in fact_sales_impact for an urgency bar chart, the positive values represent orders at highest risk of missing their delivery date and should be highlighted.

Overall, the Gold Layer was successfully developed, tested, and verified. All seven business questions can be answered, and the layer is ready to serve as the data foundation for the project's final Power BI reporting deliverable.

5.5.3.5 Power BI Dashboard Usability Review

User testing for the Power BI Dashboard was conducted by Syarifah Dania as the dashboard owner, with feedback gathered from the project leader and one other team member who

reviewed the dashboard from the perspective of a business stakeholder with no technical background in the underlying pipeline.

Testing was structured around three key interaction points: navigating between the two dashboard pages, interpreting the KPI cards and chart visuals without requiring explanation, and using the visual-level filters to understand the data context shown on each visual.

The page navigation between Inventory Health and Operational Risk was straightforward, with each page title clearly communicating the business questions it addressed. The KPI cards on each page were identified immediately as the headline numbers. Reviewers were able to state the total MC provision amount, the number of stockout risk materials, and the number of affected customers without assistance.

One usability improvement was identified during the review. The Q4 donut chart was noted as particularly effective for communicating the financial exposure breakdown, as reviewers could immediately identify Acetone as the dominant contributor to expired material provision without reading the table. The Q7 heatmap matrix was also well received, as the gradient red colouring allowed reviewers to identify the most impacted customers at a glance without counting individual cells.

One minor issue was identified on the Operational Risk page: the Q6 table initially used raw Sum of lead_time_days and Sum of stock_cover_days column references, which returned inflated totals identical to the quantity on hand issue resolved earlier. This was corrected by replacing those references with the Lead Time Fixed and Stock Cover Fixed TREATAS measures. Following the correction, all Q6 table values were confirmed correct by the reviewer.

Overall, the dashboard was found to be clear, visually engaging, and suitable for presentation to business stakeholders as a self-contained inventory risk reporting tool. The two-page structure was noted as an improvement over a three-page layout, as it reduced navigation steps while maintaining a logical separation between inventory health and operational risk content.

5.6 Summary

For the chapter 5 has implemented and validated the full PPG Recoverable Assets (RA) and Inventory Risk Management (IRM) data engineering pipeline, starting from Bronze layer ingestion and data quality checks, through Silver layer business logic for RA, Magna Carta, and stockout risk, up to the Gold layer star schema and Power BI dashboard for inventory analytics. The results of Black Box, White Box, and User Testing confirm that the pipeline correctly applies the defined business rules, produces consistent analytics-ready data, and delivers dashboards that stakeholders can interpret to support financial provisioning and operational decisions. Overall, the work in this chapter demonstrates that the proposed Medallion-based architecture on Microsoft Azure is technically feasible, reliable, and suitable as a foundation for inventory risk analysis at PPG.

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